Exact Addressing, Proof Geometry, Locality, Verifier-Safe Hilbert Search, and Formalized Scientific Hypothesis Generation

PRELIMINARY RELEASE 3.3 - REPAIRED APPENDIX AND RATIONAL CORE

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Abstract 0 · predecessor

What is established? On the explicit non-forced 3CNF family F_k, a computable residual-cover feature isolates the right first action in a proper two-action Hilbert prefix, with distribution-free locality surplus $\Lambda \geq 7/8$. Under the frozen indexed-access and charged candidate-inspection-equivalent cost model, the lazy-prefix treatment saves at least $3k - 11/2 > 0$ in expected charged search cost for every $k \geq 2$. The improvement comes from the declared structural feature and search policy, not from a reversible address change by itself. It is not a polynomial-time SAT algorithm or a general 3CNF speedup theorem.

Why can the geometry be trusted? The manuscript then reconstructs exact rational and WFF identity, finite Hilbert traversal, intrinsic verifier-valid path distance, and the calibration required to turn coordinate displacement into a certified lower bound. The geometric bridge layer treats target potentials, geodesic certificates, quotients, macro shortcuts, discrete curvature, and route multiplicity; GBG/GBG$^+$ adds rule-complete states and exact addressed histories without promoting search scheduling to verifier authority.

What follows from the core? Exact WFF-address transport preserves specified 3CNF syntax/semantics under fixed computable renamings; an explicit, selected consequence-mining catalogue records low-parent results without claiming exhaustive closure. A conjecturing case study separately refutes a strong curvature-burden conjecture, proves conditional repairs, and retains a carefully qualified fictional hypothesis-generation pilot. The closing open-problem section reconstructs next steps in proof geometry, locality, implementation, and controlled hypothesis generation.

Editorial architecture inherited from Release 2.4. To present the results before reconstructing their machinery, this edition orders the certified locality/search theorem package first and then proceeds through foundations, transports, consequences, conjectures, and open research. It removes the Central Composition Problem to its own one-section PDF; it transfers the IAE historical case study, interpretive limits, Triangle/password discussion, scientific null model, joint experiments, synthetic benchmark, and formal scientific targets to a separate companion PDF. Original section numbers have been reset; theorem labels and proof content remain tied to the original statements. Appendices and bibliography are retained, with the original historical audit explicitly attributed to Release 2.3. This editorial reorganization is not a new independent formal-verifier audit.

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Updated contents — working repair edition

The original contents pages are predecessor material. This sheet gives corrected PDF-page links.

Abstract 0 is retained; Abstracts 1 and 2 are tentative. Sections 1–15 and Appendices A–E stay in order.

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PDF page numbers are for this assembled working edition; original printed leaf numbers remain historical.

Exact Addressing — Editorial Front Matter

Abstract alternatives 0 / 1 / 2 and subsection roadmap

Working Repair Edition 3.3.1 — 12 October 2026

Status. The existing paper's page-1 abstract is Abstract 0 (predecessor), left intact. Abstracts 1 and 2 remain *tentative*, not replacements already approved for publication. All existing sections and appendices retain their order.

Reader's quick entry

Exact addresses preserve formula identity; meaningful features may improve the order in which proof candidates are examined; only legal inferences and a verifier certify results. The positive charged-search examples concern a particular 3CNF family and explicitly chosen comparators. The separate Novelty section contributes conditional combinatorial consequences, without promoting them to unverified shell +5/+6 discoveries.

Abstract 0 — predecessor

The original abstract remains on original PDF page 1. It is not silently replaced.

Tentative Abstract 1 — narrative roadmap

Result and boundary (Part I, Sections 1–4). We ask whether a reversible address map, an informative feature, and an adaptive but verifier-safe search schedule can yield a genuine charged advantage. After distinguishing the claim from a general SAT speedup (Section 1), we define locality surplus and a comparable random search baseline (Section 2). A concrete non-forced 3CNF family then admits a residual-cover classifier whose two-cell action region contains an optimally tie-broken first action (Section 3). The IAE-derived lazy-prefix construction turns this concentration into conditional cost theorems, and the supplementary two-candidate comparison makes both recognition and differential costs explicit (Section 4).

The geometry beneath the result (Part II, Sections 5–11). Exact rational identities (Section 5) and transported WFF addresses (Section 6) preserve syntax and legal proof paths but do not shorten proofs by renaming them. Finite Hilbert traversals offer coverage and scheduling (Section 7); seven comparisons among identity, navigation, target, and proof distances prevent a numerical map from being mistaken for deduction (Section 8). Rectilinear routes are compared with intrinsic proof geodesics (Section 9). Path-calibrated bridge results then formalize potentials, residuals, shortcuts, curvature, route multiplicity, and the precise scope of their claims (Section 10). GBG/GBG+ completes the operational picture with rule-complete states, exact histories, and verifier-governed board actions (Section 11).

Consequences, discovery, and next tests (Parts III–IV, Sections 12–16). Fixed satisfiability-preserving symmetries generate exact address transports and a distinct transported-locality theorem (Section 12); selected low-fan-in consequence mining yields 24 descendants, currently reaching local shell +4 (Section 13). Eureka and IAE are retained as explicitly documented *hypothesis-generation and iterative-search methods*, not as independent proof or novelty certificates; a failed curvature conjecture and a fictional pilot make this limit concrete (Section 14). The open research program identifies the missing transfer, benchmarking, and implementation conditions (Section 15). Finally, a new Novelty section records computer-assisted conditional consequences of the fully charged IAE-20 two-target benchmark without claiming certified outer-shell depth or literature priority (Section 16). The central principle is unchanged throughout: addresses guide, legal rules constrain, and verifiers certify.

Tentative Abstract 2 — three paragraphs, verbatim

Your paper asks a surprisingly simple question: **Can we give every mathematical formula an exact address, use geometry to search for proofs more intelligently, and still leave proof validity entirely in the hands of a verifier?** Its first major result is a restricted but positive example: on an explicit family of 3CNF formulas, a structural feature identifies a small, promising search region. Under carefully specified cost assumptions, searching that region can beat a uniformly random candidate order. This is not a general SAT breakthrough; it establishes that useful search locality can exist without changing the underlying logic.

The middle of the paper explains why this is mathematically legitimate. Exact addresses identify formulas without changing their meaning; Hilbert traversal organizes candidate searches; proof distance measures legal deductions rather than numerical closeness; and calibrated geometry can sometimes provide rigorous information about proof paths. GBG and GBG+ extend this perspective to proof states, histories, and scheduling. The essential distinction is that **a map can guide a search, but it cannot make an invalid inference valid**.

The final part studies consequences and discovery. Satisfiability-preserving transformations transport logical properties across addresses. Consequence Mining constructs a selected catalogue of 24 derived results, currently reaching shell +4. Eureka proposes potentially useful ideas, while IAE iteratively searches for and tests them; neither method supplies proof authority. The larger research challenge is to find navigation strategies whose benefits survive full cost accounting, independent verification, and comparison with fair alternatives.

Tentative Abstract 1 — detailed subsection waypoints

Annotated contents and extended abstract

This is the detailed reader's roadmap. Every existing subsection has a one-sentence waypoint here, without forcing dozens of sentences into the three-paragraph publication abstract. Page numbers are intentionally omitted in this extended roadmap; the working PDF includes a linked updated contents sheet, while the existing original pages retain their own printed numbering.

1. A positive charged-search result, and the boundary of its meaning

Opens with the precisely delimited positive 3CNF result and explains why it is not a general SAT theorem.

2. Locality surplus and expected search advantage

Defines locality surplus and a charged random-order baseline, separating action enrichment from address naming.

3. Audited multi-step structural-concentration theorem

Proves the family-specific non-forced 3CNF concentration theorem. **3.1 Audited fallback and total comparable cost** Freezes fallback decisions, propagation costs, identity tie-breaks, and common preprocessing.

3.2 Residual-cover lemma Identifies the two useful actions by a visible clause-cover predicate.

3.3 Feature-induced Hilbert block Explains that a structural feature, not an unassisted Hilbert curve, produces the two-cell prefix.

3.4 Distribution-free locality bound and the original benchmark Proves a family-wide surplus bound and reports the earlier nine-instance calibration.

4. IAE-derived Lazy Hilbert Exact-Address Speedup Theorem

Converts locality into model-relative cost comparisons, with separate recognition and differential-cost qualifications. **4.1 IAE provenance and release history** Treats iterative Eureka as a historical discovery route, not an additional proof rule.

4.2 Indexed Exact-Address Access Hypothesis States the lookup assumption that reversible addressing alone cannot supply.

4.3 Lazy Hilbert implementation and cost ledger Separates common cost from extra treatment work and records the later executable two-candidate comparison.

4.4 What the theorem does not claim Restricts the result to its frozen family, baseline, recognizer, and charged cost model.

5. Exact rational addressing

Builds effective rational identity codes while distinguishing dense from isolated address images.

6. WFF identity and inference-action addressing

Explains what exactly is encoded and which logic is preserved under transport. **6.1 Strengthened Exact Addressing Theorem** Shows that rule-transported bijective recoding preserves derivability, proof paths, and minimum proof lengths.

7. Two roles for Hilbert traversal

Distinguishes exhaustive finite-grid scheduling from learned feature-guided exploration. **7.1 Finite face-continuous Hilbert probes** Shows finite cell coverage and axis-parallel steps without claiming every rational candidate is thereby settled.

8. What Distance Are We Talking About?

Provides the paper's essential dictionary separating three inequivalent distances. **8.1 Exact-address distance** Separates reversible formula identity from reasoning cost.

8.2 Feature or navigation distance Introduces coordinates designed to direct search, not to certify deductions.

8.3 Logical proof distance Measures shortest legal proof routes in the formal calculus.

8.4 Target-relative logical distance Relates distances to a designated target and state.

8.5 When can geometric distance say something rigorous about proof distance? States the extra calibration needed to turn coordinate displacement into a lower bound.

8.6 Distance and Hilbert traversal Separates a path through navigation cells from a path through legal deductions.

8.7 The four questions Collects the distinct questions an address, a map, and a proof geometry answer.

9. Rectilinear routes and intrinsic proof geodesics

Compares rectilinear navigation routes to shortest verifier-valid paths.

10. Path-Calibrated Bridge Theorems for Exact Addressing

Turns the address metaphor into conditional path and cost theorems. **10.1 Self-contained path and trace setup** Specifies the directed path space, costs, and trace certificates.

10.2 Calibrated coordinates as rigorous lower bounds on proof distance Proves lower-bound rules only where coordinate changes are certified against legal edge costs.

10.3 Premise-doorway residuals and address isolation Tests when the absence of exact premises can or cannot be inferred from coordinate distance.

10.4 Target potentials, exact geodesic certification, and bottlenecks Uses admissible potentials and matching certificates to reason about optimal paths.

10.5 Approximate geodesic navigation and cartographic regret Bounds heuristic detours relative to the best legal route under stated assumptions.

10.6 Re-addressing, trading, and the separation of maps from roads Explains why changing coordinates need not change the legal route structure.

10.7 Cross-fiber distortion, quotient geometry, and certified shortcuts Studies collapsing or projecting route information without losing necessary verifier data.

10.8 Discrete curvature on theorem graphs Introduces graph curvature as a structural observable rather than a universal difficulty oracle.

10.9 A geometric dictionary for theorem space Names the geometric terms used for distinct logical objects.

10.10 Route multiplicity and cartographic signatures Counts alternative proof histories and separates raw from meaningful route multiplicity.

10.11 Effect on the Exact Addressing program States which navigation claims genuinely survive the calibration tests.

11. GBG and GBG+ as an exact operational substrate

Provides a verifier-safe operational machine underneath the map. **11.1 Rule-complete proof states and the state-lift of multipremise inference** Recasts multipremise rules on fully specified proof states.

11.2 Exact state addresses and continuation sufficiency Encodes precisely the state information needed to determine legal continuations.

11.3 Exact history addresses, endpoint projection, and route multiplicity Records proof histories without confusing their endpoint images.

11.4 GBG+: exact addressing with a board-player Adds a board-player to the addressed operational structure.

11.5 Passive board extension and projected proof-geometry invariance Proves that a passive extension preserves the declared projected geometry.

11.6 Active board actions, scheduling, and verifier sovereignty Allows scheduling interventions while preserving independent verification.

11.7 Consequences for Exact Addressing Extracts the consequences for coding, history, and multi-agent navigation.

12. Satisfiability preservation under exact WFF-address transport

Derives useful transported relations without attributing logical content to coordinates alone. **12.1 Satisfiability-Preservation Lemma** Shows that a fixed variable renaming preserves satisfiability.

12.2 Address-space transport Carries that symmetry through a reversible formula-address map.

12.3 Symmetry-Transported WFF-Address Locality Theorem Computes locality for a transported-neighbor relation, not necessarily for raw metric neighbors.

12.4 Interpretation and limitation Explains why transported locality is not a claim about independent Hilbert metric structure.

13. Consequence Mining and Interesting Theorems

Mines 24 selected descendants reaching local shell +4, rather than a universal consequence closure. **13.1 Protocol: theorem DAG, normalized retained catalogue, and parent gate** Defines selected low-parent consequences without claiming exhaustive closure.

13.2 Preliminary Release 1.10 retained shell convention Defines relative shell depth and its dependence on the frozen parent ledger.

13.3 Frozen root theorem vertices Identifies the precise source results from which consequences are mined.

13.4 Parent ledger for the retained interesting-shell theorems Records derivation parents, relative shells, and locally witnessed fan-in.

13.5 Fan-in one and two consequence catalogue Presents twenty retained one- or two-parent results.

13.6 Retained three-parent syntheses Adds four explicitly three-parent synthesis results.

13.7 Interpretation of the mined layer Explains what the selected finite catalogue does and does not certify.

14. Conjecturing, Eureka, and scientific hypothesis generation as discovery spaces

Explains discovery mechanisms and their strict boundaries. **14.1 Conjecturing precedes settlement** Separates proposing an idea from verifying it as a theorem.

14.2 Earlier theorem-space conjectures as generated hypotheses Tracks prior conjectures without granting them post hoc proof authority.

14.3 Worked conjecturing example: the Geodesic Curvature-Profile Conjecture Shows a proposed curvature rule that was refuted and replaced by conditional results.

14.4 From mathematical conjectures to scientific hypotheses Sets an evidence interface where test outcomes, not formal provability, govern scientific evaluation.

14.5 Recovered pilot: evidence-conditioned basin escape in a fictional differential-diagnosis task Treats the reconstructed fictional pilot as protocol calibration, not clinical evidence.

15. Open research directions: from certified corridors to general discovery

Records the transfer, benchmarking, implementation, and general-composition questions still unresolved.

16. Novelty: Audited Computer-Assisted Conditional Consequences

Records audited conditional computer-assisted consequences without unsupported +5/+6 promotion. **16.1 What is new, and what is not** States the frozen cost model and the distinction between mathematical consequence and verified historical novelty.

16.2 Exact stopping-position law and a conditional high-probability result Derives the exact random permutation stopping law and a lower bound for positive charged advantage.

16.3 Robustness to separately charged implementation overhead Tests when additional deterministic charges preserve the conditional gain.

16.4 A conditional randomized policy hedge Analyzes a finite randomized mixture without claiming proof-search completeness.

16.5 Quantitative bridge to the existing locality formula Relates the locality surplus to the random comparator bounds through their shared family parameter.

16.6 Proof-status ledger, shell discipline, and Eureka/IAE Distinguishes proposed computer-assisted consequences from certified +5/+6 novelty and credits prompts only as discovery methodology.

Appendices and their waypoints

A. Frozen IAE / AMLD+0 Prompt Preserves a historical prompt, whose frozen-status claims must be separated from independently documented timestamps.

B. IAE Speedup-Theorem Provenance and Release History Records the Scout hint as discovery provenance without granting it theorem authority.

C. Consequence-Mining Protocol and Release History Explains why the 24-term catalogue is selected rather than exhaustive.

D. Repair Report and Release Audit Distinguishes mathematical theorem proofs from an incompletely certified historical run.

E. Universal Proof Horizons and Exact-Address Geometry: Repaired Reconciles cost minima with geometrical search language.

E.1 Two notions of minimum and necessary interface Distinguishes least proof cost from operational search completion.

E.2 Exact re-addressing and canonical codes Makes canonical identity transport explicit.

E.3 Potentials and horizon certificates States the certificate interface required for horizon claims.

E.4 Passive GBG+, finite layers, and policies Treats proof-state layers without creating new proof authority.

E.5 Consequence ledger and audit boundary Records what follows and where verification ends.

Part I

The Certified 3CNF Result: Locality and Charged Search

1 A positive charged-search result, and the boundary of its meaning

A precise positive result is the main entry point of this release. For every $k \geq 2$, the explicitly defined family

$$F_k = \bigwedge_{i=1}^{k} (x \vee u_i \vee v_i) \ \wedge \ \bigwedge_{i=1}^{k} (\neg x \vee y \vee z_i)$$

admits a theoremhood-independent residual-cover feature whose pre-branch two-action region contains every retrospective minimum under the declared cost/tie-break system. The ensuing theorem proves locality surplus at least 7/8 for any distribution supported on the family. Under a further frozen indexed-access and candidate-inspection ledger, its lazy Hilbert-prefix specialization has a strictly positive expected net charged advantage for every $k \geq 2$.

The order is deliberately *result, foundation, further consequences.* Part I states the finite probabilistic accounting and the certified 3CNF results first. Part II steps back to exact WFF addressing, Hilbert scheduling, proof distance, calibrated bridges, and GBG/GBG$^+$ state/history representations. Part III returns to exact WFF transport, selected theorem-DAG consequences, curvature, conjecturing, and a qualified scientific-hypothesis pilot. Part IV regenerates the open research directions. The change of order is rhetorical; it does not reverse logical proof dependencies.

The framework separates four levels:

formal object and legal action $\longrightarrow$ exact identity/learned geometry $\longrightarrow$ search schedule $\longrightarrow$ verifier.

Coordinates encode and organize; a proof-system transition relation determines which moves are legal; the declared cost ledger governs search comparisons; the verifier determines admissibility. Invertible address recoding alone cannot improve optimal intrinsic proof length or certify an inference. Calibrated features and changed legal policies can organize discovery, but must not be mistaken for semantic change.

Release boundaries. The separately extracted Central Composition Problem remains open in general, although a family-specific instance is proved here. The IAE discovery chronology, Triangle/password synergy, and four scientific-method sections are preserved in a separate companion rather than silently deleted. Neither the source historical run nor the fictional diagnosis pilot constitutes a fully pre-frozen scientific evaluation. The 2.4 edit changes manuscript architecture; mathematical strength does not increase merely because sections move.

The bibliography documents the prior works underlying this synthesis, including exact rational coding [1], WFF/Hilbert correspondence [2], MIU and intrinsic proof geometry [3], calibrated path geometry [6], verifier-safe GBG/GBG$^+$ dynamics [17], and omnicartographic approximate navigation [16].

2 Locality surplus and expected search advantage

Let A be a finite candidate-action set with $|A| = N$, let $A^\star$ be the next proof-enabling action, and let $G \subseteq A$ have $|G| = g$. Write

$$p = \Pr(A^\star \in G).$$

Under the conditional-uniformity model of the Locality Advantage Theorem, searching G before its complement gives

$$E[R_G] = \frac{N + g + 1 - pN}{2},$$

while a uniformly random no-repeat ordering gives

$$E[R_{\text{random}}] = \frac{N+1}{2}.$$

Therefore

$$\boxed{E[R_{\text{random}}] - E[R_G] = \frac{N}{2}\left(p - \frac{g}{N}\right).}$$

Define

$$\Lambda(G) = p - \frac{g}{N}.$$

Then $\Lambda(G) > 0$ is precisely the condition for strict expected-rank improvement in this model [4].

For multiple disjoint regions with probability masses p_i and sizes n_i, complete-block search is optimized by nonincreasing p_i/n_i. An adaptive controller can replace the static masses by posterior masses $p_i(t)$ and compare

$$\delta_i(t) = \frac{p_i(t)}{n_i(t)},$$

while reserving a fairness quota so that incorrect estimates cannot starve regions forever. This provides a disciplined interpretation of "creativity" as adaptive reallocation rather than unstructured random deviation [4].

3 Audited multi-step structural-concentration theorem

For every integer $k \geq 2$, define

$$F_k = \bigwedge_{i=1}^{k} (x \vee u_i \vee v_i) \wedge \bigwedge_{i=1}^{k} (\neg x \vee y \vee z_i),$$

with all variables distinct. Every clause has three literals, there are no unit clauses, and unit propagation forces no assignment initially. The candidate grammar contains both truth assignments to every unassigned variable, hence

$$N(F_k) = 2(2 + 3k) = 4 + 6k.$$

The actions $(x, 1)$, $(x, 0)$, and $(y, 1)$ each delete exactly k clauses immediately. Thus immediate clause-deletion gain has a three-way maximum and does not uniquely identify the retrospectively best branch.

3.1 Audited fallback and total comparable cost

Fix the deterministic fallback order

$$y \prec x \prec u_1 \prec v_1 \prec \cdots \prec u_k \prec v_k \prec z_1 \prec \cdots \prec z_k.$$

After a candidate action, exhaust unit propagation. If clauses remain, assign the first unassigned variable still occurring in the residual formula according to this order to true, exhaust unit propagation again, and repeat until the formula is settled. This procedure terminates on every descendant of every F_k.

For a first action a, let $B(a)$ be the number of non-forced fallback decisions, let $U(a)$ be the number of unit-propagation assignments, and let $E(a) \in \mathbb{N}$ be the remaining branch-dependent charged work under any fixed reproducible implementation ledger. Use the total lexicographic cost

$$C_{F_k}(a) = (B(a), U(a), E(a))$$

ordered lexicographically. This avoids any unproved large-constant domination assumption. If two actions have exactly the same cost triple, the unique retrospective action is the one with the smaller exact syntactic identity code $\mathrm{Id}(a)$. The tie-break therefore depends only on exact identity after equality of the declared cost and never on satisfiability or a successful future branch.

All pre-branch work is also charged. Let $H(F_k)$ denote the complete state-level overhead for exact encoding, computation of every candidate's structural feature, construction of the finite Hilbert schedule, neighborhood construction, and bookkeeping. The total experiment ledger for choosing candidate a is

$$T_{F_k}(a) = H(F_k) + C_{F_k}(a).$$

Because the same full pre-branch computation $H(F_k)$ is performed before selecting any candidate, it is common to all candidates and does not change the minimizing action. No runtime speedup claim is inferred merely from omitting this common term in branch comparisons.

Direct analysis gives

$$B(x,1) = B(y,1) = 1, \qquad U(x,1) = U(y,1) = 0.$$

Indeed, after $(x,1)$ the residual is $\bigwedge_i (y \vee z_i)$ and the fallback sets $y = 1$; after $(y,1)$ the residual is $\bigwedge_i (x \vee u_i \vee v_i)$ and the fallback sets $x = 1$.

For $(x,0)$, the residual formula is $\bigwedge_i (u_i \vee v_i)$ with disjoint supports, so $B(x,0) = k \geq 2$. For $(y,0)$, the fallback sets $x = 1$ and unit propagation then assigns all z_i, giving $B(y,0) = 1$ and $U(y,0) = k > 0$. For an initial assignment to any u_i, v_i, or z_i, at least two non-forced fallback decisions are required. Therefore every lexicographic minimum of C_{F_k} lies in

$$\boxed{\{(x,1),(y,1)\}}.$$

The explicit exact-identity tie-break selects a unique action if the two cost triples coincide.

3.2 Residual-cover lemma

Define the pre-branch residual-cover feature $R_{F_k}(a)$ to be 1 exactly when, after applying only the declared one-step syntactic simplification for a, some literal occurs in every remaining clause; otherwise let it be 0. This is a theoremhood-independent syntactic computation.

For $(x,1)$ every residual clause contains y, and for $(y,1)$ every residual clause contains x. For $(x,0)$ the residual supports are disjoint. For $(y,0)$ both a positive-x family and a negative-x family remain. For assignments to u_i or v_i, at least one positive-x clause and one negative-x clause remain. For an assignment to z_i, the assumption $k \geq 2$ ensures that at least one negative-x clause remains together with the positive-x family. No single literal is common to both families. Hence

$$\boxed{R_{F_k}(a) = 1 \iff a \in \{(x,1),(y,1)\}} \qquad (k \geq 2).$$

3.3 Feature-induced Hilbert block

Sort the candidate actions by the theoremhood-independent key

$$(1 - R_{F_k}(a), \mathrm{Id}(a))$$

and assign this deterministic order to consecutive cells of a fixed finite Hilbert traversal. Then

$$G(F_k) = \{(x,1),(y,1)\}$$

is a nonempty proper contiguous block with $g(F_k) = 2$.

This step is intentionally described as *feature-induced Hilbert contiguity*. The residual-cover classifier, not the Hilbert curve by itself, creates the two-action concentration: the key explicitly places $R = 1$ actions first, and Hilbert traversal supplies a deterministic contiguous schedule for the resulting finite representation. Therefore this construction does *not* establish that an independently chosen Hilbert embedding would spontaneously reveal the same locality. That stronger exogenous-geometry question remains open.

3.4 Distribution-free locality bound and the original benchmark

Let μ be any probability distribution supported on integers $k \geq 2$, and draw F_k according to μ. The unique retrospective action selected by the audited cost and tie-break always belongs to $G(F_k)$, so $p = 1$. Since

$$\frac{g(F_k)}{N(F_k)} = \frac{2}{4+6k} = \frac{1}{2+3k} \leq \frac{1}{8},$$

we have

$$\beta = \mathbb{E}_\mu\left[\frac{g(F_k)}{N(F_k)}\right] \leq \frac{1}{8}$$

and therefore

$$\boxed{\Lambda = p - \beta \geq \frac{7}{8} > 0.}$$

Thus the proof is not intrinsically limited to the original nine-state benchmark.

For the original uniform distribution on $k = 2, \ldots, 10$,

$$\beta = \frac{1}{9}\sum_{k=2}^{10} \frac{2}{4+6k} = \frac{109736843}{1816054240} \approx 0.0604259722,$$

so

$$\boxed{\Lambda = 1 - \frac{109736843}{1816054240} \approx 0.9395740278 > 0.}$$

Multi-Step Structural Concentration Theorem (audited form)

There exists an infinite explicitly parameterized class of non-forced 3CNF search states for which immediate one-step clause-deletion gain does not uniquely identify the retrospectively best branch, yet a computable theoremhood-independent pre-branch structural feature contains every retrospectively minimum-cost action in a proper feature-induced contiguous Hilbert block with strictly positive locality surplus. Under the audited lexicographic fallback-cost model, the result holds for every $k \geq 2$; for every distribution supported on this family, $\Lambda \geq 7/8$.

Proof. The immediate-gain tie and fallback analysis above show that every cost minimum lies in $\{(x, 1), (y, 1)\}$. The residual-cover lemma proves that this set is exactly the $R = 1$ set. The deterministic key $(1 - R, \mathrm{Id})$ therefore places the entire minimizing set in the proper two-action contiguous block $G(F_k)$. Hence the uniquely tie-broken right action belongs to $G(F_k)$ for every $k \geq 2$, giving $p = 1$. Since $g(F_k)/N(F_k) \leq 1/8$, the distributional baseline satisfies $\beta \leq 1/8$, so $\Lambda \geq 7/8 > 0$. □

4 IAE-derived Lazy Hilbert Exact-Address Speedup Theorem

4.1 IAE provenance and release history

This section records a theorem produced by a subsequent Iterative Adaptive Eureka / AMLD+0 run whose frozen objective was to convert the certified F_k locality result into a fully charged net expected-search advantage. It is therefore designated an *IAE-derived theorem.* The designation records the discovery protocol, not the proof authority: the theorem is admitted here only because the statement below has a direct verifier-checkable proof from the declared hypotheses and cost model.

The first form of the theorem appeared in Preliminary Release 1.5 and used a conditional-uniformity model inside the certified two-action block. Preliminary Release 1.7 strengthens the main statement: conditional uniformity inside that block is no longer required. No claim of exhaustive external historical priority is made without a separate prior-art audit.

4.2 Indexed Exact-Address Access Hypothesis

Assume that the exact-address state representation shared by both baseline and treatment maintains an index from each legal named action to its candidate record and exact identity code. Once the state has been parsed and indexed, retrieving a specified action record or its exact identity requires one charged primitive indexed access. The construction and maintenance of the shared index are common representation costs and are not charged uniquely to the Hilbert treatment.

This assumption is stronger than exact identity alone. Exact Addressing by itself guarantees neither constant-time lookup nor a speedup; the indexed-access hypothesis is therefore stated explicitly rather than hidden inside the address notation.

4.3 Lazy Hilbert implementation and cost ledger

For the certified family F_k, the residual-cover lemma gives

$$R_{F_k}(a) = 1 \iff a \in \{(x, 1), (y, 1)\}.$$

The Hilbert-local treatment uses this proved symbolic characterization directly on the visible syntax. It does not recompute residual cover independently for all $4 + 6k$ candidates and does not materialize the complement of the Hilbert order. Instead it lazily exposes the two-cell Hilbert prefix corresponding to

$$G(F_k) = \{(x, 1), (y, 1)\}.$$

The remainder of the Hilbert schedule is constructed only if the prefix fails; under the certified model the uniquely tie-broken right action belongs to this prefix, so the treatment inspects at most two candidates before reaching it.

Freeze the following extra primitive ledger for obtaining and using the lazy prefix:

Extra treatment operation	Count
Exact-address action lookup for $(x, 1)$	1
Exact-address action lookup for $(y, 1)$	1
Exact-identity comparison for deterministic within-block order	1
Hilbert-prefix access	1
Two-action neighborhood construction	1
Iterator/bookkeeping initialization	1
Total	6

No clause or literal scan is charged in this extra ledger because the parsed exact-address state and its action index are common infrastructure available to both systems. No retrospective branch cost, satisfying assignment, winning-action label, or hidden solver output is used.

For the candidate-inspection-equivalent version of the theorem, freeze a normalized unit-cost inspection model: processing any one inspected candidate through the ordinary common search pipeline—including ordinary indexed retrieval, state update/evaluation, and verifier work for that inspection—is charged exactly one candidate-inspection-equivalent unit. This is an abstract calibrated search-cost model, not a wall-clock claim. Consequently the ordinary variable inspection costs are exactly the rank terms R_{random} and R_H; they do not cancel. Only fixed state-level setup costs that are identical for baseline and treatment cancel. Impose additionally that each treatment-specific primitive operation in the table costs at most one candidate-inspection-equivalent unit. Then

$$\boxed{C_H(k) \leq 6.}$$

Lazy Hilbert Exact-Address Speedup Theorem (distribution-free within the certified block)

For every $k \geq 2$, consider

$$F_k = \bigwedge_{i=1}^{k} (x \vee u_i \vee v_i) \wedge \bigwedge_{i=1}^{k} (\neg x \vee y \vee z_i)$$

with the candidate grammar, residual-cover theorem, feature-induced Hilbert block, indexed exact-address access hypothesis, and primitive-cost calibration stated above. Let the baseline inspect the same legal candidates in uniformly random no-repeat order, and let the Hilbert treatment search the certified two-action block first. No probability distribution is assumed over which member of that two-action block is the uniquely tie-broken right action. Then, in this frozen candidate-inspection-equivalent cost

model, the Hilbert treatment has strictly lower expected total charged search cost than the baseline for every $k \geq 2$. More precisely,

$$\boxed{E[C_{\text{baseline}}] - E[C_{\text{Hilbert}}] \geq 3k - \frac{11}{2} > 0.}$$

The speedup changes neither theoremhood, inference legality, shortest-proof length, target distance, nor verifier authority.

Proof. The candidate count is

$$N(F_k) = 4 + 6k.$$

For a fixed target in a uniformly random no-repeat ordering, its expected baseline rank is

$$E[R_{\text{random}}] = \frac{N+1}{2} = 3k + \frac{5}{2}.$$

The structural-concentration theorem certifies that the uniquely tie-broken right action is in the two-action block $G(F_k)$. Because the treatment searches that entire block first,

$$R_H \leq 2$$

deterministically, regardless of the target's within-block position. Therefore the gross expected candidate-rank advantage is bounded below by

$$E[R_{\text{random}}] - R_H \geq 3k + \frac{5}{2} - 2 = 3k + \frac{1}{2}.$$

Under the frozen lazy implementation ledger, the complete extra Hilbert-local overhead satisfies $C_H(k) \leq 6$. By the declared normalized unit-cost calibration, ordinary per-candidate processing is exactly counted by the rank terms R_{random} and R_H and therefore does not cancel. Only fixed state-level setup costs that are identical for baseline and treatment cancel. Hence

$$E[C_{\text{baseline}}] - E[C_{\text{Hilbert}}] \geq \left(3k + \frac{1}{2}\right) - 6 = 3k - \frac{11}{2}.$$

For every $k \geq 2$, $3k - 11/2 \geq 1/2 > 0$, so the treatment has a strict expected net charged advantage throughout the infinite family. The no-speedup-by-addressing corollary is not violated because the gain comes from the extra theoremhood-independent locality feature and lazy scheduling policy, not from invertible recoding alone. □

Conditional-uniformity refinement

If one additionally assumes the earlier conditional-uniformity model inside the two-action block, then

$$E[R_H] = \frac{3}{2}.$$

The exact gross expected rank advantage becomes $3k + 1$, and the preceding proof sharpens to

$$E[C_{\text{baseline}}] - E[C_{\text{Hilbert}}] \geq 3k - 5 > 0.$$

This is retained as an optional refinement, not as a hypothesis of the main speedup theorem.

Section 4 — differential-cost calibration

Status of the existing theorem

The candidate-rank calculation in Section 4 is mathematically valid in its **normalized, identifiable-target inspection model**. It is not by itself a bound on the cost of finding, certifying, or recognizing a minimum-cost 3CNF branch in an unrestricted solver.

Let S(k) be setup genuinely executed by both competitors and let D(k) be **all** extra treatment costs, measured in the same candidate-inspection-equivalent units. D(k) includes any non-shared family recognizer, syntactic shortcut, feature compilation, lookups, Hilbert-prefix scheduling, setup, and identification or stopping test. Nothing may be canceled merely because it is called an address or a shared representation.

Corrected conditional accounting

For the uniformly randomized baseline and the feature-induced two-action treatment, provided inspection of a target is recognizable at the charged cost asserted by the model, E[Rrandom] = 3k + 5/2 and R_H <= 2. Therefore

E[C_baseline] - E[C_treatment] >= 3k + 1/2 - E[D(k)],

assuming no further unmatched baseline/treatment cost terms. In particular the originally displayed margin **3k - 11/2** is justified only when E[D(k)] <= 6. The six listed primitives suffice only if they cover every non-shared operation, or if an independent argument proves that all other operations truly belong to S(k).

A sufficient strict-net-gain condition is E[D(k)] < 3k + 1/2. Without an identifiability/stopping test or differential-cost calibration, Sections 4 and 13 are conditional rank-model claims rather than proved end-to-end SAT runtime improvements.

Relation to Section 3

Section 3 may compute every candidate feature eagerly as part of a common comparison experiment. Section 4 uses a lazy prefix. These are different implementations: their preprocessing is not automatically identical. For a valid ledger, explicitly run and charge the common eager procedure in both competitors or charge the full lazy-specific difference in D(k). This leaf clarifies existing hypotheses; it does not replace the 3CNF residual-cover or locality theorems.

Section 4 – An executable two-candidate comparison

Release 3.3 addition; Section 4's original theorem remains unchanged. This insertion adds a separate, policy-relative result. The original claim that the single-target Hilbert treatment saves at least $3k - 11/2$ under a six-unit extra-cost assumption still depends on the Section 4 differential-cost conditions (leaf 11a). The theorem below does not assume that a target is recognizable on first inspection.

Frozen mathematical input. For $k \geq 2$, let

$$F_k = \bigwedge_{i=1}^{k} (x \vee u_i \vee v_i) \ \wedge \ \bigwedge_{i=1}^{k} (\neg x \vee y \vee z_i), \qquad N = 4 + 6k.$$

Section 3 proves, for the prescribed fallback and lexicographic cost $(B(a), U(a), E(a), \mathrm{Id}(a))$, that every minimizing first action belongs to

$$G_k = \{a_x = (x, 1),\ a_y = (y, 1)\}.$$

The residual-cover predicate identifies exactly G_k, but need not tell which of its two members wins the E comparison.

Two-target recognition lemma. *Given a total, deterministic, charged evaluator of E, it suffices to evaluate a_x and a_y, compare the complete lexicographic costs, and apply the exact-identity tie-break to recover the unique retrospective minimizer. Ordinary SAT success alone cannot serve as the stopping certificate: every initial legal first action on F_k extends to a satisfying assignment.*

Proof. Section 3 excludes all other actions before comparing E. The first two cost coordinates of a_x, a_y agree, so their E values and identity codes decide the winner. For any single first assignment, either complete with $x = y = 1$ (when compatible), with $x = 0, y = 1$ and all $u_i = 1$, or with $y = 0, x = 1$ and all $z_i = 1$. Thus every first action is SAT-extendable. □

Common task and explicit interfaces. Both algorithms must output the same first-action minimizer under the same fixed evaluator and exact-identity order. Both receive a canonical F_k record array with public word-size indices for a_x, a_y (or both perform and pay for the same deterministic recognizer, parser, and index construction). Both execute a common charged inspection routine which reads one action record, checks membership in G_k, and accounts for its per-inspection control and loop work. Inspection does *not* evaluate E.

Algorithms. The treatment (H) obtains the two indexed records as a lazy, feature-induced prefix, inspects both, evaluates their costs in a fixed order, compares, and verifies the selected output. The baseline (B) draws a uniformly random no-repeat order with all random sampling and permutation costs charged, inspects until it has encountered *both* a_x and a_y, then executes *the same* two evaluations, comparison, and verifier. Both programs therefore use the same correctness test and return the same answer.

This is a first-action optimization benchmark, not a theorem about the best general-purpose SAT solver.

Fully charged cost theorem and proof

Frozen charged word-RAM model. Words have $\Omega(\log k)$ bits so array indices and public action labels fit in one word. The common parser and action-array construction run and are charged in *both* executions. A total deterministic candidate-cost evaluator and an independent deterministic output verifier exist and are invoked identically in both runs; they may be computationally expensive. They are not treated as free. The verifier may exhaustively compare every candidate if necessary.

Every inspection invokes the same fixed, history-independent inspection routine. For each outsider $a \notin G_k$, its charged inspection cost is $c_k(a) \geq 1$ word-RAM operation. Each member of G_k is inspected once by each algorithm; their inspection charges therefore cancel. The baseline-only random permutation and control charges not included in inspection costs are $P_k \geq 0$. The treatment alone requires $h_k \leq 6$ additional pointer and prefix-register instructions (two pointer loads, two slot stores, cursor and length initialization). No other unmatched cost is permitted: any such cost must be charged separately and would change the displayed identity. There is no eager materialization of the full Hilbert order. These conditions specify a particular executable comparison model; they do not follow from reversible addressing alone.

Random-order stopping lemma. *For a uniformly random permutation of N actions, let R_B be the position of the later of two specified distinct actions. Then*

$$\mathbb{E}[R_B] = \frac{2(N+1)}{3}, \qquad \Pr\{a \text{ appears before the later of the two}\} = \frac{2}{3} \quad (a \notin G_k).$$

Proof. Relative order among an outsider and the two distinguished actions is uniform over six possibilities. The outsider is not inspected precisely when it is last among the three, with probability 1/3. Hence the second identity. The expected rank equals two plus the expected number of outsider inspections, or $2 + 2(N-2)/3 = 2(N+1)/3$. □

Supplementary IAE-20 Theorem (Release 3.3). *Under the preceding common-task, indexed-access and charged-operation specifications, the two algorithms return the same correctly verified first-action minimizer and satisfy, for every $k \geq 2$,*

$$\boxed{\mathbb{E}[C_B - C_H] = \mathbb{E}[P_k] + \frac{2}{3} \sum_{a \notin G_k} c_k(a) - h_k,}$$

and consequently

$$\boxed{\mathbb{E}[C_B - C_H] \geq 4k - \frac{14}{3} > 0.}$$

The bound counts the declared charged word-RAM operations, not measured seconds.

Proof. Both programs execute and pay for identical parsing, indexing, two named candidate inspections, two cost evaluations, deterministic tie-breaking, and verification. Their actual common costs cancel by equality of executed work. Only the baseline pays P_k and the inspections of outsiders visited before the second special action; only the treatment pays h_k. Each outsider has inclusion probability 2/3, independently of its fixed charge, so linearity of expectation yields the exact identity. There are $N - 2 = 6k + 2$ outsiders, each costing at least one unit, $P_k \geq 0$, and $h_k \leq 6$. Thus

$$\mathbb{E}[C_B - C_H] \geq \tfrac{2}{3}(6k+2) - 6 = 4k - \tfrac{14}{3} \geq \tfrac{10}{3} > 0.$$

No winning-action label was an algorithm input. □

Audit closure, dependency and limitations

Interpretation of the new result. This closes the need to supply an uncharged oracle for the winning label in one explicitly modeled comparison: both algorithms discover the winner by computing the two certified candidate costs, paying their costs and paying for the output verification. The randomized baseline's stopping rule is intentionally *collect both*, not the original *stop when the unique winner appears* model.

Why it does not settle the original six-unit theorem.

- The bound $h_k \leq 6$ is for fixed prefix-register instructions *after* a shared indexed representation has been constructed and charged to both. It is not an end-to-end six-unit differential-cost proof for arbitrary input representations. If target detection, family recognition, or indexing differs between competitors, the additional differential must be counted.
- A fair competitor is allowed the same public family theorem. An unrestricted baseline can also access G_k directly; no advantage over the optimal legal baseline follows from this comparison with a fixed random enumeration policy.
- The original Section 4 claim uses a different random stopping condition and remains conditional upon its own recognizable-target, complete-cost hypotheses. No change to that theorem is asserted here.
- All conclusions are specific to the frozen F_k benchmark and the explicit word-RAM unit ledger. They are not a generic 3CNF runtime claim, a polynomial-time SAT algorithm, or a wall-clock performance measurement.

Dependency ledger.

(1) Section 3's residual-cover and fallback theorems $\Rightarrow$ optimizer is in G_k and $|G_k| = 2$.

(2) A total charged cost evaluator and fixed identity order $\Rightarrow$ deterministic winner selection after evaluating two actions.

(3) The shared representation and six extra register operations $\Rightarrow$ $h_k \leq 6$, with all other matched computations charged to both.

(4) Uniform random permutation $\Rightarrow$ every outsider is inspected with probability $2/3$.

(5) These facts $\Rightarrow$ the exact expected-cost identity and strict margin.

Audit status for Release 3.3. New theorem: mathematical proof within the declared model. Original Section 4 differential-cost condition: *still open for a general complete implementation.* Appendix E and Sections 1–15 otherwise retain their Release 3.2 structure and original proof-status qualifications. No proof-assistant certificate or independent software-performance benchmark is claimed. In the theorem graph, the new result is a separately named descendant of the existing Section 3 structural lemma, not a replacement for the original Section 4 theorem.

Reproducibility. The accompanying IAE-20 source package includes a finite test harness and the longer settlement report with the complete operational interface. Finite tests support but do not establish the universal identity; the displayed proof supplies that implication.

Release change classification: additive theorem, preserved architecture; 3.2 $\mapsto$ 3.3 (+0.1), not 4.0.

Operative Section 4 Mathematical Clarification

Exact Addressing 3.3.1 — supplemental correction leaf; insert after source leaf 11d

Status. This leaf resolves ambiguity between the original single-target inspection claim and the later IAE-20 two-candidate comparison. It does not replace the Section 3 residual-cover lemma or the frozen source hypotheses.

Correction C4.1 (single-target model). The displayed Section 4 bound

$$\mathbb{E}[C_{\text{baseline}}] - \mathbb{E}[C_{\text{treatment}}] \geq 3k - \frac{11}{2} > 0$$

is established *only if* (i) the single winning action is recognized when inspected at the stated charged inspection cost; (ii) both implementations actually execute any allegedly shared preprocessing; and (iii) *all* unmatched treatment costs, including any family recognizer, classification, indexing, lookup, scheduling, and stopping-test work, total at most six candidate-inspection-equivalent units. If the extra treatment cost is $D(k)$ instead, the justified statement is

$$\mathbb{E}[C_{\text{baseline}}] - \mathbb{E}[C_{\text{treatment}}] \geq 3k + \frac{1}{2} - \mathbb{E}[D(k)],$$

under the same unit-inspection and recognizable-target assumptions. Exact addressing alone does not imply $D(k) \leq 6$.

Correction C4.2 (two-candidate comparison). The inserted IAE-20 theorem has a *different* common task and stopping rule: both sides identify and verify the lexicographically optimal first action by inspecting and evaluating the two known actions in $G_k = \{(x, 1), (y, 1)\}$. The comparator uses a uniform random order and stops when it has encountered *both* known actions. Given public common indices, equal charged parsing and evaluation, fixed history-independent outsider charges $c_k(a) \geq 1$, baseline-only charges $P_k \geq 0$, and at most $h_k \leq 6$ treatment-only register operations, the valid claim is

$$\mathbb{E}[C_B - C_H] \geq 4k - \frac{14}{3} > 0.$$

This is a result against the *specified random comparator*, not against an unrestricted competitor permitted to exploit the same known two-action certificate. It counts charged word-RAM operations, not observed wall-clock time or general SAT runtime.

Correction C4.3 (proof status). The conditional mathematical deductions are checked within their definitions. The original six-unit end-to-end premise remains a model assumption, not an empirically certified property of arbitrary implementations. The source manuscript's Appendix D historical run must not be described as fully independently verified.

Publication instruction. When reading earlier unqualified shorthand about "fully charged speedup," apply C4.1–C4.3, especially the separate recognition conditions, differential costs, and fixed-comparator restriction.

4.4 What the theorem does not claim

The theorem is family-specific, feature-induced, and model-relative. It does not show that a raw or independently frozen Hilbert geometry discovers the two-action block; the residual-cover classifier creates the successful ordering. It does not establish a speedup for arbitrary 3CNF, a polynomial-time SAT algorithm, or $P \neq NP$. It also does not survive arbitrary implementation models: eager materialization of all $4 + 6k$ Hilbert positions can cost at least one operation per candidate and thereby erase the robust gross lower bound $3k + 1/2$ (and likewise the conditional-uniformity refinement). The theorem therefore identifies *lazy indexed access* as an essential implementation hypothesis in this certified speedup result.

Part II

Foundations: Exact Identity, Navigation, and Verifier-Valid Geometry

5--6 Revised exact rational and WFF addressing

The authoritative Release 3.1 replacement of Sections 5 and 6 follows in two inserted pages (12a--12b). It includes the source's explicit surjection and least-preimage inverse, an effective bijective rational enumeration, and distinct dense versus isolated exact addresses.

5 Exact rational addressing (replacement)

We take $\mathbb{N} = \{1, 2, \dots\}$, as in Tenneson's *A Closed-Form Surjection from* $\mathbb{N}$ *onto* $\mathbb{Q}$*—With Its Pseudo-Inverse*, Version 2.2 (July 17, 2026) [1]. For each $m \geq 1$, the crescendo/decrescendo pairing produces positive integers $(a(m), b(m))$ with the explicitly proved inverse

$$m(a, b) = \frac{(a + b - 1)(a + b - 2)}{2} + a \qquad (a, b \in \mathbb{N}).$$

Put $a(0) = 0$ and $b(0) = 1$. The source's closed-form rational map is

$$r(n) = (-1)^n \frac{a(\lfloor n/2 \rfloor)}{b(\lfloor n/2 \rfloor)} \quad (n \in \mathbb{N}).$$

For a reduced rational p/q, where $p \in \mathbb{Z}$, $q \geq 1$, define the least-preimage section

$$s(p/q) = \begin{cases} 1, & p = 0, \\ 2m(p, q), & p > 0, \\ 2m(-p, q) + 1, & p < 0. \end{cases}$$

This is the pairing presentation of the explicit polynomial formula (4) in [1]. Theorems 1–2 of [1] prove both $r(\mathbb{N}) = \mathbb{Q}$ and

$$r(s(q)) = q, \qquad s(q) = \min\{n \in \mathbb{N} : r(n) = q\} \quad (q \in \mathbb{Q}).$$

Source-derived consequence. s is injective, and the canonical set $C := s(\mathbb{Q}) \subseteq \mathbb{N}$ is an exact representative system, with $s : \mathbb{Q} \to C$ bijective and inverse $r|_C$. The source does *not* assert that r is injective on all of $\mathbb{N}$; for example, $r(2) = r(10) = 1$. The displayed inverses are restricted to C.

Theorem 1 (Effective canonical-subset test and bijective rational enumeration)**.** *Let $C = \{n \in \mathbb{N} : s(r(n)) = n\}$. Then C is decidable, infinite, and each rational occurs exactly once among $\{r(c) : c \in C\}$. If $c_1 < c_2 < \cdots$ is the increasing enumeration of C, the map*

$$b_{\mathbb{Q}} : \mathbb{N} \longrightarrow \mathbb{Q}, \qquad b_{\mathbb{Q}}(j) = r(c_j)$$

is a computable bijection with computable inverse

$$b_{\mathbb{Q}}^{-1}(q) = \#\{n \leq s(q) : s(r(n)) = n\}.$$

Proof. The formulas for r and s evaluate exactly on natural numbers and reduced fractions, making the displayed predicate decidable. By minimality, $n \in C$ exactly when n is the chosen least preimage of $r(n)$. Every q has exactly one representative $s(q)$, and there are infinitely many rationals. Increasing enumeration therefore terminates at every finite index. If $s(q) = c_j$, precisely j members of C are at most $s(q)$, proving the inverse formula. □

Interval version. Set $C_{01} = \{n \in C : 0 \leq r(n) \leq 1\}$ and enumerate it increasingly as $c_1^{01} < c_2^{01} < \cdots$. Then $b_{01}(j) = r(c_j^{01})$ is a computable bijection $\mathbb{N} \leftrightarrow \mathbb{Q} \cap [0, 1]$, with inverse given by counting members of C_{01} up to $s(q)$. This follows from Theorem 1 not from a mere surjection onto the interval.

Two distinct geometries. The maps $b_{\mathbb{Q}}$ and b_{01} enumerate *dense* rational sets, whose elements are never isolated from one another. In contrast, the canonical code $s(q) \in C \subset \mathbb{N} \subset \mathbb{Q}$ lies in an integer-spaced set and is isolated by radius at least 1 from every other code. Numerical distance between canonical indices is an identity metric, not a calibrated logical-proof distance.

6 WFF identity and inference-action addressing (replacement)

Let $\mathcal{W}$ be a countably infinite *effectively decidable* set of well-formed formulas of a fixed formal language. Fix a computable bijective enumeration $E : \mathbb{N} \to \mathcal{W}$ with computable inverse, e.g. length-lexicographic enumeration of valid finite syntax strings. The following constructions retain distinct types.

- **Dense rational exact address:** $\lambda_{01} = b_{01} \circ E^{-1} : \mathcal{W} \xrightarrow{\sim} \mathbb{Q} \cap [0,1]$. The inverse is $E \circ b_{01}^{-1}$; both directions are computable.
- **Canonical sparse rational exact address:** $\lambda_C = s \circ b_{\mathbb{Q}} \circ E^{-1} : \mathcal{W} \xrightarrow{\sim} C \subset \mathbb{N} \subset \mathbb{Q}$. The inverse on C is $E \circ b_{\mathbb{Q}}^{-1} \circ r|_C$; distinct formula addresses differ by at least 1.
- **Navigation/feature coordinate:** $\Phi(a) \in (\mathbb{Q} \cap [0,1])^d$ is separately declared and need not be injective. Its geometric proximity carries no intrinsic theoremhood or proof-distance guarantee.

Corollary 2 (Exact identity is independent of rational density)**.** *Both λ_{01} and λ_C yield computably reversible formula identities, but $\lambda_{01}(\mathcal{W})$ is dense in $[0,1]$ and $\lambda_C(\mathcal{W})$ is discrete. Thus reversible addressing by itself neither forces nor forbids positive isolation radii. Any isolation or distance-based proof-search claim must specify which address image and metric it uses.*

Proof. Both composites are computable bijections onto their stated images. Every point in $\mathbb{Q} \cap [0,1]$ is approached by other rational points, while distinct elements of $C \subset \mathbb{N}$ differ by at least one. □

These are representations of formula identity only. The rule-tagged premise-slot address space from the original Section 6, $\bigsqcup_{r \in R}(\{r\} \times (\mathbb{Q} \cap [0,1])^{\mathrm{ar}(r)})$, remains unchanged. No primitive-inference, proof-cost, or verifier result follows from the surjection paper without the separate transported-rule and cost hypotheses of Section 6.1. Computing $r, s, b_{\mathbb{Q}}, b_{01}$ or maintaining indexed lookup has a cost that must be charged when an algorithmic efficiency comparison is asserted.

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[1] B. Tenneson, *A Closed-Form Surjection from $\mathbb{N}$ onto $\mathbb{Q}$—With Its Pseudo-Inverse*, Version 2.2, July 17, 2026. https://btenneson.github.io/pub/math.LO_Logic/closed_form_surjection_n_to_q/. Source: surjectivity and least-preimage proofs; the computable first-occurrence bijections and sparse/dense contrast here are new deductions from those stated results.

Section 6 continues on pages 12a--12b

Original Section 6 text on this leaf is superseded by the immediately preceding insert. Section 6.1 (Strengthened Exact Addressing Theorem) begins below and remains unchanged.

6.1 Strengthened Exact Addressing Theorem

The preceding correspondence becomes substantially stronger when the address map is treated not merely as a naming convention but as an exact transport of a fixed deductive presentation.

Exact Addressing Theorem (strengthened recoding-invariance form)

Let F be a fixed deductive presentation on a WFF space W. Let

$$t : W \longrightarrow W'$$

be a bijective computable syntactic recoding with computable inverse, and let F^t be the transported presentation obtained by carrying every primitive inference rule of F through t. Thus a primitive step from premises Γ to conclusion φ is legal in F exactly when the corresponding step from $t[\Gamma]$ to $t(\varphi)$ is legal in F^t.

Let $G = (S, E, \ell)$ be the labeled operational graph of complete proof-search states, and let

$$e : S \longrightarrow S'$$

be an injective complete-state encoding. Define the transported operational graph on $e[S]$ by

$$e(s) \to e(s') \quad \Longleftrightarrow \quad s \to s',$$

and transport all target, observation, ownership, and verifier-relevant labels exactly. Then:

1. **Derivability is preserved.**

$$\Gamma \vdash_F \varphi \quad \Longleftrightarrow \quad t[\Gamma] \vdash_{F^t} t(\varphi).$$

2. **Proof paths are preserved step for step.** Every finite proof in F transports to a proof of the same number of primitive steps in F^t, and conversely. Hence exact shortest-proof length is invariant:

$$\|t(\varphi)\|_{F^t}(t[\Gamma]) = \|\varphi\|_F(\Gamma).$$

3. **Exact breadth-first proof layers are preserved.** If $L_F(\Gamma, m)$ denotes the set of formulas whose minimum proof length from Γ is exactly m, then

$$t[L_F(\Gamma, m)] = L_{F^t}(t[\Gamma], m)$$

for every finite m.

4. **The complete-state transition system is preserved.** The map e is a labeled graph isomorphism from G onto its encoded image. Therefore finite labeled paths correspond edge for edge. In the unit-cost setting, target distances are equal; if owner and target labels are among the transported labels, forced-settlement horizons are equal as well.

5. **Navigation policy is logically separable from validity.** A scheduler may permute or adaptively switch among legal candidate-selection policies without changing which accepted certificates are valid, provided every chosen transition remains legal and the same sound verifier is applied. Fairness remains a separate requirement when eventual discovery of every finite proof is desired.

Proof. The derivability claim follows by transporting each proof line through the bijection t and using the transported rule in both directions. Because each primitive step is matched by exactly one transported primitive step, proof lengths are preserved in both directions; taking minima gives the shortest-proof equality, and the exact-layer identity follows immediately. For complete states, injectivity of e gives an inverse on the encoded image, while the transported edge and label definitions give a two-way labeled-edge correspondence. Hence paths, unit edge counts, targets, and any transported ownership structure are preserved. Finally, changing only the order in which already-legal transitions are proposed does not alter the verifier's acceptance relation. These are the recoding and policy-separation consequences collected in the low-fan-in repository, here assembled as one exact-addressing statement [7]. □

No-speedup-by-addressing corollary

Under the theorem's hypotheses, an invertible exact recoding by itself cannot shorten a shortest proof, move a theorem to a shallower breadth-first proof layer, or create a shorter unit-cost target path. Any discovery advantage attributed to an exact-address system must therefore come from additional structure—for example, an informative feature geometry, a different search schedule, reusable certified lemmas, or a changed proof presentation—rather than from identity recoding alone.

This corollary sharpens the paper's central separation. Exact addressing can preserve the proof problem exactly while still providing a reproducible substrate on which a separate navigation geometry is defined. Numerical closeness of exact addresses remains non-semantic unless an independent calibration or locality theorem is proved.

Operational DAG bridge

For the append-only derivation-state representation used in One-Step DAG analysis, each accepted transition appends one formula. Consequently the state graph is acyclic and a reachable exact sequence

state has a uniquely determined depth from the root. Moreover, any concrete proof path gives an explicit upper bound on theorem-graph distance. Together with the theorem above, this supplies a clean directed graph on which exact identities can be transported without confusing graph depth with raw coordinate distance [7].

7 Two roles for Hilbert traversal

There are two conceptually different uses of Hilbert order. A *generative* Hilbert traversal can exhaust a finite rule-action grid, supporting coverage when repeated fairly. A *geometric* Hilbert traversal can order an already defined candidate set by a learned or engineered feature geometry. The first primarily serves coverage; the second can yield speedup only if the geometry is informative [4].

Continuous Hilbert surjectivity alone is not an executable completeness argument. Effective proof search instead requires finite shells or finite grids together with a fairness condition ensuring eventual visitation of each required finite action once its premises are available.

7.1 Finite face-continuous Hilbert probes

The following finite-grid statements are imported from the repaired MAPQUEST foundations and reproved here in the notation needed by the present manuscript [15]. They concern the geometry of the navigation schedule, not the legality of proof steps. The proofs below were checked directly from the stated hypotheses.

Definition 7.1 (Face-continuous finite Hilbert approximant)**.** Fix a dimension $k \geq 1$, a resolution $p \geq 1$, and an orientation/state parameter o. A face-continuous finite Hilbert approximant is a bijection

$$\mathcal{H}_{k,p,o} : \{0, 1, \ldots, 2^{pk} - 1\} \longrightarrow \{0, 1, \ldots, 2^p - 1\}^k$$

such that

$$\|\mathcal{H}_{k,p,o}(h+1) - \mathcal{H}_{k,p,o}(h)\|_1 = 1$$

for every $0 \leq h < 2^{pk} - 1$. For a lattice cell j, write

$$c_p(j) = 2^{-p}\left(j + \frac{1}{2}\mathbf{1}\right)$$

for its normalized cell center.

Theorem 7.2 (Axis-Parallel Probe Theorem)**.** *Fix one k-dimensional rule component, one resolution p, and one orientation o. Let*

$$a(h) = \mathcal{H}_{k,p,o}(h).$$

Then for every primitive probe transition there exist a unique coordinate $i_h \in \{1, \ldots, k\}$ and a sign $\varepsilon_h \in \{-1, +1\}$ such that

$$a(h+1) = a(h) + \varepsilon_h e_{i_h}, \qquad \|a(h+1) - a(h)\|_2 = 1.$$

For normalized cell centers $x(h) = c_p(a(h))$,

$$x(h+1) - x(h) = \varepsilon_h 2^{-p} e_{i_h}, \qquad \|x(h+1) - x(h)\|_2 = 2^{-p}.$$

Thus each primitive face-continuous Hilbert move changes exactly one premise-slot cell coordinate.

Proof. The difference vector $a(h+1)-a(h)$ has integer coordinates and L^1 norm one. Hence exactly one coordinate is nonzero, and that coordinate equals $+1$ or -1. Multiplying by 2^{-p} gives the normalized cell-center displacement. □

Proved by Sol.

Theorem 7.3 (Exact finite-cell coverage)**.** *A face-continuous finite Hilbert approximant in dimension k and resolution p visits each of its*

$$M_{k,p} = 2^{pk}$$

finite cells exactly once. Consequently a complete local sweep has $M_{k,p}$ cell visits and $M_{k,p}-1$ primitive axis-parallel transitions.

Proof. Bijectivity gives one and only one visit to each of the 2^{pk} cells. A path through $M_{k,p}$ visited vertices has $M_{k,p}-1$ successive transitions, and Theorem 7.2 makes each transition axis-parallel. □

Proved by Sol.

The coverage statement is deliberately about cells, not exact candidate settlement. A finite cell can contain many rational candidates; exact candidate identity remains the responsibility of the exact-address layer.

8 What Distance Are We Talking About?

The word *distance* appears naturally once formulas and inference actions are assigned numerical or geometric coordinates. But several mathematically different notions of distance arise in this framework, and they must not be conflated. In particular, an exact identity address, a navigation or feature geometry, and the cost of a legal derivation answer different questions.

The central distinction is

$$\boxed{\text{identity distance} \neq \text{navigation distance} \neq \text{logical proof distance.}}$$

The distinctions are not merely terminological. The first describes where an object is encoded, the second describes how a search procedure chooses to organize objects or actions, and the third describes the actual deductive work required by a fixed formal presentation.

8.1 Exact-address distance

An exact address is first of all an **identity coordinate**. Its purpose is to specify a formal object reversibly.

In the MIU construction, every WFF g receives a binary address

$$A(g) \in \mathbb{N}_{>0},$$

and the address map is bijective. Because the address uniquely determines the WFF, one may define

$$d_A(g,h) = |A(g) - A(h)|.$$

This is a genuine metric: in particular,

$$d_A(g,h) = 0 \iff g = h.$$

The metric is therefore mathematically valid as a metric on WFF identity. Its interpretation, however, must remain modest. A small value of $d_A(g,h)$ means that the two exact integer addresses are numerically close. It does **not**, by definition alone, mean that g and h have similar proofs, share useful lemmas, or are separated by only a few inference steps. The MIU paper explicitly treats the exact identity address and logical proof distance as distinct typed objects and assumes no numerical relation between them without an additional theorem.

The same caution applies to the more general WFF–Hilbert correspondence. There, formulas may be placed in an exact rational address space so that an address can be translated back to a concrete WFF. The correspondence supplies exact navigation and recovery, but exact addressability by itself does not establish proof-theoretic locality. The strengthened Exact Addressing Theorem above adds a different positive fact: when the syntax, rules, and operational transitions are transported faithfully, exact recoding preserves theoremhood and intrinsic step-count structure exactly. Preservation is therefore stronger than mere recoverability, but it still does not make numerical address separation a proof metric.

Thus exact addressing answers the question

$$\boxed{\text{Which formal object is this?}}$$

not yet

$$\boxed{\text{How hard is it to prove from another object?}}$$

8.2 Feature or navigation distance

A second notion of distance arises when formal objects or inference actions are deliberately placed in a geometry intended to assist search.

In the MIU model, the count-pair

$$w(g) = (i(g), u(g))$$

may be assigned a scalar height

$$E(w(g)),$$

giving the height distance

$$d_E(g,h) = |E(w(g)) - E(w(h))|.$$

Unlike the exact address, the count-pair forgets symbol order. Distinct WFFs can therefore occupy the same height, so d_E is generally a **pseudometric**, rather than a metric on individual WFFs.

The more general proof-search framework uses the analogous but richer idea of a feature representation

$$\Phi(a) \in (\mathbb{Q} \cap [0,1])^D$$

for an inference action a. Such coordinates may incorporate dependencies, symbol structure, graph relations, proof depth, learned scores, verifier history, or other selected features. Importantly, the framework keeps this learned geometry separate from the exact address of the action.

Once a feature map has been chosen, one may equip its image with some geometric distance, for example

$$d_\Phi(a, b) = \|\Phi(a) - \Phi(b)\|,$$

for a specified norm or other declared distance function.

But this geometry is **chosen for navigation**. Unless a calibration theorem has been proved, one cannot infer that

$$d_\Phi(a, b) \text{ small}$$

means

$$d_P(a, b) \text{ small.}$$

The geometry may nevertheless be valuable without approximating shortest-proof distance. The locality theory requires something different: that geometrically selected regions concentrate useful proof actions.

If G is a local region of a candidate space A, its locality surplus is

$$\Lambda(G) = P(A^\star \in G) - \frac{|G|}{|A|}.$$

A region is search-informative when

$$\Lambda(G) > 0.$$

Under the finite model of the Locality Advantage Theorem,

$$E[R_{\text{random}}] - E[R_G] = \frac{|A|}{2}\Lambda(G).$$

Thus a feature geometry can be useful because it places **more next-step proof mass than chance would predict** inside regions searched early, even if geometric distance is not an accurate numerical estimate of shortest logical distance.

This distinction is important:

$$\boxed{\text{proof-locality enrichment} \neq \text{proof-distance approximation.}}$$

8.3 Logical proof distance

The third notion is deductive rather than representational.

Fix a deductive presentation P and a nonnegative proof-cost model. Define

$$d_P(g, h) = \inf\{\text{cost}_P(\pi) : \pi : g \rightsquigarrow h\}.$$

If h cannot be derived from g, set

$$d_P(g, h) = +\infty.$$

With ordinary unit rule costs, $d_P(g, h)$ is the minimum number of legal inference steps whenever h is reachable from g.

This is the quantity that measures genuine deductive work in the chosen presentation.

It is also fundamentally different from the preceding coordinate distances because logical reachability can be directed. Exact-address and scalar-height distances satisfy

$$d_A(g, h) = d_A(h, g)$$

and

$$d_E(g,h) = d_E(h,g),$$

whereas it may happen that

$$d_P(g,h) < \infty$$

but

$$d_P(h,g) = +\infty.$$

The MIU paper therefore treats d_P as a **directed extended distance** rather than an ordinary symmetric metric. Consequently, no symmetric coordinate metric can in general coincide with the full directed proof distance on all ordered pairs whenever one-way derivability occurs.

8.4 Target-relative logical distance

For proof search it is often more natural to fix the target T and ask how much logical work remains.

Define

$$V_T(g) = d_P(g,T).$$

On states from which T is reachable, $V_T(g)$ gives the exact remaining deductive cost in the chosen presentation.

If states having the same target distance are identified,

$$g \sim_T h \iff V_T(g) = V_T(h),$$

then

$$\rho_T([g],[h]) = |V_T(g) - V_T(h)|$$

is a genuine metric on the resulting quotient space.

This gives a useful conceptual ideal for search:

$$\boxed{V_T(g) = \text{true remaining proof cost.}}$$

Of course, if V_T were already known exactly, much of the proof-search problem would already have been solved. The practical role of a learned geometry Φ is therefore not to assume access to V_T, but to construct computable coordinates that may correlate with useful aspects of the unknown proof landscape.

8.5 When can geometric distance say something rigorous about proof distance?

A geometric or height distance can constrain logical distance when an explicit **calibration theorem** is available.

Suppose every primitive legal unit-cost inference step satisfies

$$d_E(g,g') \le L$$

for some finite constant $L > 0$.

Then any derivation from g_1 to g_2 must contain enough steps to cover their total height separation. Consequently,

$$\boxed{d_P(g_1, g_2) \geq \frac{d_E(g_1, g_2)}{L}.}$$

Under unit costs,

$$\boxed{d_P(g_1, g_2) \geq \left\lceil \frac{d_E(g_1, g_2)}{L} \right\rceil.}$$

This is a genuine bridge from a chosen geometry to logical proof cost—but only because an additional bounded-step hypothesis has been proved.

The distinction matters because such a global constant need not exist. In the MIU example, Rule II can produce unbounded displacement for nonconstant linear heights over the full reachable system, so the simple global calibration fails for those choices of E.

A geometry can nevertheless remain useful locally. If S_0 is a restricted state region and

$$L_{S_0} = \sup\{d_E(g, g') : g \to g',\ g, g' \in S_0\} < \infty,$$

then the restricted logical distance satisfies

$$\boxed{d_P^{S_0}(g_1, g_2) \geq \frac{d_E(g_1, g_2)}{L_{S_0}}.}$$

Here $d_P^{S_0}$ considers only derivations that remain inside S_0. This qualification is necessary because a shorter unrestricted derivation might leave the calibrated region.

This local form is especially relevant to Hilbert-guided proof search, because the scheduler already works with selected neighborhoods or regions. A global geometry need not model the entire proof graph accurately in order to provide a useful calibrated structure inside particular search regions.

8.6 Distance and Hilbert traversal

Hilbert traversal introduces one further potential source of confusion. The Hilbert curve supplies an **ordering of movement through a geometric space**. It does not define logical distance.

If candidate actions have feature coordinates

$$\Phi(a),$$

then Hilbert order may arrange geometrically nearby cells consecutively. This can reduce transition churn or exploit locally concentrated proof mass. But the curve itself does not imply that two consecutively visited candidates are close in logical proof distance.

Thus the relevant chain is not

$$\text{nearby address} \Rightarrow \text{nearby proof}.$$

Rather, it is

exact identity
↓
chosen search geometry
↓
empirical or theorem-backed locality
↓
Hilbert-local scheduling
↓
verifier-checked inference.

Any further arrow from geometric separation to actual logical distance requires a separate calibration result.

8.7 The four questions

The terminology can therefore be kept clear by asking four different questions.

Identity: Which formal object or inference action is this?

$$A(g), \qquad \mathrm{Addr}(a)$$

answer this question.

Geometry: Where has the search representation placed it?

$$E(w(g)), \qquad \Phi(a)$$

answer this question.

Locality: Does a geometrically selected neighborhood contain more useful proof mass than its size alone would predict?

$$\Lambda(G) = P(A^\star \in G) - \frac{|G|}{N}$$

answers this question.

Logic: What is the least cost of a legal derivation in the selected proof system?

$$d_P(g, h)$$

answers this question.

These four quantities may interact, but none should be identified with another without proof.

The discipline adopted in this paper is therefore:

address first, geometry second, calibration separately, logic last.

Exact coordinates provide reversible identity. Learned or engineered coordinates provide navigation. Positive locality surplus establishes that a navigation geometry is useful for search under the stated probability model. Calibration theorems may sometimes connect geometric displacement to lower

bounds on deductive work. The verifier and the deductive presentation determine actual logical reachability and proof cost.

That separation allows Hilbert-addressed proof search to use geometry aggressively without claiming more from that geometry than has actually been established.

9 Rectilinear routes and intrinsic proof geodesics

MAPQUEST distinguishes navigation displacement from proof distance. For a chosen proof-space coordinate map $x : \mathrm{Proofs}(A) \to \mathbb{R}^d$, define the rectilinear endpoint displacement

$$D_{\mathrm{rect}}(P, Q) = \|x(Q) - x(P)\|_1.$$

If there exists an admissible verifier-valid proof-transformation path from P to Q whose coordinate trajectory is monotone and whose charged length equals this L^1 displacement, then that route is one competitor in the definition of intrinsic geodesic distance. Therefore

$$\boxed{d_{\mathrm{geo}}(P, Q) \leq D_{\mathrm{rect}}(P, Q).}$$

The inequality is an upper bound on the geodesic, not a maximum over all proof paths. The shortcut gap

$$S_{\mathrm{geo}}(P, Q) = D_{\mathrm{rect}}(P, Q) - d_{\mathrm{geo}}(P, Q)$$

is nonnegative under the realizability hypothesis and is positive exactly when the admissible proof geometry contains a shorter route than the canonical rectilinear realization [6].

A converse lower bound requires calibration. Suppose admissible edges have nonnegative cost $w(R, R')$, every zero-cost edge has zero coordinate displacement, and the set of positive-cost admissible edges satisfies

$$0 < \Lambda_A := \sup_{\substack{R \to R' \\ w(R,R') > 0}} \frac{\|x(R') - x(R)\|_1}{w(R, R')} < \infty.$$

The restriction to positive-cost edges makes the ratio well-defined; zero-cost edges contribute no coordinate displacement by hypothesis. Then every admissible path obeys

$$\frac{D_{\mathrm{rect}}(P, Q)}{\Lambda_A} \leq d_{\mathrm{geo}}(P, Q).$$

Combining both sides gives

$$\boxed{\frac{D_{\mathrm{rect}}(P, Q)}{\Lambda_A} \leq d_{\mathrm{geo}}(P, Q) \leq D_{\mathrm{rect}}(P, Q).}$$

When $\Lambda_A = 1$, the coordinate route is exactly geodesic. These statements illustrate the principle developed in the preceding distance section: coordinate geometry becomes proof geometry only after an explicit admissibility and cost bridge is proved [6].

10 Path-Calibrated Bridge Theorems for Exact Addressing

This section makes explicit the strongest mathematical connection between exact addressing and intrinsic proof geometry. Exact addressing supplies recoverable identity and can transport a proof

problem without changing it. Intrinsic proof distance, however, is defined by verifier-valid paths and their charged costs. The bridge between the two arises only after an address-derived or feature-derived coordinate is lifted to proof states and calibrated against legal edge costs. The results below are included in full, with their proofs, so the present manuscript is self-contained on this bridge.

The relationship can be summarized as

$$\boxed{\begin{array}{c}\text{exact identity} \longrightarrow \text{state coordinate} \xrightarrow{\text{calibration}} \text{proof-distance lower bound} \\ \longrightarrow \text{target potential / geodesic certificate}\end{array}}$$

The reverse implication does not hold automatically: fixed addresses can coexist with different legal roads and therefore different intrinsic proof distances.

10.1 Self-contained path and trace setup

Definition 10.1 (Presented system and proof states)**.** Fix a presented formal system

$$F = (\Sigma, \Gamma, R, V),$$

with WFF set $W_F = \mathrm{WFF}(F)$. Let S_F be a declared proof-state space appropriate to the calculus. In a monotone certificate-accumulating calculus, a state may be a trusted set $B \subseteq W_F$. In a nonmonotone calculus, the state may instead be a sequent, multiset, configuration, or other verifier-sufficient object.

Definition 10.2 (Legal elementary move)**.** A legal elementary move is a verifier-backed transition

$$x \xrightarrow{e} y,$$

where the edge label e records enough data to replay the inference: at least its rule tag and, when needed, premises, substitution data, witnesses, or instantiations.

Definition 10.3 (Verifier-valid path family)**.** A finite verifier-valid path from x to y is an ordered sequence

$$\gamma = (x_0, e_1, x_1, \ldots, e_n, x_n), \qquad x_0 = x,\ x_n = y,$$

with every $x_{j-1} \xrightarrow{e_j} x_j$ legal. Write

$$\mathcal{P}_F(x, y)$$

for the set of all such finite paths, including the empty path when $x = y$.

Definition 10.4 (Path cost and intrinsic distance)**.** Let $w_F(e) \in [0, \infty)$ be a declared nonnegative edge cost. Define

$$\mathrm{cost}_F(\gamma) = \sum_{j=1}^{n} w_F(e_j),$$

and

$$d_F(x, y) = \inf_{\gamma \in \mathcal{P}_F(x,y)} \mathrm{cost}_F(\gamma),$$

with $d_F(x, y) = \infty$ if $\mathcal{P}_F(x, y) = \varnothing$. When the formal system is fixed, write $w(e) = w_F(e)$. Because zero-cost edges and directionality are allowed, d_F is in general an extended directed path pseudodistance rather than a symmetric metric.

Definition 10.5 (Exact WFF address and WFF height)**.** An exact address is an injective map

$$a_F : W_F \hookrightarrow D_F$$

into a chosen address set D_F for which such an injection exists. In this paper, *exact* means set-theoretically exact: WFF identity is recoverable from the address on $a_F(W_F)$. When operational encoding and decoding are required, call the address *effectively exact* if a_F and its inverse $a_F^{-1} : a_F(W_F) \to W_F$ are computable on their declared representations. No rational or Euclidean structure is required. A WFF height is any declared map

$$h_F : W_F \to \mathbb{R}.$$

For the MIU experiment we will use

$$h(g) = E(i(g), u(g)),$$

where $i(g)$ and $u(g)$ count the symbols I and U.

Definition 10.6 (Conclusion, address, height, and rule traces)**.** Assume each legal edge e_j has a distinguished newly certified conclusion $C_j \in W_F$ whenever that notion is meaningful. For

$$\gamma = (x_0, e_1, x_1, \ldots, e_n, x_n),$$

define

$$C(\gamma) = (C_1, \ldots, C_n),$$
$$A_a(\gamma) = \big(a_F(C_1), \ldots, a_F(C_n)\big),$$
$$H_h(\gamma) = \big(h_F(C_1), \ldots, h_F(C_n)\big),$$

and the rule trace

$$R(\gamma) = (r_1, \ldots, r_n).$$

In MIU the state itself can be taken to be a WFF, so one may instead trace the full WFF sequence $(g_0, \ldots, g_n)$ directly.

Theorem 10.7 (Exact re-addressing preserves WFF identity)**.** *If $a : W_F \hookrightarrow D$ is injective, then a is a bijection from W_F onto its image $a(W_F)$, with inverse $a^{-1} : a(W_F) \to W_F$.*

Proof. Every injection is bijective onto its image. □

Theorem 10.8 (Address-trace existence)**.** *Every verifier-valid path whose edges have distinguished conclusions has a unique ordered address trace under a fixed exact address map.*

Proof. Each conclusion has exactly one address under the function a. □

Theorem 10.9 (Exact-address trace reconstruction)**.** *Let $a : W_F \hookrightarrow D$ be exact. If two verifier-valid paths γ and η have the same ordered address trace,*

$$A_a(\gamma) = A_a(\eta),$$

then they have the same ordered conclusion trace:

$$C(\gamma) = C(\eta).$$

Thus exact addressing is lossless at the level of the ordered conclusion sequence, even though it need not recover rule tags, witnesses, or other edge data that are not encoded in the conclusions.

Proof. By Theorem 10.7, a^{-1} exists on $a(W_F)$. Apply a^{-1} coordinatewise to the common address trace. □

Theorem 10.10 (State-lift requirement)**.** *In a multipremise proof system, an address or height on individual WFFs does not by itself define a coordinate on proof states. To invoke a state-level coordinate-distance theorem, one must separately define a state map $H : S_F \to X$ or an equivalent rule-slot construction.*

Proof. A proof state contains more information than any single WFF. A function on WFFs has no canonical value on an arbitrary state until an aggregation or state representation is declared. □

10.2 Calibrated coordinates as rigorous lower bounds on proof distance

The following results formalize the heuristic lower-bound discussion in the preceding distance and route sections. They apply to any declared state coordinate, including one built from exact addresses, heights, learned features, or a product of several such coordinates, provided the stated edge calibration is proved.

Definition 10.11 (State coordinate and edge slope)**.** Let $H : S_F \to \mathbb{R}^d$ be a declared state coordinate. Assume every zero-cost edge has zero H-displacement. Define

$$L_{H,w} = \sup_{e:x\to y,\ w(e)>0} \frac{\|H(x) - H(y)\|_1}{w(e)}.$$

The supremum is taken in $[0, \infty]$, with the convention $\sup \varnothing = 0$ when there are no positive-cost edges.

Definition 10.12 (Point-to-set distance and coordinate distance)**.** For a nonempty set $C \subseteq S_F$, define

$$d_F(x, C) = \inf_{t\in C} d_F(x, t).$$

For a nonempty set $A \subseteq \mathbb{R}^d$, define

$$\mathrm{dist}_1(z, A) = \inf_{a\in A} \|z - a\|_1.$$

Thus $\mathrm{dist}_1(H(x), H(C))$ is a coordinate-space quantity, whereas $d_F(x, C)$ is an intrinsic path-cost quantity.

Definition 10.13 (Consistent target potential)**.** Fix a nonempty target set $C_T \subseteq S_F$. A function $p : S_F \to [0, \infty]$ is a consistent target potential if

$$p(t) = 0 \qquad (t \in C_T)$$

and every legal edge $x \xrightarrow{e} y$ satisfies

$$p(x) \le w(e) + p(y).$$

When a family of potentials is combined below, all members are understood to use the same target set and edge-cost convention.

Theorem 10.14 (Pathwise coordinate-displacement bound)**.** *If $L_{H,w} < \infty$, then every finite legal path*

$$\gamma : x = x_0 \to \cdots \to x_n = y$$

satisfies

$$\|H(x) - H(y)\|_1 \le L_{H,w} \,\mathrm{cost}_F(\gamma).$$

Proof. Apply the triangle inequality to the coordinate increments and then the edgewise slope bound to each edge. □

Theorem 10.15 (Endpoint lower bound)**.** *Under the same hypotheses and $L_{H,w} > 0$,*

$$d_F(x, y) \geq \frac{\|H(x) - H(y)\|_1}{L_{H,w}}.$$

Proof. The preceding inequality holds for every legal path. Take the infimum of path costs. □

Theorem 10.16 (Target-set lower bound)**.** *For a nonempty target set $C_T \subseteq S_F$, under the preceding hypotheses with $0 < L_{H,w} < \infty$,*

$$d_F(x, C_T) \geq \frac{\mathrm{dist}_1(H(x), H(C_T))}{L_{H,w}}.$$

Proof. Apply the endpoint lower bound to each $t \in C_T$ and take infima over targets. □

Theorem 10.17 (Fixed-target geometric proof-distance specialization)**.** *Let $G = (S, E)$ be a unit-cost verifier-valid proof-state graph with fixed target state x_T. Let*

$$\Xi : S \to [0, 1]^d$$

be any declared state representation and define its maximum one-step displacement

$$\delta_\Xi = \sup\big(\{\|\Xi(x) - \Xi(y)\|_1 : (x, y) \in E\} \cup \{0\}\big).$$

Then every state x satisfies

$$\|\Xi(x) - \Xi(x_T)\|_1 \leq \delta_\Xi \, d_G(x, x_T)$$

in the extended-real sense. If $\delta_\Xi > 0$ and $d_G(x, x_T) < \infty$, then

$$d_G(x, x_T) \geq \frac{\|\Xi(x) - \Xi(x_T)\|_1}{\delta_\Xi}.$$

Proof. This is Theorem 10.14 with unit edge cost and $L_{\Xi,w} = \delta_\Xi$, followed by Theorem 10.15. If the target is unreachable, the first inequality is immediate in the extended-real convention. □

Proved by Sol. The fixed-target formulation is imported from the authorś geometric proof-distance note [12]; the proof here is the direct specialization of the more general calibrated theorem already established in this manuscript.

Theorem 10.18 (Calibrated coordinate target potential)**.** *Assume $0 < L_{H,w} < \infty$ and define*

$$p_H(x) = \frac{\mathrm{dist}_1(H(x), H(C_T))}{L_{H,w}}.$$

Then p_H is a consistent target potential.

Proof. For $t \in C_T$, $H(t) \in H(C_T)$, so $p_H(t) = 0$. If $x \xrightarrow{e} y$ is legal, point-to-set distance and the edge-slope bound give

$$\mathrm{dist}_1(H(x), H(C_T)) \leq \|H(x) - H(y)\|_1 + \mathrm{dist}_1(H(y), H(C_T)) \leq L_{H,w} w(e) + \mathrm{dist}_1(H(y), H(C_T)).$$

Divide by $L_{H,w}$. □

Theorem 10.19 (Pointwise supremum of consistent potentials)**.** *Let $\{p_i\}_{i\in I}$ be consistent target potentials and suppose $p(x) = \sup_i p_i(x)$ is finite on the declared domain. Then p is a consistent target potential.*

Proof. For every edge $x \xrightarrow{e} y$, $p_i(x) \le w(e) + p_i(y)$ for all i. Taking suprema gives $p(x) \le w(e) + p(y)$, and the common target value remains 0. □

Theorem 10.20 (Finite coordinate-ensemble potential)**.** *Let $H_1, \dots, H_m$ be state coordinates with finite positive calibration constants $L_1, \dots, L_m$ for the same edge-cost model and target set. Define*

$$p_{\max}(x) = \max_{1\le i\le m} \frac{\mathrm{dist}_1(H_i(x), H_i(C_T))}{L_i}.$$

Then $p_{\max}$ is a consistent target potential and hence a certified lower bound on every target-reaching path cost.

Proof. Each coordinate term is a consistent target potential by Theorem 10.18. Their finite maximum is their pointwise supremum, so Theorem 10.19 proves consistency. Repeatedly applying the consistency inequality along any target-reaching path gives $p_{\max}(x)$ no larger than that path's cost. □

Theorem 10.21 (Saturated-path exactness criterion)**.** *Suppose $0 < L_{H,w} < \infty$ and a legal path $\gamma : x_0 \to \cdots \to x_n$ satisfies both*

$$\|H(x_0) - H(x_n)\|_1 = \sum_{j=0}^{n-1} \|H(x_j) - H(x_{j+1})\|_1$$

and

$$\|H(x_j) - H(x_{j+1})\|_1 = L_{H,w} w(x_j, x_{j+1})$$

for every edge. Then γ is a minimum-cost path and

$$d_F(x_0, x_n) = \mathrm{cost}_F(\gamma) = \frac{\|H(x_0) - H(x_n)\|_1}{L_{H,w}}.$$

Proof. The two saturation equalities give

$$\|H(x_0) - H(x_n)\|_1 = L_{H,w} \,\mathrm{cost}_F(\gamma).$$

The endpoint lower bound, Theorem 10.15, therefore yields

$$d_F(x_0, x_n) \ge \frac{\|H(x_0) - H(x_n)\|_1}{L_{H,w}} = \mathrm{cost}_F(\gamma).$$

Because γ itself is a legal path from x_0 to x_n, the defining infimum gives $d_F(x_0, x_n) \le \mathrm{cost}_F(\gamma)$. Hence equality holds throughout, so γ is geodesic. □

Theorem 10.22 (Local calibration theorem)**.** *Let $D \subseteq S_F$ be a declared subgraph and suppose the edge slope is bounded there by $L_D < \infty$. Then the corresponding coordinate-displacement lower bound is valid for every legal path that remains inside D whenever $L_D > 0$.*

Proof. The global proof of the pathwise bound uses only the edges actually traversed. □

Theorem 10.23 (Bounded-coordinate obstruction)**.** *Suppose $H(S_F)$ has finite diameter M while the intrinsic directed distances in a family of reachable pairs are unbounded. Then there is no constant $c > 0$ such that*

$$c\, d_F(x, y) \le \|H(x) - H(y)\|_1$$

for all those pairs.

Proof. The right-hand side is at most M, whereas the left-hand side is unbounded. □

10.3 Repaired doorway and isolation theorems

The original Section 10.3 is replaced in full on inserted pages 28a--28b. The corrected formulation distinguishes dense rational addresses, isolated canonical codes, and the closure of the trusted bank image.

10.3 Premise-doorway residuals and address isolation (replacement)

Fix a set $\mathcal{W}$ of well-formed formulas, a nonempty trusted bank $B \subseteq \mathcal{W}$, and a finite candidate $c = (P_1, \dots, P_k) \Rightarrow T$ with $k \geq 1$. For an injective exact rational address $\lambda : \mathcal{W} \hookrightarrow \mathbb{Q}$, define

$$D_\lambda(B, c) = \sum_{i=1}^{k} \operatorname{dist}(\lambda(P_i), \lambda(B)), \qquad \delta_\lambda(P) = \inf_{A \in \mathcal{W} \setminus \{P\}} |\lambda(P) - \lambda(A)|, \qquad \delta_c = \min_{1 \leq i \leq k} \delta_\lambda(P_i).$$

Distances are Euclidean distances on $\mathbb{Q} \subset \mathbb{R}$; an infimum may be zero even when its point is absent from the bank. The original Theorem 10.24 remains valid *as a conditional theorem*, but its isolation hypothesis cannot hold for an address map onto all $\mathbb{Q}$ or $\mathbb{Q} \cap [0, 1]$.

Theorem 1 (Corrected doorway criterion; original Theorem 10.24)**.** *If $\delta_c > 0$ and $D_\lambda(B, c) < \delta_c$, then $P_i \in B$ for all i. In particular, under $\delta_c > 0$,*

$$D_\lambda(B, c) = 0 \quad \Longleftrightarrow \quad \{P_1, \dots, P_k\} \subseteq B.$$

Proof. If a required premise $P_j \notin B$, all $A \in B$ are different from P_j, and hence $|\lambda(P_j) - \lambda(A)| \geq \delta_\lambda(P_j) \geq \delta_c$. Its summand alone forces $D_\lambda(B, c) \geq \delta_c$, a contradiction. If all premises belong to B, all summands are zero; the converse follows by the same argument. □

Proposition 2 (Exact zero-residual characterization without isolation)**.** *For* every *injective address map, with arbitrary nonempty B,*

$$D_\lambda(B, c) = 0 \quad \Longleftrightarrow \quad \lambda(P_i) \in \overline{\lambda(B)} \text{ for all } i,$$

where the closure is taken in $\mathbb{R}$. Thus, when λ is injective and B is finite (or $\lambda(B)$ is closed relative to $\lambda(\mathcal{W})$), the same zero-residual membership equivalence holds without requiring $\delta_c > 0$.

Proof. The sum consists of finitely many nonnegative terms, so it is zero exactly when every summand is zero. The defining property of metric-space closure gives $\operatorname{dist}(x, S) = 0 \Leftrightarrow x \in \overline{S}$. If B is finite, $\lambda(B)$ is closed, and injectivity equates membership of $\lambda(P_i)$ in $\lambda(B)$ with membership of P_i in B. The relative-closedness variant gives the same conclusion for points in $\lambda(\mathcal{W})$. □

Proposition 3 (Dense-address counterexample)**.** *Suppose $\lambda : \mathcal{W} \to \mathbb{Q} \cap [0, 1]$ is a bijection and $\mathcal{W}$ is infinite. Then $\delta_\lambda(P) = 0$ for every P. Moreover, for any P and $B = \mathcal{W} \setminus \{P\}$, the candidate $c = (P) \Rightarrow T$ has $P \notin B$ yet $D_\lambda(B, c) = 0$.*

Proof. Every rational point in the unit interval is a limit of different rational points there. Removing one point leaves a dense subset, so $\lambda(P)$ lies in the closure of $\lambda(B)$ while it is not in the bank image. Both assertions follow. □

Theorem 4 (Constructive positive-isolation repair from the surjection paper)**.** *Use the effective first-occurrence enumeration $c_1 < c_2 < \cdots$ of the least-preimage set $C = s(\mathbb{Q}) \subset \mathbb{N}$ established in the replacements to Sections 5–6. Fix a computable bijection $E : \mathbb{N} \to \mathcal{W}$ and set*

$$\lambda_C(P) = c_{E^{-1}(P)} \in C \subseteq \mathbb{N} \subseteq \mathbb{Q}.$$

Then λ_C is computably invertible on its image, $\delta_{\lambda_C}(P) \geq 1$ for every P, and for every nonempty bank B and finite premise tuple c,

$$D_{\lambda_C}(B, c) < 1 \quad \Longleftrightarrow \quad D_{\lambda_C}(B, c) = 0 \quad \Longleftrightarrow \quad \{P_1, \dots, P_k\} \subseteq B.$$

Proof. The codes c_j are distinct positive integers with an effective increasing enumeration and rank inverse. Consequently their pairwise distance is at least one. If a premise is missing, its distance to every bank code is at least one and the corresponding infimum is at least one. If all premises occur in the bank, the residual vanishes. Nonnegativity completes the equivalence. □

Scope and cost. The isolated code is a different exact address map from the dense rational bijection; one must not claim a positive isolation radius for the latter. Constructing and searching canonical codes, maintaining a bank index, and checking support for bank membership carry costs. The doorway criterion is a rigorous identity/membership fact, not a measured search speedup or a lower bound on verifier-valid proof distance.

Proposition 5 (Retained address-distortion result; original Theorem 10.25)**.** *If B is finite and nonempty, $\mathcal{W}$ is at most countable, and c has a missing premise, there are injective rational address maps λ, μ for which $D_\lambda(B, c) \neq D_\mu(B, c)$, while the intrinsic proof-state distance is unchanged when the transition system and edge costs are held fixed.*

Proof. Give the finitely many bank formulas distinct rational codes. Keep those codes and all premise codes other than one missing premise P fixed. Choose two different codes for P with different distances to the finite bank-image (possible by taking one close to, and the other sufficiently far from, all bank codes). All occurrences of P in the candidate change their nonnegative summands in the same direction. Extend the finite partial injections to $\mathcal{W} \to \mathbb{Q}$; a countable complement admits such an extension. The verifier-valid road graph and costs never change, so neither does its intrinsic distance. □

Source foundation: B. Tenneson, *A Closed-Form Surjection from* $\mathbb{N}$ *onto* $\mathbb{Q}$*—With Its Pseudo-Inverse*, Version 2.2 (2026), https://btenneson.github.io/pub/math.LO_Logic/closed_form_surjection_n_to_q/. This paper proves $r \circ s = \mathrm{id}_{\mathbb{Q}}$ and least-preimage minimality; doorway theorems and the constructive isolated-code corollary here are derived in the present manuscript, not asserted in that source.

Section 10.3 replacement concluded on pages 28a--28b

Original Theorem 10.25 is replaced by the retained and re-proved address-distortion theorem in the inserted Section 10.3. The original Section 10.4 starts below.

10.4 Target potentials, exact geodesic certification, and bottlenecks

Definition 10.26 (Target value)**.** For a nonempty target-state set C_T, define

$$V_T(x) = d_F(x, C_T).$$

Theorem 10.27 (Bellman inequality)**.** *For every legal edge $x \xrightarrow{e} y$,*

$$V_T(x) \le w(e) + V_T(y).$$

Proof. Concatenate the edge e with an arbitrarily near-optimal path from y to the target and take an infimum. □

Theorem 10.28 (Target value is the greatest consistent target potential)**.** *On the target-reachable basin, V_T is a consistent target potential, and every consistent target potential p satisfies*

$$p(x) \le V_T(x).$$

Consequently,

$$V_T(x) = \sup\{p(x) : p \text{ is a consistent target potential}\}.$$

Proof. The Bellman inequality and $V_T(t) = 0$ on C_T show that V_T is consistent. For any other consistent p and any finite target-reaching path $x = x_0 \to \cdots \to x_n \in C_T$, repeated consistency gives

$$p(x) \le \sum_{j=1}^{n} w(x_{j-1}, x_j) + p(x_n) = \mathrm{cost}_F(\gamma).$$

Taking the infimum over target-reaching paths yields $p(x) \le V_T(x)$. Since V_T itself belongs to the family, the supremum equals V_T. □

Theorem 10.29 (Potential-saturation geodesic certificate)**.** *Let p be any consistent target potential. If a verified target-reaching path γ from x has finite cost U and*

$$p(x) = U,$$

then

$$V_T(x) = U,$$

so γ is a minimum-cost target path.

Proof. Theorem 10.28 gives $p(x) \le V_T(x)$, while the verified path gives $V_T(x) \le U = p(x)$. All three quantities are equal. □

Theorem 10.30 (Certified target-annulus theorem)**.** *If $\ell(x)$ is any certified lower bound on $V_T(x)$ and γ is any verified target-reaching path of cost U, then*

$$\ell(x) \le V_T(x) \le U.$$

Thus x lies in the logical annulus

$$\mathcal{A}_T(\ell(x), U) = \{z : \ell(x) \le V_T(z) \le U\}.$$

Proof. The lower inequality is the hypothesis. The verified path is one admissible target route, so the infimum target cost is at most U. □

Theorem 10.31 (Unit-cost exact-shell criterion)**.** *In a unit-cost geometry, if*

$$\lceil \ell(x) \rceil = U$$

and a verified U-step target path exists, then

$$V_T(x) = U.$$

Proof. The integer-valued target distance lies between $\lceil \ell(x) \rceil$ and U, which are equal. □

Theorem 10.32 (Separator distance decomposition)**.** *Let $A \subseteq S_F$ intersect every x-to-C_T path. Then, in the extended nonnegative reals,*

$$V_T(x) = \inf_{a \in A} \big(d_F(x, a) + V_T(a) \big).$$

Proof. Every target path meets some $a \in A$, so its cost is at least $d_F(x, a) + V_T(a)$ in the infimum sense. Conversely, concatenating near-optimal x-to-a and a-to-target paths gives the opposite inequality. □

Theorem 10.33 (Multi-coordinate separator lower bound)**.** *Under the hypotheses of Theorem 10.20, let A be an x-to-C_T separator and define*

$$D_{\max}(x, a) = \max_{1 \le i \le m} \frac{\|H_i(x) - H_i(a)\|_1}{L_i}.$$

Then

$$\boxed{V_T(x) \ge L_A(x) := \inf_{a \in A} \big(D_{\max}(x, a) + p_{\max}(a) \big).}$$

Thus a bottleneck can combine endpoint displacement to the separator with a certified residual target potential beyond the separator.

Proof. For each $a \in A$, the endpoint lower bound applied to every coordinate gives

$$d_F(x, a) \geq D_{\max}(x, a).$$

Theorem 10.20 gives $V_T(a) \geq p_{\max}(a)$. Substitute both inequalities into Theorem 10.32 and take the infimum over $a \in A$. □

Theorem 10.34 (Unit-cost multi-coordinate bottleneck shell certificate)**.** *In a unit-cost geometry, let $L_A(x)$ be the separator lower bound from Theorem 10.33. If a verified U-step target path exists and*

$$\lceil L_A(x) \rceil = U,$$

then that path is shortest and

$$V_T(x) = U.$$

Proof. Theorem 10.33 gives $L_A(x) \leq V_T(x)$. Apply the unit-cost exact-shell criterion, Theorem 10.31, with $\ell = L_A$. □

10.5 Approximate geodesic navigation and cartographic regret

The preceding results treat exact lower bounds and exact geodesic certification. The next package quantifies what happens when continuation values are only approximately known. The statements are imported from the authorś Omnicartography development and reproved here from the definitions [16]. No derivation-history terminology is needed for the statements themselves.

Fix a nonempty settlement set C_T and write

$$H(x) = d_F(x, C_T).$$

At every state under consideration assume the minimum continuation value is attained. For a legal successor y of x, define

$$Q_x(y) = w(x, y) + H(y), \qquad S(x, y) = Q_x(y) - H(x).$$

Thus $S(x, y) \geq 0$, and $S(x, y) = 0$ exactly when y is an optimal first continuation. When the optimum is unique, write it as $y^\star$ and define the directional gap

$$\Delta(x) = \inf_{z \neq y^\star} \big(Q_x(z) - Q_x(y^\star)\big) \geq 0.$$

Uniqueness alone need not force $\Delta(x) > 0$ when the successor set is infinite. Whenever a positive separation is required below, it is imposed explicitly as the additional hypothesis $\Delta(x) > 0$ (and it holds automatically for a finite candidate set with a unique optimum). Suppose estimated continuation values satisfy

$$|\widehat{Q}_x(y) - Q_x(y)| \leq \varepsilon$$

on the candidate set being used.

Theorem 10.35 (Approximate-Minimizer Local-Regret Bound)**.** *Let $\widehat{y}$ minimize $\widehat{Q}_x$ on a candidate set containing a true optimal successor. Then*

$$0 \leq S(x, \widehat{y}) \leq 2\varepsilon.$$

The constant 2 is uniformly best possible if an estimated tie may be resolved arbitrarily.

Proof. Let $y^\star$ be a true optimal successor in the candidate set. Then

$$Q_x(\widehat{y}) \le \widehat{Q}_x(\widehat{y}) + \varepsilon \le \widehat{Q}_x(y^\star) + \varepsilon \le Q_x(y^\star) + 2\varepsilon = H(x) + 2\varepsilon.$$

Subtracting $H(x)$ gives the upper bound, while nonnegativity is the Bellman inequality. Sharpness follows from two true continuation values $H(x)$ and $H(x) + 2\varepsilon$ perturbed by $+\varepsilon$ and $-\varepsilon$, which creates an estimated tie. □

Proved by Sol.

Theorem 10.36 (Cumulative Approximate-Geodesic Bound)**.** *Consider a finite legal path*

$$x_0 \to x_1 \to \cdots \to x_k.$$

At step i, suppose x_{i+1} is chosen as an estimated minimizer from a candidate set containing a true optimal successor, with continuation-value error at most ε_i. Then

$$\sum_{i=0}^{k-1} w(x_i, x_{i+1}) + H(x_k) - H(x_0) \le 2\sum_{i=0}^{k-1} \varepsilon_i.$$

Proof. Theorem 10.35 gives

$$S(x_i, x_{i+1}) \le 2\varepsilon_i.$$

But

$$S(x_i, x_{i+1}) = w(x_i, x_{i+1}) + H(x_{i+1}) - H(x_i).$$

Summing telescopes the horizon terms. □

Proved by Sol.

Theorem 10.37 (Wrong-Move Slack Barrier)**.** *If x has a unique optimal successor $y^\star$ with directional gap $\Delta(x) > 0$, then every nonoptimal successor z satisfies*

$$S(x, z) \ge \Delta(x).$$

Consequently,

$$S(x, y) < \Delta(x) \quad \Longrightarrow \quad y = y^\star.$$

Proof. Because $Q_x(y^\star) = H(x)$,

$$S(x, z) = Q_x(z) - Q_x(y^\star).$$

The directional gap is the infimum of these quantities over nonoptimal successors, so each is at least $\Delta(x)$. □

Proved by Sol.

Theorem 10.38 (Selective-Exactness Regret Decomposition)**.** *In the setting of Theorem 10.36, suppose that at a set of steps the optimum is unique with directional gap $\Delta_i > 0$ and*

$$2\varepsilon_i < \Delta_i.$$

Let U be the remaining unresolved steps. Then every separated step has zero slack and

$$\sum_{i=0}^{k-1} w(x_i, x_{i+1}) + H(x_k) - H(x_0) \le 2\sum_{i \in U} \varepsilon_i.$$

If $x_k \in C_T$, the same right-hand side bounds the routeś additive excess over $H(x_0)$.

Proof. At a separated step, Theorem 10.35 gives $S_i \le 2\varepsilon_i < \Delta_i$. Theorem 10.37 therefore forces the chosen successor to be the unique optimum, so $S_i = 0$. At unresolved steps the local-regret theorem gives $S_i \le 2\varepsilon_i$. Summing and telescoping proves the claim. □

Proved by Sol.

Theorem 10.39 (Relative Approximation Criterion)**.** *Assume $H(x_0) > 0$ and the path of Theorem 10.36 ends in settlement. If for some $\rho \ge 0$,*

$$2\sum_{i=0}^{k-1} \varepsilon_i \le \rho H(x_0),$$

then its total cost C satisfies

$$C \le (1+\rho)H(x_0).$$

Proof. Settlement gives $H(x_k) = 0$. Substitute this into Theorem 10.36 and apply the stated aggregate error budget. □

Proved by Sol.

Theorem 10.40 (BANK-Shortcut Relative-Precision Effect)**.** *Suppose a verifier-safe BANK update reduces a positive starting horizon from H to H' with*

$$0 < H' \le H,$$

and an approximate navigation procedure on the updated geometry has additive regret at most $E \ge 0$. Then

$$C \le H' + E = \left(1 + \frac{E}{H'}\right) H'.$$

For fixed E,

$$1 + \frac{E}{H'} \ge 1 + \frac{E}{H}.$$

Thus an absolute shortcut can reduce optimal distance while making the same absolute cartographic error weaker as a relative guarantee. To retain factor $1+\rho$, it is sufficient to require $E \le \rho H'$.

Proof. The first display is the definition of an additive-regret guarantee relative to the updated optimum. Since $0 < H' \le H$, one has $1/H' \ge 1/H$, proving the comparison. The final sufficient condition is exactly $E/H' \le \rho$. □

Proved by Sol.

Theorem 10.41 (Shortcut-Induced Resolution Rescaling)**.** *Assume a bit-resolution model $\varepsilon_i \le 2^{-r_i}$ and suppose a verifier-safe BANK update contracts the starting horizon to*

$$H' = \alpha H, \qquad 0 < \alpha \le 1.$$

For the same target relative factor $1+\rho$, the aggregate sufficient allowance contracts from

$$\sum_i 2^{-r_i} \le \frac{\rho H}{2}$$

to

$$\sum_i 2^{-r_i'} \le \frac{\rho H'}{2} = \alpha \frac{\rho H}{2}.$$

A sufficient uniform rescaling is

$$r_i' \ge r_i + \log_2 \frac{1}{\alpha} = r_i + \log_2 \frac{H}{H'}.$$

For integer resolutions, rounding the added quantity upward is sufficient.

Proof. Let $\delta = \log_2(1/\alpha)$. Increasing each r_i by δ multiplies every error proxy 2^{-r_i} by $2^{-\delta} = \alpha$. Hence any schedule satisfying the old aggregate allowance satisfies the contracted allowance after the uniform increase. Upward integer rounding only decreases the error proxies further. □

Proved by Sol.

These seven results extend exact geodesic certification into an approximate-navigation calculus. The central point is that local bounded error has a directly auditable global cost, while states whose directional gaps exceed twice the local error are exact and consume no regret budget.

10.6 Re-addressing, trading, and the separation of maps from roads

These final results explain how the strengthened Exact Addressing Theorem fits the path geometry. Exact re-addressing can leave intrinsic geometry fixed; a cost-preserving operational transport is an isometry; and merely preserving formulas or coordinates is weaker than preserving the road system.

Theorem 10.42 (Pure re-addressing invariance)**.** *If the proof-state graph and edge costs are unchanged while only the exact WFF address map is replaced, then the verifier-valid path family and intrinsic distance d_F are unchanged.*

Proof. Neither path legality nor edge cost refers to the external WFF address. □

Theorem 10.43 (Cost-preserving trading isometry)**.** *Suppose $\Phi : S_P \to S_Q$ is a bijection carrying legal edges bijectively to legal edges, preserving edge costs and targets. Then*

$$d_Q(\Phi x, \Phi y) = d_P(x, y)$$

for all states x, y.

Proof. Every P-path corresponds to a Q-path of identical cost and vice versa, so the two infima agree. □

Theorem 10.44 (Target-value transport under isometric trading)**.** *Under the preceding hypotheses,*

$$V_T^Q(\Phi x) = V_T^P(x).$$

Proof. The target-compatible bijection identifies all target-reaching path costs exactly. □

Theorem 10.45 (Consequence equivalence is weaker than metric equivalence)**.** *Two presentations may have the same theorem set while carrying different path families, branching patterns, proof horizons, or intrinsic distances.*

Proof. Adding verifier-expandable macro rules can preserve theoremhood while shortening search-step paths; conversely, compiling a primitive rule into a multi-step simulation can lengthen them. □

Theorem 10.46 (Fixed landscape, movable roads)**.** *If a trade preserves the WFF set, exact WFF addresses, and height map but changes legal primitive or macro transitions, then all WFF address and height values remain fixed while logical path distance may change.*

Proof. The address and height functions depend only on WFF identity; path distance depends on the legal transition system and its costs. □

Theorem 10.47 (Path-first determination of intrinsic UTG)**.** *Fix a state set S, a verifier-valid path family $\mathcal{P}$, and an additive nonnegative path-cost functional C. Then the intrinsic directed distance*

$$d(x, y) = \inf_{\gamma \in \mathcal{P}(x,y)} C(\gamma)$$

is completely determined by these data and is independent of any additional WFF address or height map.

Proof. The defining infimum refers only to the path family and path cost. □

Theorem 10.48 (Same path geometry, different maps)**.** *Two address/height systems placed on the same $(S, \mathcal{P}, C)$ have identical intrinsic logical distance even if their coordinate traces differ arbitrarily.*

Proof. Apply the preceding result to both coordinate choices. □

Theorem 10.49 (Same map, different path geometry)**.** *Two presentations may share the same WFF set, exact addresses, and heights while having different intrinsic distances if their legal path families or path costs differ.*

Proof. The intrinsic distance is determined by paths and costs, not by the shared coordinate maps. □

Theorem 10.50 (No automatic single-coordinate principle)**.** *Let $q : S_F \to \mathbb{R}$ be an arbitrary scalar coordinate. The mere existence of q does not imply that q determines WFF identity, rule applicability, path legality, path cost, target distance, separator structure, and translation provenance simultaneously. Any such recovery requires additional compatibility hypotheses.*

Proof. These objects depend on distinct data: WFF identity requires an injective representation; rule applicability and path legality depend on R and V; path cost depends on w; target distance additionally depends on the target set; separator structure depends on the path graph; and provenance depends on the declared translations. Holding q fixed while changing any of those independent data can change the corresponding structure. Therefore no simultaneous recovery follows from the scalar-coordinate assumption alone. □

10.7 Cross-fiber distortion, quotient geometry, and certified shortcuts

The following results make "universal theorem geometry" literal in two directions: they compare proof distances across formal-system fibers and show when a quotient can safely compress a proof-state graph. The cross-fiber statements are imported from *Universal Theorem Geometry I* [11]; the quotient and shortcut statements are imported from the *Trading Theorem* manuscript [14]. The proofs below are direct reproofs under the displayed hypotheses.

Definition 10.51 (Proof-preserving translation and simulation constant)**.** Let F and G be presented formal systems. A proof-preserving translation consists of a formula translation together with a state map

$$\widehat{\tau} : S_F \to S_G$$

that maps every legal F derivation to a legal G derivation of the translated conclusion. It has edge-simulation constant $K \geq 0$ if every legal F edge $e : B \to B'$ can be simulated in G by a path from $\widehat{\tau}(B)$ to $\widehat{\tau}(B')$ of cost at most $K w_F(e)$.

Theorem 10.52 (Lipschitz transport of proof distance)**.** *If $\widehat{\tau} : F \to G$ is proof-preserving with finite edge-simulation constant K, then for all $B, C \in S_F$ with $d_F(B, C) < \infty$,*

$$d_G\big(\widehat{\tau}(B), \widehat{\tau}(C)\big) \leq K d_F(B, C).$$

Proof. For every $\epsilon > 0$, choose an F path P from B to C with

$$\operatorname{cost}_F(P) < d_F(B, C) + \epsilon.$$

Replace each edge of P by its simulating G path. Concatenation produces a G path of cost at most

$$K \operatorname{cost}_F(P) < K d_F(B, C) + K\epsilon.$$

Take the infimum over G paths and let $\epsilon \downarrow 0$. □

Proved by Sol.

Theorem 10.53 (Bi-Lipschitz and isometric transport)**.** *Let $X \subseteq S_F$ and $Y \subseteq S_G$, and suppose $\widehat{\tau} : X \to Y$ is a bijection with edge-simulation constant K. Suppose its inverse is induced by a proof-preserving reverse translation with finite constant $L > 0$. Then*

$$\frac{1}{L} d_F(B, C) \leq d_G\big(\widehat{\tau}(B), \widehat{\tau}(C)\big) \leq K d_F(B, C)$$

for all finite-distance pairs $B, C \in X$. If $K = L = 1$, the translation is an isometry of the directed proof distances on X and Y.

Proof. The upper bound is Theorem 10.52. Apply that theorem to the reverse translation:

$$d_F(B, C) \leq L d_G\big(\widehat{\tau}(B), \widehat{\tau}(C)\big).$$

Divide by L. □

Proved by Sol.

Definition 10.54 (Verifier-safe search quotient)**.** Let S be a proof-state graph with target set T. A search quotient is a map $q : S \to Q$ such that every original edge $s \to t$ induces a quotient edge $q(s) \to q(t)$. For weighted search, assign quotient-edge costs satisfying

$$w_Q(q(s), q(t)) \leq w(s, t)$$

for every original edge.

Theorem 10.55 (Quotient Obstruction Theorem)**.** *If no quotient path exists from $q(s_0)$ to any class in $q(T)$, then no original path exists from s_0 to T.*

Section 10.7 — proof-interface clarification

Theorem 10.52: the zero-constant case

The theorem allows an edge-simulation constant K >= 0. Its proof should use a **non-strict bound** on the simulated path: cost_G(sim(P)) <= K cost_F(P) <= K(d_F(B,C) + epsilon). Then let epsilon tend to zero. This remains valid when K = 0; the original strict inequality in that case was an avoidable proof-writing error.

Theorem 10.53: inverse paths and ambient distances

For the reverse estimate, the proof-preserving reverse translation must apply to the **entire G-paths used to calculate the ambient G-distance**, or the theorem must explicitly use distances restricted to the subgraphs X and Y with simulators staying within those subgraphs. A bijection only on endpoint sets X and Y does not by itself justify the lower bound when shortest paths leave Y.

A sufficient corrected hypothesis is: forward and reverse simulations are defined on every legal edge of the relevant full state graphs, map corresponding endpoints inversely, and have the displayed finite multiplicative cost bounds. Under that hypothesis, applying the two simulation inequalities gives d_F/L <= d_G <= K d_F for the pairs considered.

Terminology

Since inference reachability can be asymmetric and edge weights can vanish, intrinsic logical path distance is generally a **directed extended pseudodistance**, not a symmetric separating metric. Undirected graph distance d_u used for curvature is a different object. The label-geodesic result of Section 11.4 has the same directed-distance character.

Proof. Any original path $s_0 \to s_1 \to \cdots \to s_m \in T$ would project vertex by vertex to a quotient path ending in $q(T)$, contradicting quotient unreachability. □

Proved by Sol.

Theorem 10.56 (Quotient-distance lower bound)**.** *Under the edge-relaxation inequality above,*

$$d_Q(q(s), q(T)) \leq d_F(s, T).$$

Hence quotient distance is an admissible lower bound on intrinsic target distance.

Proof. Every original target-reaching path projects to a quotient target-reaching path whose cost is no larger edge by edge. Take infima. □

Proved by Sol.

Theorem 10.57 (Exact Geodesic Quotient Theorem)**.** *Assume, in addition, cost-preserving path lifting toward T: every quotient path from $q(s)$ to $q(T)$ lifts to an original path from s to an actual target with the same cost. Then*

$$d_Q(q(s), q(T)) = d_F(s, T)$$

for every relevant state s.

Proof. Theorem 10.56 gives $d_Q \leq d_F$. Conversely, for every $\epsilon > 0$ choose a quotient target-reaching path of cost less than $d_Q + \epsilon$. Cost-preserving lifting produces an original target-reaching path of the same cost, so $d_F \leq d_Q + \epsilon$. Letting $\epsilon \downarrow 0$ gives $d_F \leq d_Q$, hence equality. □

Proved by Sol.

Theorem 10.58 (Settlement-congruence reduction)**.** *Let $\sim$ be an equivalence relation on proof states such that: (i) target membership is constant on each equivalence class; (ii) whenever $s \sim t$ and $s \to s'$, there exists $t \to t'$ with $s' \sim t'$; and (iii) the same matching property holds with s and t reversed. Then a target is reachable in the original graph if and only if a target class is reachable in the quotient $S/\sim$. Consequently, passing to the quotient preserves settlement completeness.*

Proof. Original paths project to quotient paths. Conversely, lift a quotient path one class at a time using the matching-transition property. The terminal lifted state is a target because target membership is class invariant. □

Retaining a single concrete representative per reachable class is a stronger implementation claim and does not follow from bisimulation alone: it additionally requires a declared canonicalization or redirect mechanism that realizes quotient transitions from the selected representatives. *Proved by Sol.*

Theorem 10.59 (Verified-Lemma Macro Shortcut Theorem)**.** *Suppose a verified derivation segment D establishes a lemma λ at primitive cost m, and a BANK reference to λ has declared search cost $b \leq m$. Assume the macro is enabled only in proof contexts in which every hypothesis, side condition, and provenance item required to replay D is available, and that BANK retains a verifier-checkable expansion of D with that support. Add a macro edge for the BANK reference. Then:*

1. the theorem set is unchanged;

2. *the shortest proof horizon cannot increase;*

3. *if a certified proof of cost L contains r disjoint re-derivations of D, replacing them by BANK references yields a certified proof of cost at most*

$$L - r(m - b).$$

Proof. The macro proves nothing new because, whenever it is enabled, its stored support and expansion replay the already verified derivation D in the current admissible context. Thus theoremhood is unchanged. All old proof paths remain available, hence the shortest horizon cannot increase. Each disjoint replacement saves $m - b$, and r replacements save $r(m - b)$. □

Proved by Sol.

Together these results separate three notions that are easy to conflate: a change of coordinates, a distortion-controlled translation between formal systems, and a quotient or macro operation that genuinely changes the represented search graph while preserving a declared certification property.

10.8 Discrete curvature on theorem graphs

The path metric tells how expensive it is to travel through the verifier-defined road system. Curvature asks a different question: how do the neighborhoods around a legal inference edge overlap, separate, or expand? This release adopts a concrete baseline from discrete metric geometry, recorded for this project in the unit-edge curvature note [9].

Let $G = (V, E)$ be a theorem or proof-state graph with unit edge weights. Proof legality may remain directed, but for the curvature calculation below let G_u denote the underlying undirected graph. Write d_u for the graph metric on G_u. When legal directed path length is needed later, write $d_{\text{legal}}(s, t)$ for the minimum number of legal directed edges from s to t (with value $+\infty$ if no such path exists). For each $x \in V$, let m_x be the uniform probability measure on the neighbors of x in G_u.

Definition 10.60 (Ollivier–Ricci theorem-graph curvature)**.** For adjacent vertices $x \sim y$, define

$$\boxed{\kappa(x, y) = 1 - \frac{W_1(m_x, m_y)}{d_u(x, y)}}$$

where W_1 is Wasserstein–1 distance computed with the undirected graph metric d_u. For a unit edge, $d_u(x, y) = 1$, so

$$\boxed{\kappa(x, y) = 1 - W_1(m_x, m_y).}$$

The sign separates three local regimes:

$\kappa(x, y) > 0$	neighbor distributions are comparatively close or overlapping;
$\kappa(x, y) \approx 0$	the local region is approximately flat at this scale;
$\kappa(x, y) < 0$	the neighborhoods spread apart, indicating local expansion or branching.

The interpretation is local. Negative curvature is not, by definition, a hard proof or a certified bottleneck; positive curvature is not, by definition, an easy proof. Any connection to global proof length or search difficulty requires a separate theorem or experiment.

For a legal proof path

$$P = (T_0, T_1, \ldots, T_n),$$

its curvature profile is the sequence

$$\kappa(T_0, T_1), \ldots, \kappa(T_{n-1}, T_n).$$

A useful path functional introduced in the curvature-profile settlement is the cumulative negative-curvature burden

$$\boxed{B_-(P) = \sum_{i=0}^{n-1} \max\{0, -\kappa(T_i, T_{i+1})\}.} \tag{1}$$

This quantity will be used in Section 14 as a worked example of conjecture generation followed by AMLD+0 settlement.

10.9 A geometric dictionary for theorem space

The framework can now be read as a discrete metric and differential geometry of theorem space. The analogy is structural rather than literal: theorem graphs are discrete objects, not smooth manifolds.

Geometric notion	Theorem-space analogue
Point	theorem, formula, or rule-complete proof state
Curve or path	verifier-legal inference path; in a GBG presentation, a move-labeled match-proof
Path length	charged cost of a proof path; ordinary ply count in the unit-cost GBG specialization
Intrinsic metric	minimum legal proof cost $d_F(u, v)$
Geodesic	minimum-cost verifier-accepted proof path
Coordinate or address	exact identity or declared feature representation
Coordinate distance	separation in an addressing or navigation representation
Local neighborhood	adjacent or nearby theorem/proof states under declared legal moves
Potential function	target-relative admissible proof potential
Separator	certified set through which every relevant proof route must pass
Bottleneck or gateway	narrow separator or low-width passage in theorem space
Curvature	discrete graph curvature, here the Ollivier–Ricci quantity κ
Curvature along a path	the curvature sequence encountered along a proof
Local geometry	neighborhood overlap, branching, curvature, and local path structure
Global geometry	distances, geodesics, separators, connectivity, and theorem-graph structure
Extrinsic/navigation geometry	feature embeddings, Hilbert ordering, learned representations, and other search maps

The dictionary reinforces the central restriction of this manuscript. Intrinsic proof geometry is determined by legal inference structure and the declared cost model. Exact addresses, learned embeddings, Hilbert coordinates, and other navigation representations may assist search, while curvature describes local neighborhood transport geometry. None is automatically identical to intrinsic proof

distance. A bridge theorem or calibration result is required before distances, curvature, or local structure in one representation may be interpreted as a theorem about another.

10.10 Route multiplicity and cartographic signatures

Intrinsic distance records the cost of a best route, but it does not record how many essentially different verified routes are known, how much downstream traffic passes through a state, how severe a bottleneck is, or how sensitive apparent accessibility is to the chosen chart. The following definitions are imported from *Proof-State Cartography* [13] and are included because they add genuinely different coordinates to theorem geometry.

Let $\mathcal{R}_\Omega(T)$ be the represented verifier-accepted routes reaching T in a finite mapped corpus or bounded regime Ω. Fix a sound declared route-equivalence relation $\sim$ that may, for example, identify histories differing only by certified commutations of independent actions.

Definition 10.61 (Essential route class and observed multiplicity)**.** An essential route is an equivalence class in

$$\mathcal{R}_\Omega(T)/\!\sim.$$

The observed essential proof multiplicity is

$$\mu_\Omega(T) = \left|\mathcal{R}_\Omega(T)/\!\sim\right|.$$

Unless the mapped regime is independently complete, $\mu_\Omega(T) = 1$ means one essential route is represented in Ω; it does not assert absolute proof uniqueness.

For a GBG presentation of the same operational state graph, the unquotiented move-labeled history count is exact rather than merely observed. If $M_n(s_0, T)$ denotes the number of n-ply match-proofs from the distinguished start s_0 to endpoint T, then Section 11 proves that exact state addressing together with exact move labels reconstructs every such history. Thus $M_n(s_0, T)$ is the cardinality of the corresponding set of exact addressed history traces. The essential multiplicity $\mu_\Omega(T)$ remains a further quotient by the declared route-equivalence relation and therefore answers a different question.

For a mapped proof state v, let $H_0(v)$ denote a declared proof horizon from the frozen trusted base, let $\tau(v)$ be a frozen traffic/reuse statistic, let $\beta(v)$ be a frozen bottleneck statistic, and let $\rho(v)$ measure representation sensitivity across a declared family of admissible charts. One example is

$$\rho(v) = \sup_{q,q'} |A_q(v) - A_{q'}(v)|$$

for a frozen accessibility statistic A_q.

Definition 10.62 (Cartographic signature)**.** Define

$$\Sigma_{\mathrm{cart}}(v) = \big(H_0(v), \mu(v), \tau(v), \beta(v), \rho(v)\big).$$

This is a heterogeneous structural signature, not a claim that all five coordinates arise from one metric.

Theorem 10.63 (Verified Multiplicity Increase)**.** *If the mapped regime currently contains m distinct essential route classes to T, and a verifier-accepted route P is added whose class differs from all m represented classes, then the observed essential proof multiplicity after insertion is at least $m + 1$.*

Proof. Verifier acceptance places P in the represented route set. Since its equivalence class is new, adjoining it increases the cardinality of the quotient route set by at least one. □

Proved by Sol.

This coordinate package supports terrain distinctions that distance alone cannot express: a state can be close but structurally narrow, far but highly redundant, central but easy to bypass, or apparently difficult only under a particular representation. Such labels are descriptive unless their numerical thresholds are frozen for a concrete corpus.

10.11 Effect on the Exact Addressing program

The added theorems sharpen the claims already central to this paper. Exact address distance is a genuine identity geometry but not automatically a proof metric. A chosen geometry becomes proof-theoretically informative when legal edges have a finite displacement-to-cost calibration; the resulting coordinate potential is then a certified lower bound on remaining proof cost. If that lower bound meets the cost of a verified path—directly, after rounding in the unit-cost case, or after separator refinement—the path is certified shortest. Address isolation can turn a sufficiently small premise-slot residual into a premise-eligibility certificate, while arbitrary re-addressing can still distort raw residuals without changing intrinsic proof distance. Approximate continuation values now carry a quantified regret calculus, including exact steps whenever the directional gap dominates the local error. Cross-fiber simulation constants become Lipschitz distortion constants, and sufficiently faithful quotients can preserve exact geodesic distance while certified macro edges can shorten horizons without changing theoremhood. Discrete curvature supplies a local terrain descriptor distinct from path length, while route multiplicity, traffic, bottleneck importance, and representation sensitivity provide additional nonmetric cartographic coordinates. Thus Exact Addressing can support a richer universal theorem geometry without conflating coordinates, quotients, curvature, route multiplicity, or learned navigation maps with the verifier-defined road system.

11 GBG and GBG$^+$ as an exact operational substrate

Preliminary Release 2.1 brings the generalized-board-game substrate directly into the operational layer of Exact Addressing. The purpose is not to replace the deductive presentation by game terminology. It is to make explicit a useful fact already implicit in the proof-state treatment: once a state contains everything needed to determine its legal immediate continuations, the evolution of proof search may be represented as a move-labeled transition system. In the GBG foundation, a rule-complete state is the formal object, a legal move is a partial unary inference rule on complete states, finite reachability is theoremhood relative to the distinguished start, and a legal move-labeled history is a match-proof [17].

This section uses that substrate to sharpen four questions that recur throughout the manuscript: what exact object should receive an operational address, what information is required to reconstruct a proof history, how route multiplicity differs from endpoint identity, and when a board-player extension changes the intrinsic road system rather than merely adding administrative structure.

11.1 Rule-complete proof states and the state-lift of multipremise inference

A primitive mathematical rule may have several premises even though a GBG move rule is unary on a complete state. There is no conflict. The unary object is the *complete proof state*, not an isolated premise. A state may contain a trusted BANK, a sequent, a multiset of open goals, a derivation prefix, a frontier, verifier status, or any other finite data required to determine legal immediate continuations.

Definition 11.1 (GBG presentation of proof infrastructure)**.** Let S be a rule-complete proof-state space with distinguished start s_0. Let A be a set of canonical move labels. For each $\mu \in A$, let

$$r_\mu : S \rightharpoonup S$$

be a partial unary transition rule whose domain is exactly the states at which move μ is legal. The tuple

$$\mathcal{G} = (S, s_0, A, \{r_\mu\}_{\mu \in A})$$

is a GBG presentation of the proof infrastructure when every legal operational transition is represented by exactly one declared move label and every declared move has a uniquely determined successor.

Theorem 11.2 (Proof-path/match-proof correspondence)**.** *For a GBG presentation of a verifier-valid operational proof system, finite move-labeled match-proofs beginning at s_0 are in one-to-one correspondence with finite verifier-valid operational paths beginning at s_0.*

Proof. A match-proof is, by definition, a sequence

$$(s_0, \mu_0, s_1, \ldots, \mu_{n-1}, s_n)$$

with $s_{i+1} = r_{\mu_i}(s_i)$ at every step. Since each r_{μ_i} is exactly a declared legal operational transition, this is a verifier-valid labeled path. Conversely, every verifier-valid labeled transition is represented by exactly one canonical move label, so a finite labeled operational path determines a unique match-proof. □

Reproved here from the GBG axioms.

The theorem gives a precise state-level realization of the manuscript's earlier state-lift warning. A WFF may be one coordinate of a proof state, but the legal road system lives on rule-complete states whenever future legality depends on BANK contents, open assumptions, scheduler state, verifier status, or other operational metadata.

11.2 Exact state addresses and continuation sufficiency

Let

$$\alpha : S \longrightarrow D$$

be an injective exact address map on the rule-complete state space. When computation matters, assume α and its inverse on $\alpha(S)$ are computable.

Theorem 11.3 (Addressed continuation sufficiency)**.** *If $\alpha(s) = \alpha(t)$ for two rule-complete states, then $s = t$. Consequently s and t have exactly the same applicable move labels and, for each applicable move label μ, exactly the same successor.*

Proof. Injectivity gives $s = t$. Rule completeness makes move applicability and successor values functions of the complete state, so equal states have equal legal continuation structure. □

Thus an exact address of a rule-complete state is stronger operationally than an exact address of one WFF occurring inside that state. Decoding the complete-state address recovers enough information to determine every legal immediate continuation. Conversely, if two operational states with different legal continuation structures are assigned the same supposed "exact operational address," then that address is not exact for the operational state space.

Theorem 11.4 (Operational-address adequacy criterion)**.** *Let $q : S \to Q$ be any state representation. If there exist $s, t \in S$ with $q(s) = q(t)$ but with different applicable move-label sets or different successors for some common applicable label, then q cannot be an exact address of the rule-complete state space.*

Proof. An exact address must be injective on S. The assumed pair is operationally distinct because its legal continuation structure differs, so identifying the pair violates injectivity on rule-complete states. □

This criterion leaves feature maps free to be many-to-one. It applies only to the exact operational address. A learned navigation map Φ may intentionally forget information, provided the exact state identity remains separately recoverable.

11.3 Exact history addresses, endpoint projection, and route multiplicity

For a move-labeled match-proof

$$\pi = (s_0, \mu_0, s_1, \ldots, \mu_{n-1}, s_n),$$

define its exact addressed history trace by

$$\mathsf{Hist}_\alpha(\pi) = \big(\alpha(s_0), \mu_0, \alpha(s_1), \ldots, \mu_{n-1}, \alpha(s_n)\big).$$

The move labels are retained because a conclusion-only address trace need not reconstruct witnesses, substitutions, or other edge data.

Theorem 11.5 (Exact addressed-history reconstruction)**.** *If α is exact on rule-complete states and move labels are canonical, then*

$$\mathsf{Hist}_\alpha(\pi) = \mathsf{Hist}_\alpha(\eta) \quad \Longrightarrow \quad \pi = \eta.$$

Hence exact addressed history traces are lossless representations of move-labeled match-proofs.

Proof. Apply α^{-1} to each state address in the common trace. The complete state sequence is recovered exactly, and the canonical move labels already occur explicitly in the trace. □

The endpoint map need not be injective. Distinct match-proofs can terminate at the same rule-complete state. Let $M_n(s_0, T)$ be the number of n-move labeled histories from s_0 to endpoint T when finite.

Theorem 11.6 (Exact history-count/address-count identity)**.** *Under the preceding hypotheses,*

$$M_n(s_0, T) = \left|\{\mathsf{Hist}_\alpha(\pi) : \pi \text{ is an } n\text{-move match-proof from } s_0 \text{ to } T\}\right|.$$

Proof. The addressed-history map is injective by Theorem 11.5 and is defined on every such match-proof. Therefore it is a bijection from the set being counted to its image. □

This theorem separates two quantities that are easy to conflate: the number of theorem states reached and the number of distinct histories proving those states. Exact addressing identifies endpoints without collapsing their derivational multiplicity. The essential route multiplicity $\mu_\Omega(T)$ of the route-multiplicity subsection of Section 10 may then be obtained by imposing a further declared equivalence relation on these exactly represented histories.

11.4 GBG$^+$: exact addressing with a board-player

GBG$^+$ adds a distinguished participant

$$P_0 = B^\star$$

while requiring every board action to remain inside the declared transition system [17]. Scheduling, memory writes, certification, handoff, federation, reflection, and controlled mutation are not extra-system interventions: when present, they are themselves represented by legal transitions of an enlarged rule-complete state.

For an ATP or multi-agent settlement architecture, a typical enlarged state may therefore have the form

$$S = (q, M_1, \dots, M_N, M_B, \sigma, \Pi, \eta),$$

where q is the current proof/search state, M_i are local memories, M_B is trusted board memory, σ is scheduler state, Π records provenance/certificates, and η collects any additional rule-relevant metadata. The exact-address discipline now has a sharp consequence: if any coordinate can change legal immediate continuations, an exact operational state address must distinguish its relevant values. A navigation representation may omit such information; the exact state address may not.

The GBG$^+$ player-label operator is

$$F(r) = \max(0, r-1), \qquad r \in \{0, 1, \dots, N\}.$$

It acts on player labels, not on theorem states, unless an additional structure explicitly lifts it.

Theorem 11.7 (Board-label unit geodesic)**.** *Equip the directed player-label graph with the edges $r \to F(r)$ for $r > 0$, each of unit cost. Then*

$$F^k(r) = \max(0, r-k),$$

and the directed distance from r to the unique fixed label 0 is exactly

$$d_{\text{label}}(r, 0) = r.$$

The canonical orbit $r, F(r), \dots, 0$ is therefore a unit-speed geodesic in the player-label graph.

Proof. The iterate formula follows by induction. Each nonzero step lowers the label by exactly one, so the displayed orbit reaches 0 in r unit-cost moves. No directed path using these edges can lower the label by more than one per move, so fewer than r moves cannot reach 0. □

This is an exact directed-distance statement about the player-label graph. It is not a theorem about proof distance. The two geometries remain separately typed.

11.5 Passive board extension and projected proof-geometry invariance

The GBG$^+$ foundation constructs a canonical passive extension $E(G)$ of an ordinary player-indexed GBG G. The extension adjoins an administrative bit and a board-only action ε that leaves the underlying ordinary state unchanged; ordinary moves reset the administrative bit. A history projection $\bar{U}$ erases the administrative coordinate and occurrences of ε, preserving the ordinary move sequence and ordinary ply count [17].

For proof geometry, assign projected edge cost

$$w_+(e) = \begin{cases} 1, & e \text{ is a lifted ordinary move,} \\ 0, & e = \varepsilon. \end{cases}$$

Let d_G denote ordinary unit-cost path distance in G, and let $d_{E,+}$ be minimum path cost in the passive extension under w_+.

Theorem 11.8 (Passive GBG$^+$ projection isometry)**.** *For ordinary states s, t and their tagged copies in the canonical passive extension,*

$$\boxed{d_{E,+}(\langle s, 0\rangle, \{\langle t, 0\rangle, \langle t, 1\rangle\}) = d_G(s, t).}$$

More generally, $\bar{U}$ preserves the projected cost of every passive history, and every ordinary history has a lifted passive history of the same projected cost.

Proof. Projecting any passive history deletes only ε-moves, each of projected cost zero, and retains the ordinary move sequence, each of cost one. Hence projection preserves total projected cost and gives $d_G(s, t) \leq d_{E,+}$. Conversely, lift any ordinary path from s to t using only the corresponding ordinary moves in the passive extension and no ε-move. The lifted path has exactly the same cost, giving $d_{E,+} \leq d_G(s, t)$. □

Theorem 11.9 (Passive settlement preservation)**.** *If the target or settlement condition depends only on the underlying ordinary state and is retained through the tagged passive extension, then ordinary reachability and finite settlement by ordinary-move count are preserved under E and $\bar{U}$.*

Proof. Every ordinary legal history lifts, while every passive history projects to an ordinary legal history with the same ordinary move sequence. The passive construction retains ordinary terminal and settlement conditions on tagged copies. Therefore existence of a finite target-reaching history, together with its ordinary-move length, is preserved in both directions. □

The zero cost assigned to ε is essential to the isometry statement. If every administrative move were charged unit cost, the enlarged metric would be a different metric. This is another instance of the paper's general rule that geometry depends on both the legal road system and the declared cost model.

11.6 Active board actions, scheduling, and verifier sovereignty

The passive theorem identifies the conservative baseline. Active GBG$^+$ packages may intentionally change operational dynamics by adding legal scheduler, verifier, memory, handoff, federation, or mutation actions. Such actions can alter discovery chronology, route multiplicity, and charged search cost. They do not acquire epistemic authority merely because the board-player selected them.

Theorem 11.10 (GBG$^+$ scheduler/verifier separation)**.** *Suppose an enlarged GBG$^+$ proof-search system satisfies all of the following:*

1. *scheduling chooses only currently enabled participants;*
2. *every proof-producing action is a declared legal transition;*
3. *trusted certification requires acceptance by the declared verifier;*
4. *board memory and handoff preserve the provenance required by the verifier.*

Then changing the legal scheduling policy can change which legal candidate is attempted first and can change discovery cost, but it cannot by scheduling alone make an invalid certificate valid or an illegal proof transition legal.

Proof. By scheduler legality, scheduling selects only among enabled participants and does not create new legal proof actions. By the certification hypothesis, trusted status depends on verifier acceptance rather than scheduler choice. Provenance-preserving memory and handoff do not alter the certified semantic content without a separately declared transformation. Therefore scheduling can change order and cost but not the verifier's acceptance relation or the declared transition legality. □

This is the GBG$^+$ specialization of the policy/validity separation already used in the strengthened Exact Addressing Theorem. The operational advantage of the board-player, if any, must therefore be measured in navigation, reuse, communication, or settlement cost rather than treated as a change in theoremhood by fiat.

11.7 Consequences for Exact Addressing

The GBG/GBG$^+$ integration changes the operational vocabulary of this manuscript in a useful but controlled way.

1. A WFF address answers exact symbolic identity. A rule-complete state address answers exact operational identity. They are different typed objects.
2. A move-labeled match-proof is an exact operational history. State addresses plus canonical move labels reconstruct that history losslessly.
3. Endpoint identity does not determine history identity. Many exact histories may reach the same exact state.
4. Route multiplicity can therefore be measured on exact addressed histories before any quotient by an "essential-route" equivalence.
5. A passive board-player adds administrative structure without changing the projected ordinary proof geometry under the declared zero-cost administrative metric.
6. An active board-player can change the road system only through explicitly declared legal transitions. When it does, the resulting geometry is a new operational geometry and must be analyzed as such.
7. Exact state addressing does not itself supply a useful navigation geometry. The paper's existing distinction between identity, navigation, locality, and logic remains in force.

Thus GBG$^+$ does not replace Exact Addressing. It supplies a formally rule-complete operational substrate on which exact state addresses, exact histories, route multiplicity, scheduling, certification, and settlement can be typed without ambiguity. The map remains separate from the road system; GBG$^+$ makes the road system and its administrative extensions explicit.

Part III

Transport, Theorem Consequences, and Conjecturing

12 Satisfiability preservation under exact WFF-address transport

The exact-address framework permits a second, logically distinct connection between WFF addressing and 3-SAT. This connection does not assert that numerically nearby rational or Hilbert coordinates have similar satisfiability status. Instead, it transports a fixed syntax-level symmetry through the exact address bijection and proves that satisfiability is preserved by the resulting address-space transformation.

Fix an effectively enumerable propositional WFF class containing 3CNF, and let

$$\lambda : \mathrm{WFF} \longrightarrow \mathcal{A}$$

be a fixed exact computable bijection with computable inverse. Exact identity and navigation geometry remain separate: the point $\lambda(F)$ identifies F, while additional relations on addresses require independent justification [2].

Let

$$X = \{x_1, x_2, \ldots\}, \qquad Y = \{y_1, y_2, \ldots\}$$

be disjoint variable namespaces. Define the computable involutive variable renaming τ by

$$\tau(x_i) = y_i, \qquad \tau(y_i) = x_i,$$

extended homomorphically to literals, clauses, and 3CNF formulas. Thus τ^2 is the identity and τ changes only variable names, never polarity, connective structure, or clause structure.

12.1 Satisfiability-Preservation Lemma

For every 3CNF formula F,

$$\boxed{F \text{ is SAT} \iff \tau(F) \text{ is SAT}}$$

and therefore

$$\boxed{F \text{ is UNSAT} \iff \tau(F) \text{ is UNSAT.}}$$

Proof. Suppose F is satisfiable and let v be a satisfying assignment for the variables occurring in F. Define the renamed assignment v^τ on the variables of $\tau(F)$ by

$$v^\tau(\tau(z)) = v(z)$$

for every variable z occurring in F. Each literal of F and its renamed literal in $\tau(F)$ receive the same truth value under v and v^τ, respectively. Hence every clause satisfied by v is carried to a clause satisfied by v^τ, so $\tau(F)$ is satisfiable. Applying the same argument to the involution τ gives the converse. The UNSAT equivalence is the logical complement of the SAT equivalence. □

The proof is certificate-preserving as well: a satisfying assignment is carried to a satisfying assignment by variable renaming, while any verifier-checkable refutation can be renamed step-by-step whenever the fixed proof system is invariant under uniform variable renaming.

12.2 Address-space transport

Transport τ through the exact WFF address map by defining

$$\boxed{T_\lambda = \lambda \circ \tau \circ \lambda^{-1}}$$

on the addressed 3CNF subspace. For an address $a = \lambda(F)$, define the transported one-neighbor relation

$$\boxed{N_\lambda(a) = \{T_\lambda(a)\} = \{\lambda(\tau(F))\}.}$$

The computation

$$\lambda(F) \xrightarrow{\lambda^{-1}} F \xrightarrow{\tau} \tau(F) \xrightarrow{\lambda} \lambda(\tau(F))$$

uses only exact decoding, a fixed syntactic renaming, and exact re-encoding. It neither queries nor encodes the SAT/UNSAT label.

Let

$$Y(F) = \mathbf{1}_{\{F \text{ is SAT}\}}.$$

By the Satisfiability-Preservation Lemma,

$$\boxed{Y(\tau(F)) = Y(F)}$$

for every 3CNF formula F. Consequently, whenever G is the transported address-neighbor of F,

Equivalently, the unique transported neighbor G = tau(F) satisfies Y(G) = Y(F) identically.

12.3 Symmetry-Transported WFF-Address Locality Theorem

Let $\mathcal{F} \subseteq$ 3CNF be any infinite family closed under τ, and let μ be any frozen probability distribution on $\mathcal{F}$ for which

$$p = \Pr_{F \sim \mu}(F \text{ is SAT})$$

satisfies $0 < p < 1$. Draw the baseline neighbor G_{rand} independently from μ. Then the transported address-neighborhood has SAT-label agreement probability 1, whereas the independent random-neighbor baseline has agreement probability

$$\Pr\big(Y(G_{\text{rand}}) = Y(F)\big) = p^2 + (1-p)^2.$$

Therefore its SAT-locality surplus is

$$\boxed{\Lambda_{\text{SAT}} = 1 - \big(p^2 + (1-p)^2\big) = 2p(1-p) > 0.}$$

Thus exact WFF addresses can support a computable, satisfiability-preserving neighborhood relation with strictly positive SAT-relevant locality relative to the frozen independent-random baseline.

Proof. The Satisfiability-Preservation Lemma gives $Y(\tau(F)) = Y(F)$ for every F, so transported-neighbor agreement is 1. For independent $F, G_{\text{rand}} \sim \mu$, agreement occurs when both are SAT or both are UNSAT, giving $p^2 + (1-p)^2$. Since $0 < p < 1$,

$$1 - \left(p^2 + (1-p)^2\right) = 2p(1-p) > 0.$$

All address operations are computable because λ, λ^{-1}, and τ are computable. □

An explicit mixed unbounded class is obtained by taking any infinite collection $\mathcal{G}$ of 3CNF formulas over the X namespace that contains at least one satisfiable and one unsatisfiable formula and setting

$$\mathcal{F} = \mathcal{G} \cup \tau(\mathcal{G}).$$

For example, a satisfiable member may be any 3CNF whose clauses all contain the positive literal x_1, while an unsatisfiable member is the conjunction of all eight sign patterns on three variables x_1, x_2, x_3. Applying τ supplies disjoint renamed copies without changing satisfiability.

12.4 Interpretation and limitation

The theorem proves

3CNF syntax $\longrightarrow$ exact WFF addresses $\longrightarrow$ satisfiability-preserving address relation $\longrightarrow$ positive SAT-label locality

What is preserved is exact SAT/UNSAT status under the transported symmetry. What is *not* proved is a metric statement such as

$$\|\lambda(F) - \lambda(G)\| \text{ small} \Longrightarrow Y(F) = Y(G) \text{ with elevated probability.}$$

The bare coordinates remain identity labels unless a separate geometry-to-semantics bridge is proved. Nor does the theorem decide whether F is SAT: it says that F and its transported neighbor have the same unknown status. No polynomial-time SAT algorithm or general SAT speedup follows from this preservation result alone.

13 Consequence Mining and Interesting Theorems

13.1 Protocol: theorem DAG, normalized retained catalogue, and parent gate

The manuscript treats its proved theorem-level results as vertices of a frozen directed acyclic theorem graph. Source theorems are root vertices. A derived theorem T may be retained together with an explicit theorem-level parent set P_T sufficient for the displayed proof of T.

Literal logical consequence closure is not a usable mining object: from any theorem one can generate infinitely many arbitrary weakenings, duplicated conjunctions, equivalent rephrasings, and irrelevant disjunctions. Preliminary Release 1.6 used the phrase "normalized consequence closure" for a finite mined list. The Release 1.7 audit repairs that language: the object below is a *selected normalized consequence catalogue*, not an exhaustive closure. The frozen admissibility rules for retained derivations are:

1. Decompose a source theorem only when the manuscript itself states independently reusable clauses or named subresults.

2. Permit specialization to the certified F_k family, composition of proved implications or equalities, exact transport through a proved bijection or involution, elementary algebra on explicit formulas and bounds, limit-taking for already explicit sequences, and conjunction of at most three parent results when the conjunction yields a new semantic statement.

3. Identify logical equivalents and suppress mere syntactic renamings, arbitrary weakenings, duplicated conjunctions, tautological disjunctions, and numerical bounds weaker than a retained bound.

4. For each retained theorem, record one explicit locally irredundant parent set P_T: every listed parent is actually used in the displayed derivation. Define the *retained derivation fan-in*

$$d^-_{\mathrm{ret}}(T) := |P_T|.$$

   This witnesses the upper bound $d^-_{\min}(T) \le d^-_{\mathrm{ret}}(T)$ whenever a global minimum is meaningful, but no exhaustive search over all admissible derivations is claimed here.

5. A non-root derived theorem is called *interesting in the retained catalogue* when

$$\boxed{1 \le d^-_{\mathrm{ret}}(T) \le 3.}$$

The operation of generating and retaining such low-parent derivations is called *Consequence Mining*. The present catalogue uses a parent gate of $g = 3$. The catalogue is finite because retention is explicit and finite, not because all normalized consequences have been exhaustively enumerated.

13.2 Preliminary Release 1.10 retained shell convention

For the local consequence catalogue, freeze the root theorem vertices listed below at shell 0. If a retained consequence T has displayed retained parent set P_T, define

$$\boxed{s(T) = 1 + \max_{A \in P_T} s(A).}$$

The shell coordinate is relative to this frozen retained parent ledger. It is not a claim about logical strength, historical novelty, minimum proof length, or globally minimum fan-in. Shells +1 through +6 are eligible to be called *interesting shells* only after trivial restatements and weakenings have been excluded. The label *exciting shell* is reserved for a retained outer-shell result (+5 or +6) that also supplies substantive cross-branch synthesis, nonredundancy, or a new semantic separation. Every promoted theorem below carries its shell number and its nearest retained ancestor/descendant vertices.

For the local CM graph the resulting shell profile is

+1	CM-1, 2, 3, 4, 9, 10, 11, 13, 15, 16, 17, 18
+2	CM-5, 6, 12, 14, 19, 20
+3	CM-7, 8, 22, 23, 24
+4	CM-21.

Thus the local retained catalogue reaches shell +4; its +1–+4 consequences are interesting-shell results. Preliminary Release 1.12 keeps this section internal to the present manuscript and removes the formerly imported cross-paper six-shell chain. The terms "interesting shell" and "exciting shell" therefore describe only the retained local convention here; no outer +5 or +6 result is claimed in the present catalogue.

13.3 Frozen root theorem vertices

The following root identifiers are used only to make the parent ledger compact.

ID	Root theorem-level content
EA	Strengthened Exact Addressing Theorem: faithful invertible recoding preserves derivability, proof paths, exact shortest-proof length, exact breadth-first layers, and the labeled operational transition system.
NSA	No-speedup-by-addressing corollary: invertible exact recoding alone cannot shorten shortest proof length, exact proof layer, or unit-cost target distance.
ODB	Operational DAG bridge: append-only exact derivation-state transitions are acyclic, reachable exact sequence states have unique root depth, and concrete proof paths upper-bound theorem-graph distance.
LA	Locality Advantage identities under their stated conditional-uniformity model.
CNT	Certified candidate-count identity for the family: $N(F_k) = 4 + 6k$.
MSC	Multi-Step Structural Concentration Theorem and residual-cover construction: $G(F_k) = \{(x, 1), (y, 1)\}$, $g = 2$, the uniquely tie-broken right action lies in $G(F_k)$, and $\Lambda \geq 7/8$ for distributions supported on $k \geq 2$.
OH	Lazy indexed-access overhead bound: under the frozen primitive calibration, $C_H(k) \leq 6$.
LHS	Release 1.7 Lazy Hilbert Exact-Address Speedup Theorem: without a within-block probability assumption, $R_H \leq 2$ and the Hilbert treatment has expected charged advantage at least $3k - 11/2 > 0$ for every $k \geq 2$, while theoremhood, legality, shortest-proof length, target distance, and verifier authority remain unchanged.
SATP	Satisfiability-Preservation Lemma for the fixed involutive variable renaming τ.
AT	Exact WFF-address transport $T_\lambda = \lambda \circ \tau \circ \lambda^{-1}$.
STL	Symmetry-Transported WFF-Address Locality Theorem: transported-neighbor label agreement is 1 and $\Lambda_{\mathrm{SAT}} = 2p(1-p) > 0$ on the stated mixed-family regime.

13.4 Parent ledger for the retained interesting-shell theorems

Under the frozen admissibility grammar above, twenty-four nontrivial interesting vertices are retained. The shell column is computed from the displayed retained parents; the fan-in column is the fan-in of that displayed retained derivation, not a claim of global minimum fan-in.

ID	Interesting-shell theorem	Retained parents	shell	d^-_{ret}
CM-1	Exact theorem-DAG depth transport	EA, ODB	+1	2
CM-2	Pointwise locality formula $\Lambda_k = (3k + 1)/(3k + 2) \geq 7/8$	CNT, MSC	+1	2
CM-3	Deterministic Hilbert-first rank bound $R_H \leq 2$	MSC	+1	1
CM-4	Random-baseline expected rank $E[R_{\mathrm{random}}] = 3k + 5/2$	CNT, LA	+1	2
CM-5	Robust gross expected-rank advantage at least $3k + 1/2$	CM-3, CM-4	+2	2
CM-6	Constant non-shared treatment cost $R_H + C_H \leq 8$	CM-3, OH	+2	2

ID	Interesting-shell theorem	Retained parents	shell	d^-_{ret}
CM-7	Vanishing overhead fraction $C_H/(3k + 1/2) \to 0$	CM-5, OH	+3	2
CM-8	Distribution-free sufficient speedup threshold $C_H(k) < 3k + 1/2$	CM-5	+3	1
CM-9	Unbounded certified net advantage	LHS	+1	1
CM-10	Net-advantage/candidate-space liminf at least $1/2$	LHS, CNT	+1	2
CM-11	Discovery-cost/proof-distance orthogonality	LHS	+1	1
CM-12	Non-recoding source of the certified speedup	NSA, CM-11	+2	2
CM-13	Exact satisfiability-preserving address involution	SATP, AT	+1	2
CM-14	SAT label constancy on transported address orbits	CM-13	+2	1
CM-15	Maximum transported SAT-locality surplus is $1/2$	STL	+1	1
CM-16	Class-imbalance attenuation of transported SAT locality	STL	+1	1
CM-17	Positive SAT-label locality without assuming address-metric proximity	AT, STL	+1	2
CM-18	Positive SAT locality with unchanged shortest certificate length	EA, STL	+1	2
CM-19	Pointwise locality surplus tends to 1	CM-2	+2	1
CM-20	Random-baseline expected rank diverges	CM-4	+2	1
CM-21	Asymptotic localization with bounded search rank and negligible overhead	CM-19, CM-3, CM-7	+4	3
CM-22	Bounded-versus-linear discovery separation with invariant theorem-DAG geometry	CM-1, CM-20, CM-6	+3	3
CM-23	Vanishing search-fraction without proof shortening	CNT, CM-6, CM-11	+3	3
CM-24	Balanced transported-orbit maximal locality with certificate invariance	CM-14, CM-15, CM-18	+3	3

13.5 Fan-in one and two consequence catalogue

[interesting shell] Theorem 1 (CM-1: Exact theorem-DAG depth transport (shell +1))**.** *Under EA and ODB, exact recoding preserves the rooted theorem-state DAG up to labeled graph isomorphism. In particular, acyclicity, reachable root depth, and every concrete proof-path upper bound on theorem-graph distance are invariant under the encoding.*

Shell genealogy. **interesting descendant**; retained consequence shell +1. Nearest retained ancestor theorem vertices: EA, ODB. Immediate retained descendant vertices: CM-22 (Theorem 22). Retained fan-in: 2.

[interesting shell] Theorem 2 (CM-2: Pointwise locality formula (shell +1))**.** *For every $k \geq 2$,*

$$\Lambda_k = 1 - \frac{2}{4 + 6k} = \frac{3k + 1}{3k + 2} \geq \frac{7}{8}.$$

Shell genealogy. **interesting descendant**; retained consequence shell +1. Nearest retained ancestor theorem vertices: CNT, MSC. Immediate retained descendant vertices: CM-19 (Theorem 19). Retained fan-in: 2.

[interesting shell] Theorem 3 (CM-3: Deterministic Hilbert-first rank bound (shell +1))**.** *For every $k \geq 2$, the uniquely tie-broken right action belongs to the certified two-action block searched first. Hence*

$$R_H \leq 2$$

without any probability assumption on its within-block position.

Shell genealogy. **interesting descendant**; retained consequence shell +1. Nearest retained ancestor theorem vertices: MSC. Immediate retained descendant vertices: CM-5 (Theorem 5), CM-6 (Theorem 6), CM-21 (Theorem 21). Retained fan-in: 1.

[interesting shell] Theorem 4 (CM-4: Random-baseline expected rank (shell +1))**.** *For every $k \geq 2$, a fixed target placed in a uniformly random no-repeat ordering has*

$$E[R_{\text{random}}] = \frac{N(F_k) + 1}{2} = 3k + \frac{5}{2}.$$

Shell genealogy. **interesting descendant**; retained consequence shell +1. Nearest retained ancestor theorem vertices: CNT, LA. Immediate retained descendant vertices: CM-5 (Theorem 5), CM-20 (Theorem 20). Retained fan-in: 2.

[interesting shell] Theorem 5 (CM-5: Robust gross expected-rank advantage (shell +2))**.** *Before treatment-specific overhead,*

$$E[R_{\text{random}}] - R_H \geq 3k + \frac{1}{2}.$$

Shell genealogy. **interesting synthesis descendant**; retained consequence shell +2. Nearest retained ancestor theorem vertices: CM-3 (Theorem 3), CM-4 (Theorem 4). Immediate retained descendant vertices: CM-7 (Theorem 7), CM-8 (Theorem 8). Retained fan-in: 2.

[interesting shell] Theorem 6 (CM-6: Constant non-shared treatment cost (shell +2))**.** *Under the frozen candidate-inspection-equivalent calibration,*

$$R_H + C_H(k) \leq 2 + 6 = 8$$

for every $k \geq 2$.

Shell genealogy. **interesting synthesis descendant**; retained consequence shell +2. Nearest retained ancestor theorem vertices: CM-3 (Theorem 3), OH. Immediate retained descendant vertices: CM-22 (Theorem 22), CM-23 (Theorem 23). Retained fan-in: 2.

[interesting shell] Theorem 7 (CM-7: Vanishing overhead fraction (shell +3))**.** *Relative to the robust gross expected-rank lower bound,*

$$0 \leq \frac{C_H(k)}{3k + 1/2} \leq \frac{6}{3k + 1/2} \longrightarrow 0.$$

Shell genealogy. **interesting synthesis descendant**; retained consequence shell +3. Nearest retained ancestor theorem vertices: CM-5 (Theorem 5), OH. Immediate retained descendant vertices: CM-21 (Theorem 21). Retained fan-in: 2.

[interesting shell] Theorem 8 (CM-8: Distribution-free sufficient speedup threshold (shell +3))**.** *Keeping the same F_k block and candidate-inspection-equivalent calibration, with only fixed common state-level setup costs canceled, but allowing arbitrary nonnegative treatment overhead $C_H(k)$, strict expected net speedup is guaranteed whenever*

$$C_H(k) < 3k + \frac{1}{2}.$$

This is a sufficient threshold from the worst-case within-block rank bound; it is not asserted to be necessary.

Shell genealogy. **interesting descendant**; retained consequence shell +3. Nearest retained ancestor theorem vertices: CM-5 (Theorem 5). Immediate retained descendant vertices: none in the frozen local catalogue. Retained fan-in: 1.

[interesting shell] Theorem 9 (CM-9: Unbounded certified net advantage (shell +1))**.** *The certified lower bound satisfies*

$$E[C_{\text{baseline}}] - E[C_{\text{Hilbert}}] \geq 3k - \frac{11}{2} \longrightarrow \infty.$$

Shell genealogy. **interesting descendant**; retained consequence shell +1. Nearest retained ancestor theorem vertices: LHS. Immediate retained descendant vertices: none in the frozen local catalogue. Retained fan-in: 1.

[interesting shell] Theorem 10 (CM-10: Asymptotic half-space net-saving lower bound (shell +1))**.** *Relative to legal candidate-space size,*

$$\frac{E[C_{\text{baseline}}] - E[C_{\text{Hilbert}}]}{N(F_k)} \geq \frac{3k - 11/2}{6k + 4}.$$

Since the right-hand side tends to $1/2$*,*

$$\liminf_{k\to\infty} \frac{E[C_{\text{baseline}}] - E[C_{\text{Hilbert}}]}{N(F_k)} \geq \frac{1}{2}.$$

This is a lower-bound statement in the frozen candidate-equivalent cost model, not a claim that the normalized advantage itself has a limit or an arbitrary wall-clock claim.

Shell genealogy. **interesting descendant**; retained consequence shell +1. Nearest retained ancestor theorem vertices: LHS, CNT. Immediate retained descendant vertices: none in the frozen local catalogue. Retained fan-in: 2.

[interesting shell] Theorem 11 (CM-11: Discovery-cost/proof-distance orthogonality (shell +1))**.** *There exists an infinite explicit 3CNF family on which expected total charged discovery cost is strictly reduced by the certified Hilbert treatment while theoremhood, inference legality, shortest-proof length, unit-cost target distance, and verifier authority remain unchanged.*

Shell genealogy. **interesting semantic-separation descendant**; retained consequence shell +1. Nearest retained ancestor theorem vertices: LHS. Immediate retained descendant vertices: CM-12 (Theorem 12), CM-23 (Theorem 23). Retained fan-in: 1.

[interesting shell] Theorem 12 (CM-12: Non-recoding source of the certified speedup (shell +2))**.** *In the frozen F_k experiment, the strict expected charged advantage cannot be attributed to invertible exact identity recoding alone. It arises from additional locality, scheduling, and lazy indexed-access structure.*

Shell genealogy. **interesting synthesis descendant**; retained consequence shell +2. Nearest retained ancestor theorem vertices: NSA, CM-11 (Theorem 11). Immediate retained descendant vertices: none in the frozen local catalogue. Retained fan-in: 2.

[interesting shell] Theorem 13 (CM-13: Exact satisfiability-preserving address involution (shell +1))**.** *On the addressed 3CNF subspace,*

$$T_\lambda^2 = \mathrm{id} \qquad \textit{and} \qquad Y(T_\lambda(a)) = Y(a),$$

where Y is the SAT indicator.

Shell genealogy. **interesting descendant**; retained consequence shell +1. Nearest retained ancestor theorem vertices: SATP, AT. Immediate retained descendant vertices: CM-14 (Theorem 14). Retained fan-in: 2.

[interesting shell] Theorem 14 (CM-14: SAT-label constancy on transported address orbits (shell +2))**.** *Every orbit of T_λ has size at most two, and the SAT/UNSAT label is constant on each orbit.*

Shell genealogy. **interesting descendant**; retained consequence shell +2. Nearest retained ancestor theorem vertices: CM-13 (Theorem 13). Immediate retained descendant vertices: CM-24 (Theorem 24). Retained fan-in: 1.

[interesting shell] Theorem 15 (CM-15: Maximum transported SAT-locality surplus (shell +1))**.** *Under the hypotheses of STL,*

$$0 < \Lambda_{\mathrm{SAT}} = 2p(1-p) \le \frac{1}{2},$$

with equality exactly at $p = 1/2$.

Shell genealogy. **interesting descendant**; retained consequence shell +1. Nearest retained ancestor theorem vertices: STL. Immediate retained descendant vertices: CM-24 (Theorem 24). Retained fan-in: 1.

[interesting shell] Theorem 16 (CM-16: Class-imbalance attenuation (shell +1))**.** *For the transported SAT-locality construction,*

$$\Lambda_{\mathrm{SAT}} \longrightarrow 0 \quad \textit{as} \quad p \to 0^+ \textit{ or } p \to 1^-.$$

Shell genealogy. **interesting descendant**; retained consequence shell +1. Nearest retained ancestor theorem vertices: STL. Immediate retained descendant vertices: none in the frozen local catalogue. Retained fan-in: 1.

[interesting shell] Theorem 17 (CM-17: Positive SAT-label locality without assuming address-metric proximity (shell +1))**.** *The exact-address transported-neighbor relation has strictly positive SAT-label locality in the stated mixed-family regime, and the proof requires no numerical-closeness hypothesis on the exact rational/Hilbert coordinates.*

Shell genealogy. **interesting descendant**; retained consequence shell +1. Nearest retained ancestor theorem vertices: AT, STL. Immediate retained descendant vertices: none in the frozen local catalogue. Retained fan-in: 2.

[interesting shell] Theorem 18 (CM-18: Positive SAT locality with unchanged shortest certificate length (shell +1))**.** *For any proof or refutation presentation satisfying EA and invariant under the uniform renaming τ, transported formulas have equal shortest certificate length. On every τ-closed mixed family covered by STL, the induced transported-neighbor relation nevertheless has strictly positive SAT-label locality surplus.*

Shell genealogy. **interesting cross-branch descendant**; retained consequence shell +1. Nearest retained ancestor theorem vertices: EA, STL. Immediate retained descendant vertices: CM-24 (Theorem 24). Retained fan-in: 2.

[interesting shell] Theorem 19 (CM-19: Pointwise locality approaches its maximum (shell +2))**.** *For the certified family,*

$$\Lambda_k = \frac{3k+1}{3k+2} \longrightarrow 1.$$

Shell genealogy. **interesting descendant**; retained consequence shell +2. Nearest retained ancestor theorem vertices: CM-2 (Theorem 2). Immediate retained descendant vertices: CM-21 (Theorem 21). Retained fan-in: 1.

[interesting shell] Theorem 20 (CM-20: Random-baseline rank divergence (shell +2))**.** *For the certified family,*

$$E[R_{\text{random}}] = 3k + \frac{5}{2} \longrightarrow \infty.$$

Shell genealogy. **interesting descendant**; retained consequence shell +2. Nearest retained ancestor theorem vertices: CM-4 (Theorem 4). Immediate retained descendant vertices: CM-22 (Theorem 22). Retained fan-in: 1.

13.6 Retained three-parent syntheses

The following four consequences have explicit three-parent derivations under the frozen grammar. Preliminary Release 1.7 does not claim that three is their globally minimum fan-in; establishing that stronger statement would require exhaustive derivation search or a separate minimality proof.

[interesting shell] Theorem 21 (CM-21: Asymptotic Localization Theorem (shell +4))**.** *For the certified F_k family,*

$$\Lambda_k \longrightarrow 1, \qquad R_H \leq 2, \qquad \frac{C_H(k)}{3k+1/2} \longrightarrow 0.$$

Thus, as the candidate space grows, the certified Hilbert-local region approaches maximal locality surplus, the target rank stays uniformly bounded, and the treatment overhead becomes negligible relative to the robust gross expected-rank lower bound.

Shell genealogy. **interesting outer local synthesis descendant**; retained consequence shell +4. Nearest retained ancestor theorem vertices: CM-19 (Theorem 19), CM-3 (Theorem 3), CM-7 (Theorem 7). Immediate retained descendant vertices: none in the frozen local catalogue. Retained fan-in: 3.

[interesting shell] Theorem 22 (CM-22: Invariant-Geometry Discovery Separation Theorem (shell +3))**.** *Across the infinite family F_k,*

$$E[R_{\text{random}}] = 3k + \frac{5}{2} \longrightarrow \infty, \qquad R_H + C_H(k) \leq 8,$$

For each fixed k, exact recoding preserves the rooted theorem-state DAG geometry covered by CM-1, and the baseline and Hilbert scheduling treatments do not change that underlying proof problem. Hence an unbounded operational discovery-cost separation across the family can coexist with per-instance invariance of the underlying theorem-DAG geometry.

Shell genealogy. **interesting cross-branch synthesis descendant**; retained consequence shell +3. Nearest retained ancestor theorem vertices: CM-1 (Theorem 1), CM-20 (Theorem 20), CM-6 (Theorem 6). Immediate retained descendant vertices: none in the frozen local catalogue. Retained fan-in: 3.

[interesting shell] Theorem 23 (CM-23: Vanishing Search-Fraction Without Proof Shortening (shell +3))**.** *For the certified Hilbert treatment,*

$$\frac{R_H + C_H(k)}{N(F_k)} \leq \frac{8}{6k+4} \longrightarrow 0,$$

while shortest-proof length and unit-cost target distance remain unchanged. Thus an asymptotically vanishing fraction of the candidate-space scale can suffice for discovery in the frozen model without making the intrinsic proof shorter.

Shell genealogy. **interesting cross-branch synthesis descendant**; retained consequence shell +3. Nearest retained ancestor theorem vertices: CNT, CM-6 (Theorem 6), CM-11 (Theorem 11). Immediate retained descendant vertices: none in the frozen local catalogue. Retained fan-in: 3.

[interesting shell] Theorem 24 (CM-24: Balanced Orbit Locality With Certificate Invariance (shell +3))**.** *On a balanced transported family with $p = 1/2$, the exact address transport decomposes the relevant subspace into SAT-label-constant one- and two-element orbits, the transported SAT-locality surplus attains its maximum*

$$\Lambda_{\text{SAT}} = \frac{1}{2},$$

and shortest transported certificate length remains invariant under the stated proof-transport hypotheses.

Shell genealogy. **interesting cross-branch synthesis descendant**; retained consequence shell +3. Nearest retained ancestor theorem vertices: CM-14 (Theorem 14), CM-15 (Theorem 15), CM-18 (Theorem 18). Immediate retained descendant vertices: none in the frozen local catalogue. Retained fan-in: 3.

13.7 Interpretation of the mined layer

Consequence Mining does not increase the deductive strength of the source theory. It exposes theorem-level structure already latent in the proved vertices and makes useful dependency geometry explicit. The parent gate is a discovery and organization device: a displayed parent set of size at most three certifies the existence of a low-parent derivation in the frozen presentation. It is not a claim that the theorem is globally easy, historically new, of minimum proof length, or of globally minimum fan-in under every alternate theorem packaging.

The Release 1.7 audit itself illustrates why this distinction matters. CM-19 and CM-20 are immediate one-parent limit consequences of CM-2 and CM-4 and were omitted from the earlier twenty-two-item list despite the earlier grammar permitting limit-taking. Their addition repairs the catalogue, but does not turn the revised twenty-four-item list into an exhaustive closure. Exhaustive low-fan-in enumeration would require a mechanically complete generation procedure, canonical normal form, stopping rule, and derivation-minimality check.

The most important structural lesson of the present catalogue is the separation between identity, intrinsic proof geometry, and discovery cost. CM-12, CM-22, and CM-23 show in increasingly strong forms that exact proof structure may remain invariant while a separately certified locality-and-scheduling mechanism changes discovery cost. On the address-transport side, CM-18 and CM-24 show that semantic label locality can coexist with invariant certificate length. These consequences sharpen the manuscript's central rule: exact addressing supplies identity; informative geometry and scheduling, when independently certified and fully charged, supply the operational advantage.

14 Conjecturing, Eureka, and scientific hypothesis generation as discovery spaces

14.1 Conjecturing precedes settlement

A hypothesis is not the end of the discovery problem; it is a candidate starting point for settlement. In mathematics write the current certified knowledge state as K_t, a generated conjecture as c, and the eventual settlement state as x. The discovery lifecycle is

$$\boxed{K_t \longrightarrow \text{Eureka / conjecturing} \longrightarrow c \longrightarrow \text{AMLD+0} \longrightarrow \text{BANK/verifier} \longrightarrow x.}$$

This separates two operations that are easy to conflate:

$$\textbf{Conjecturing: } K_t \longrightarrow c, \qquad \textbf{Settlement: } c \longrightarrow x.$$

AMLD+0 already supplies the downstream mathematical settlement architecture: legal search is charged, the verifier remains authoritative, and only verifier-supported results enter BANK. The missing upstream question is where a useful c comes from.

A Eureka Scout supplies one candidate structural idea η_t. It is not automatically a theorem or even a well-formed conjecture. A compilation step must turn it into an admissible candidate:

$$K_t \xrightarrow{\text{Eureka}} \eta_t \xrightarrow{\text{Formalize}} c_t.$$

The central epistemic rule is

$$\boxed{\text{Eureka proposal} \neq \text{certified knowledge.}}$$

DATA MIND or another locality-aware ranking layer may prioritize among admissible hints or conjectures, but ranking is not certification. For a candidate hint η and public discovery state R_t, one may estimate

$$q(\eta \mid R_t) = \Pr(\eta \text{ is a useful next Eureka hint} \mid R_t),$$

and for a candidate region G use a locality surplus

$$\Lambda(G \mid R_t) = \Pr(\eta^\star \in G \mid R_t) - \frac{|G|}{N}.$$

These quantities organize exploration; they do not make a conjecture true.

14.2 Earlier theorem-space conjectures as generated hypotheses

An earlier theorem-space program in this project explicitly treated theorem proving as navigation through a graph of formulas and legal inference relations, with shortest proofs interpreted as geodesic or near-geodesic paths and learned models used as navigation aids [8]. It proposed four conjectures that can now be read as concrete mathematical hypotheses generated from the theorem-space viewpoint:

c_1. **Theorem-Space Geometry Conjecture.** Large theorem spaces possess measurable geometric structure that may be learned from proof corpora.

c_2. **Geodesic Proof Conjecture.** Shortest proofs correspond to geodesics under an appropriate theorem-space metric.

c_3. **ATP Specialization Conjecture.** Distinct automated theorem-proving systems specialize in different theorem-space regions.

c_4. **Universal Navigation Conjecture.** A sufficiently trained navigator can generalize across multiple formal systems and theorem libraries.

The present manuscript sharpens the geometry underlying these hypotheses. In particular, intrinsic proof distance is separated from learned or feature geometry. Under the path metric used here, a shortest proof is geodesic by definition, so the nontrivial form of c_2 is no longer "are shortest paths geodesics?" but whether an independently defined coordinate, learned representation, curvature profile, or other observable geometry predicts, bounds, or recovers intrinsic proof geodesics. Likewise c_4 becomes a transfer question: what theorem-space structure is sufficiently representation-stable to support navigation advantage across formal systems without claiming a universal decision procedure?

14.3 Worked conjecturing example: the Geodesic Curvature-Profile Conjecture

The curvature construction in Section 10.8 immediately generated a new hypothesis. For a legal path P, let $B_-(P)$ be the cumulative negative-curvature burden from Equation (1). The strong candidate conjecture was

$$\boxed{P^\star \text{ geodesic from } s \text{ to } t \quad \Longrightarrow \quad B_-(P^\star) \le B_-(P)} \tag{2}$$

for every legal s–t path P.

Exactly one Eureka hint was recorded before AMLD+0 acted on the conjecture:

> Search first for a "curvature detour gadget": make the short $s \to t$ route pass through locally expanding edges, while a longer route is embedded in overlapping or flatter neighborhoods that reduce negative curvature. Then shrink the gadget until no smaller counterexample survives.

The hint entered only as discovery provenance. It was not placed in BANK until independently checked consequences were available.

Theorem 14.1 (Geodesic curvature-burden counterexample)**.** *The strong conjecture in Equation* (2) *is false. There exists a six-vertex directed theorem graph with a unique two-edge geodesic P_g and a three-edge legal detour P_ℓ such that*

$$\boxed{B_-(P_g) = \frac{1}{3} > 0 = B_-(P_\ell).}$$

Proof. Take vertices

$$V = \{s, a, t, u, v, \ell\}$$

and underlying undirected edges

$$\{s,a\}, \{a,t\}, \{t,v\}, \{v,u\}, \{u,s\}, \{a,\ell\}.$$

Thus the underlying graph is the five-cycle $s - a - t - v - u - s$ with one leaf attached at a. Direct the legal proof edges as

$$s \to a \to t, \qquad s \to u \to v \to t, \qquad a \to \ell.$$

Hence

$$P_g = (s, a, t), \quad |P_g| = 2, \qquad P_\ell = (s, u, v, t), \quad |P_\ell| = 3,$$

and P_g is the unique two-edge s–t path.

For edge $s - a$,

$$m_s = \frac{1}{2}(\delta_a + \delta_u), \qquad m_a = \frac{1}{3}(\delta_s + \delta_t + \delta_\ell).$$

An optimal transport has cost $7/6$, so

$$W_1(m_s, m_a) = \frac{7}{6}, \qquad \kappa(s, a) = -\frac{1}{6}.$$

By reflection symmetry, $\kappa(a, t) = -1/6$. Therefore

$$B_-(P_g) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}.$$

For each detour edge (s, u), (u, v), and (v, t), the neighborhood measures admit transport cost exactly 1, hence $\kappa = 0$ on all three edges and

$$B_-(P_\ell) = 0.$$

Thus the shorter legal path has strictly larger negative-curvature burden. □

The six-vertex graph is the smallest counterexample found in the settlement run; global minimality is not claimed without an independently replayable exhaustive certificate.

Theorem 14.2 (Leaf-amplified failure family)**.** *Replace the single leaf at a by $r \geq 2$ leaves while keeping the same five-cycle and directed s–t routes. Then*

$$\kappa(s,a) = \kappa(a,t) = -\frac{r-1}{r+2},$$

so

$$\boxed{B_-(P_g) = \frac{2(r-1)}{r+2}, \qquad B_-(P_\ell) = 0,}$$

and $B_-(P_g) \to 2$ as $r \to \infty$.

Proof. At s, $m_s = \frac{1}{2}(\delta_a + \delta_u)$, while at a the measure m_a assigns mass $1/(r+2)$ to each of s, t and the r leaves. Every destination in the support of m_a is at undirected distance 1 from a, so transporting the mass $1/2$ at a costs at least $1/2$. For the mass $1/2$ at u, the cheapest available destination capacities have distances 1 to s, 2 to t, and 3 to every leaf. Because $r \geq 2$, the s- and t-demands total $2/(r+2) \leq 1/2$; hence any coupling costs at least

$$\frac{1}{2} + \frac{1}{r+2} + \frac{2}{r+2} + 3\left(\frac{1}{2} - \frac{2}{r+2}\right) = \frac{2r+1}{r+2}.$$

This lower bound is attained by sending the u-mass first to s and t up to their required masses, sending its remaining mass to leaves, and using the a-mass to fill the remaining leaf demand. Therefore

$$W_1(m_s, m_a) = \frac{2r+1}{r+2}, \qquad \kappa(s,a) = 1 - W_1(m_s, m_a) = -\frac{r-1}{r+2}.$$

Reflection symmetry gives the same value on (a,t). The detour edges retain curvature 0, exactly as in Theorem 14.1, so the displayed burden formulas follow. □

The family identifies the precise obstruction:

off-path neighborhood structure can change curvature without changing proof distance.

Adding leaves changes the neighborhood measure at a, hence the Wasserstein transport and edge curvature, without changing either legal s–t path or its length. Local curvature therefore cannot universally characterize global geodesicity.

Theorem 14.3 (Universal negative-curvature burden bound)**.** *For every edge in an unweighted graph,*

$$-2 \leq \kappa(x,y) \leq 1.$$

Consequently every n-edge legal path satisfies

$$\boxed{0 \leq B_-(P) \leq 2n.}$$

In particular, for a geodesic $P^\star$,

$$B_-(P^\star) \leq 2d_{\text{legal}}(s,t).$$

Proof. For adjacent x, y, every $z \in N(x)$ and $w \in N(y)$ are joined by the walk $z - x - y - w$, so $d(z,w) \leq 3$. Hence $0 \leq W_1(m_x, m_y) \leq 3$ and therefore $-2 \leq \kappa(x,y) \leq 1$. Each summand in B_- is at most 2, and summing over the path gives the result. □

The failed conjecture also shows what kind of bridge is missing. Define

$$b(x, y) = \max\{0, -\kappa(x, y)\}.$$

Theorem 14.4 (Curvature–Potential Geodesic Bridge)**.** *Fix s, t. Suppose there exist $\alpha > 0$ and a potential $\phi : V \to \mathbb{R}$ such that every legal directed edge $x \to y$ satisfies*

$$\boxed{b(x, y) \geq \alpha + \phi(x) - \phi(y),}$$

and equality holds on every edge of a geodesic $P^\star$ from s to t. Then

$$\boxed{B_-(P^\star) \leq B_-(P)}$$

for every legal s–t path P.

Proof. For any legal path $P = (x_0 = s, \ldots, x_n = t)$, summing the edge inequality telescopes the potential terms:

$$B_-(P) \geq \alpha|P| + \phi(s) - \phi(t).$$

Since $|P| \geq d_{\text{legal}}(s, t)$,

$$B_-(P) \geq \alpha d_{\text{legal}}(s, t) + \phi(s) - \phi(t).$$

Because $P^\star$ is a legal directed geodesic, $|P^\star| = d_{\text{legal}}(s, t)$. Equality on every edge of $P^\star$ therefore gives equality with the right-hand side, proving the claim. □

This is a sufficient bridge, not a claim of logical minimality. The broader question survives in a weaker and experimentally meaningful form:

Do curvature profiles statistically predict useful or short proof routes?

The settlement does not answer that question. It does, however, provide a complete example of the Section 14 lifecycle: an existing geometric framework generated a conjecture; one Eureka hint suggested a structural attack; AMLD+0 refuted the strong hypothesis; BANK retained the counterexample, obstruction, universal bound, and repaired bridge theorem; and the failed conjecture became a sharper research question [10].

14.4 From mathematical conjectures to scientific hypotheses

The same upstream architecture extends naturally to scientific discovery, but the downstream authority changes. Let $\mathcal{H}$ be a frozen finite or effectively enumerable class of admissible scientific hypotheses. A Eureka hint η_t may be compiled into a testable $H_t \in \mathcal{H}$, but the generator is not permitted to certify its own hypothesis:

$$R_t \xrightarrow{\text{Eureka}} \eta_t \xrightarrow{\text{Formalize}} H_t \xrightarrow{\text{Predict/Test}} \text{observation} \xrightarrow{\text{Update}} R_{t+1}.$$

Each $H \in \mathcal{H}$ receives an exact reproducible identity $\text{Addr}(H)$, while a separately declared feature map

$$\Phi_{\mathcal{H}} : \mathcal{H} \to \mathcal{X}$$

places hypotheses in a navigation space whose coordinates may encode mechanisms, assumptions, predicted observables, parameter regimes, or other protocol-frozen features. Exact identity and navigation geometry remain distinct.

Let $\mathcal{E}$ be a class of admissible experiments or observations. An evidence interface

$$\mathsf{Obs} : \mathcal{E} \times \mathcal{W} \longrightarrow \mathcal{O}$$

returns an observation from a hidden world or frozen synthetic environment $\mathcal{W}$, and a predeclared update rule

$$U : (\mathcal{H}_t, e_t, o_t) \longmapsto \mathcal{H}_{t+1}$$

contracts or reweights the currently viable hypothesis set. The evidence interface plays an empirical role analogous to a trust boundary, but empirical support, falsification, likelihood change, and statistical evidence are not deductive theoremhood. The scientific analogue of the mathematical rule is therefore

$$\boxed{\text{Eureka-generated hypothesis} \neq \text{evidence.}}$$

For this paper, the hypothesis scope is broad in scientific subject matter but narrow in admissibility. A candidate hypothesis should have a finite or reproducibly computable representation, exact identity, domain interpretation, at least one declared prediction or exclusion under the permitted experiment language, independent evaluability through the evidence interface, and no access to hidden outcome information. Eureka hints may be looser; not every hint becomes an admissible scientific hypothesis.

14.5 Recovered pilot: evidence-conditioned basin escape in a fictional differential-diagnosis task

A nearly lost development record preserves an early dry run of the scientific-hypothesis architecture. The source was a remembered fictional diagnostic scenario; it is not a clinical case, a medical dataset, or evidence of autonomous diagnostic competence. The generator was not told the reference diagnosis. The clue presenter, however, knew the fictional answer and supplied evidence incrementally. The mathematically relevant object is therefore the search protocol, not the medical status of the story.

The public evidence sequence recorded in the backup included sudden unilateral mid-thigh pain of extreme severity, no radiation, a largely normal-looking leg, preserved distal pulses, no trauma or exertion, inability to bear weight, diffuse tenderness, pain with both contraction and stretching, and no swelling, firmness, tense compartment, or anticoagulant use. Ordinary differential generation initially remained inside familiar vascular, compartment, thrombotic, musculoskeletal, neurologic, and infectious explanation families. The run was then judged to have entered a search basin: additional ordinary proposals were varying the same mechanism families without explaining the residual pattern.

At that point exactly one mechanism-changing instruction was issued: generate one new mechanistic hypothesis outside the ordinary explanations and provide one finding or test that would discriminate it. The generated hypothesis was localized skeletal-muscle infarction / ischemic myonecrosis, with muscle-focused imaging proposed as a discriminator. Only afterward was the fuller fictional reference mechanism revealed, in the form

$$\text{vascular lesion} \longrightarrow \text{thrombosis/obstruction} \longrightarrow \text{ischemia} \longrightarrow \text{muscle infarction} \longrightarrow \text{necrosis}.$$

Thus the generated hypothesis matched the tissue-level pathological mechanism in the later reference but did not contain the full upstream vascular explanation.

This motivates a distinction that should be explicit in scientific hypothesis generation. For a frozen public history $\mathcal{R}_t$, let $\sigma_t(H)$ denote the prediction or likelihood signature of hypothesis H on the observations admitted by $\mathcal{R}_t$. Write

$$H_1 \equiv_t H_2 \quad \Longleftrightarrow \quad \sigma_t(H_1) = \sigma_t(H_2).$$

Separately, freeze an explanatory-refinement relation $\prec$ before evaluation. Informally, $H_1 \prec H_2$ means that H_2 retains the mechanism asserted by H_1 while adding at least one further causal or mechanistic layer. This relation is not inferred merely from which hypothesis later turns out to be favored.

Theorem 14.5 (Evidence fit does not imply explanatory completeness)**.** *Fix a public history $\mathcal{R}_t$. Suppose $H_1 \equiv_t H_2$ and $H_1 \prec H_2$. Then no evaluator whose input is restricted to the observational signature $\sigma_t(H)$ can distinguish H_1 from H_2 at time t. In particular, agreement with the evidence available at time t cannot by itself certify that H_1 is explanatorily maximal.*

Proof. Every such evaluator has the form $f_t(\sigma_t(H))$ for some rule f_t. Because $\sigma_t(H_1) = \sigma_t(H_2)$,

$$f_t(\sigma_t(H_1)) = f_t(\sigma_t(H_2)).$$

Hence the current evidence-only input cannot distinguish the two hypotheses, even though the frozen refinement relation declares H_2 strictly more explanatory. Distinguishing them requires an additional discriminating observation, experiment, or an independently declared structural preference not contained in σ_t. □

The recovered run illustrates the weaker practical principle behind the theorem:

$$\boxed{\text{fit to the currently supplied evidence } \not\Rightarrow \text{ explanatory completeness.}}$$

A correct mechanism-level hypothesis can still omit a causally upstream explanation.

Audit status of the recovered run. The run is best classified as *controller-blind but presenter-not-blind*. It is stronger than an example in which the answer is supplied to the generator, because the reference answer was withheld from the generator until after the mechanism-changing proposal. It is weaker than a fully frozen held-out benchmark, because the clue presenter knew the target, the evidence schedule was not shown to have been precommitted before the interaction, and the scenario was reconstructed from fiction and memory. The run is therefore retained as a protocol-calibration pilot and discovery provenance, not as a clinical validation or as a statistically controlled scientific result.

Part IV

Research Program and Audit

15 Open research directions: from certified corridors to general discovery

The present theorems provide a concrete positive example and a calibrated geometry, not a universal navigator. The following directions are open and require frozen assumptions and charged, verifier-safe tests.

1. **General composition outside F_k.** Identify minimal sufficient conditions under which computable exact addressing, a verifier-conservative state representation, an exogenous informative feature map,

fair scheduling, and complete charged overhead yield positive net search advantage beyond the certified F_k family. The unresolved abstract form is preserved separately in *The Central Composition Problem*.

2. **Calibrated geometry to useful locality.** Determine when a coordinate potential that lower-bounds legal proof cost also yields a proper action neighborhood with positive mass surplus. A certified metric bound does not automatically identify the next useful action.

3. **Independence of metric fidelity and action enrichment.** Construct examples separating accurate proof-distance estimates from prediction of the right next action; alternatively identify restrictive hypotheses that couple the two.

4. **Shortcut-sensitive proof geodesics.** Give sufficient conditions under which a rectilinear route substantially exceeds the intrinsic shortest legal path, and determine whether a detectable gap can support verifier-safe shortcut proposals with all mining costs included.

5. **Transfer of the residual-cover theorem.** Test the visible residual-cover invariant and alternative structural features on independently frozen 3CNF families, with a predeclared benchmark and fallback, without retrospective retuning or hidden solvers.

6. **Exchangeability and opaque search.** Prove the weakest symmetry/exchangeability conditions under which mere re-addressing cannot beat a symmetry-respecting password-style baseline. Determine which certified, verifier-observable settlement event breaks that symmetry. The broader Triangle exposition is housed in the IAE and Hypothesis-Generation companion.

7. **Settlement fibers and single-collapse interventions.** Quantify variation in branching, optimal proof length, navigation locality, and total discovery cost inside certified settlement fibers. Test whether one independently verified frontier contraction causes predeclared action enrichment rather than post hoc re-ranking.

8. **Raw-address SAT locality.** Forbid syntax relations transported from an original formula and test whether the inherited rational/Hilbert metric alone predicts SAT-relevant structure on a frozen mixed benchmark. This distinguishes intrinsic information in a representation from locality artificially introduced by a feature.

9. **Stable, verifier-checkable consequence mining.** Extend the selected finite theorem-DAG catalogue under alternative frozen normalization conventions and premise grammars. Determine whether minimal fan-in, redundancy removal, and retained shell assignments survive changes in theorem granularity. Do not claim exhaustive closure without a verified exhaustive procedure.

10. **Curvature as predictor, not as proof distance.** The strong minimum-negative-curvature-burden conjecture is refuted by Theorem 14.1. Test weaker predictive claims for bottlenecks, corridors, route multiplicity, or charged search difficulty on frozen theorem graphs.

11. **Transferable universal navigation.** Declare a finite or effectively enumerable class of formal systems, common structural features, a fixed transfer metric, and costed baselines; test whether navigation knowledge transfers to held-out systems while their verifiers remain sovereign.

12. **Reproducible AMLD+0 implementation.** Supply reader-ready installation and startup instructions, including verifier setup, Player-0 schedules, Scout/Graduate Board activation rules, common BANK support provenance, full cost accounting, and a smallest reproducible example. A protocol description is not an installed system.

13. **Held-out scientific hypothesis generation.** Extend the fictional differential-diagnosis pilot only as a safety-constrained research benchmark. Precommit the evidence schedule; keep hypothesis generation, evidence presentation, causal-discriminator design, and external adjudication separate; and prohibit autonomous clinical claims. Its detailed null model, experiment grammar, synthetic benchmark, and targets now belong to the companion PDF.

A useful next paper should select one target, freeze its formal system, its claims of novelty and baseline, and its certificate requirements, and report both proven consequences and failed candidate hypotheses. The central open challenge is not to draw ever richer maps but to show exactly when a map improves a verifier-safe route at lower total charged cost.

Release 3.0 continuity note. The following appendices preserve the source-era technical record, including the complete frozen IAE prompt and the historical 2.3 release audit. Those historical statements describe their original release and should not be read as a new 3.0 verification certificate. The original Section 11 case study and related context are published separately as the IAE/scientific-hypothesis companion.

A Frozen IAE / AMLD+0 Prompt

The following prompt is reproduced verbatim from the fixed run input.

```
3CNF HILBERT-NEIGHBORHOOD MULTI-STEP ACTION-CONCENTRATION CONJECTURE

Let (W_{\mathrm{3CNF}}) be the class of 3CNF well-formed formulas, with simplified search states allowed to lie
↪  in (W_{\leq \mathrm{3CNF}}).

Assume the following frozen structure.

H1. Exact identity

Every (F\in W_{\leq \mathrm{3CNF}}) has an exact computable syntactic identity code, and the navigation
↪  representation preserves enough information to reconstruct (F) exactly.

H2. Structural locality

The encoding contains a theoremhood-independent structural component built from computable features of the CNF,
↪  including clause structure, variable/literal incidence, clause sizes, and any other frozen syntactic
↪  invariants declared before the run.

H3. Hilbert navigation

The structural representation may be placed in a finite-dimensional rational feature space and traversed using
↪  finite Hilbert ordering.

Hilbert traversal is a navigation/scheduling device only. It does not alter satisfiability, inference legality,
↪  or verifier authority.

H4. One-step SAT moves

A legal one-step SAT transformation consists of assigning one variable, deleting clauses satisfied by that
↪  assignment, deleting the falsified literal from remaining affected clauses, and performing only declared
↪  canonical simplifications.

H5. Useful locality

``Nearby'' does not merely mean small numerical address distance.

A Hilbert neighborhood is useful only if it has operational search significance, for example by concentrating
↪  useful next actions, producing positive locality surplus, reducing transition churn, or enabling reuse of
↪  previously computed or verifier-certified state.
```

16 Novelty: Audited Computer-Assisted Conditional Consequences

Status. New Section 16 for the working corrected edition. The frozen retained catalogue CM-1–CM-24 of Section 13 is unchanged. The four results below are proved under explicitly named model assumptions, but their historical originality, globally minimal parent sets, and "exciting shell +5/+6" designations are *not* certified. No result below asserts an arbitrary-3CNF speedup or wall-clock advantage.

16.1 What is new, and what is not

The term *Novelty* is used here to organize potentially useful *computer-assisted consequences*, not to infer that a prompt, a numerical calculation, or an authored parent path establishes historical first discovery. The source result is the supplementary IAE-20 two-candidate comparison (source leaves 11b–11d). This is a different problem from the original Section 4 *recognizable single winner* comparison; the Section 4 clarification leaf is controlling for its differential-cost assumptions.

Fix an integer $k \geq 2$ and the non-forced formula family

$$F_k = \bigwedge_{i=1}^{k} (x \vee u_i \vee v_i) \ \wedge \ \bigwedge_{i=1}^{k} (\neg x \vee y \vee z_i), \qquad N = 6k + 4,$$

with $G_k = \{(x, 1), (y, 1)\}$. Section 3 shows that the lexicographic first-action minimizer lies in G_k, relative to the fixed fallback and cost evaluator. In the IAE-20 comparison both algorithms identify the minimizer by evaluating and verifying both named actions. The guided procedure inspects these two actions directly; the comparison procedure inspects a uniformly random permutation of N actions until it encounters both of them.

Frozen differential ledger. Both procedures are supplied with the same charged parser, action array, public member indices, deterministic candidate-cost evaluator, and verifier; they actually execute and pay for the matched operations. Every outsider $a \notin G_k$ inspected by the random comparator has a deterministic, history-independent word-RAM charge $c_k(a) \geq 1$. The baseline may pay additional nonnegative cost P_k. The guided procedure alone pays $h_k \leq 6$ instructions; if another unmatched operation is present it must be added to this differential ledger and no longer falls under the displayed six-unit bound. The same random comparator is used as a reference in all probability statements.

Let Q_k be the outsiders seen before the comparator's second distinguished action and let $J = |Q_k|$. Define the coupled charged-cost difference $\Delta_k = C_B - C_H$. The frozen cost identity gives

$$\Delta_k = P_k + \sum_{a \in Q_k} c_k(a) - h_k \ \geq \ J - 6. \tag{1}$$

This is a *paired* comparison of two methods on one frozen instance, not a generic solver-agnostic notion of gain.

16.2 Exact stopping-position law and a conditional high-probability result

Theorem 16.1 (Exact stopping-position distribution)**.** *For $0 \leq j \leq N - 2$, and $0 \leq t \leq N - 2$,*

$$\mathbb{P}(J = j) = \frac{j + 1}{\binom{N}{2}}, \qquad \mathbb{P}(J \leq t) = \frac{\binom{t+2}{2}}{\binom{N}{2}}, \qquad \mathbb{E}[J] = \frac{2(N - 2)}{3}.$$

Proof. The two distinguished actions occupy a uniformly chosen unordered pair of positions in $\{1, \ldots, N\}$. The later position is $R = J + 2$. When $R = j + 2$, the earlier position may be any one of $j + 1$ smaller positions; summing those counts through $t + 2$ yields $\binom{t+2}{2}$. By symmetry, for any outsider a, the chance that a lies before the later distinguished action is $2/3$, since it is last among a and the two distinguished actions with probability $1/3$. Summing outsider indicators yields $\mathbb{E}[J] = 2(N - 2)/3$. □

Theorem 16.2 (N-25: high-probability and expected charged advantage)**.** *Under the frozen differential ledger above,*

$$\mathbb{P}(\Delta_k > 0) \geq 1 - \frac{56}{N(N - 1)}, \qquad \mathbb{E}[\Delta_k] \geq 4k - \frac{14}{3} > 0. \tag{2}$$

The probability lower bound tends to 1 *as* $k \to \infty$*. The bound is for the random comparator specified above, not for an optimally designed competing algorithm.*

Proof. From (1), $J \geq 7$ implies $\Delta_k \geq 1 > 0$. By Theorem 16.1, $\mathbb{P}(J \leq 6) = \binom{8}{2}/\binom{N}{2} = 56/[N(N-1)]$. Hence the probability bound. Taking expectations in (1) and substituting $N = 6k+4$ gives $\mathbb{E}[\Delta_k] \geq 2(N-2)/3-6 = 4k-14/3$. The latter is positive for all $k \geq 2$. □

Audit distinction. $1 - 56/[N(N-1)]$ is the *exact probability of the sufficient event* $J \geq 7$, not necessarily the exact probability of $\Delta_k > 0$. Higher outsider charges, higher baseline overhead, or lower treatment overhead can make strict gain occur more often. Finite test enumerations check the combinatorial law; they do not measure actual runtime or certify priority.

16.3 Robustness to separately charged implementation overhead

Theorem 16.3 (N-26: deterministic overhead robustness)**.** *Suppose the guided algorithm alone incurs an additional* deterministic integer *cost* $e_k \geq 0$*. If* $8 + e_k \leq N$*, then*

$$\mathbb{P}(\Delta_k - e_k > 0) \geq 1 - \frac{\binom{8+e_k}{2}}{\binom{N}{2}}, \qquad \mathbb{E}[\Delta_k - e_k] \geq 4k - \frac{14}{3} - e_k. \tag{3}$$

If $e_k = o(k)$ *as* $k \to \infty$*, the first lower bound approaches* 1 *and the second diverges to* $+\infty$*. For values* $e_k > N - 8$*, the probability claim is replaced by the trivial bound* 0*; no negative number should be advertised as a useful bound.*

Proof. Equation (1) gives $\Delta_k - e_k \geq J - 6 - e_k$. For integer costs, $J \geq 7 + e_k$ is sufficient for a strict saving. Applying Theorem 16.1 with $t = 6 + e_k$ gives the tail bound under the stated range condition. The expectation follows from Theorem 16.2. Because $N = 6k + 4$, $e_k = o(k)$ implies $\binom{8+e_k}{2}/\binom{N}{2} = o(1)$ and $4k - 14/3 - e_k \to +\infty$. □

Scope. Deterministic integer overhead and the fixed comparable word-RAM ledger are material hypotheses. Additional feature-recognition, indexing, data movement, or verification work is not free merely because it is called geometric.

16.4 A conditional randomized policy hedge

Proposition 16.4 (N-27: mixture against a coupled baseline)**.** *Let a coin, independent of the comparator's random permutation, choose the guided algorithm with probability* $\alpha \in [0, 1]$ *and otherwise choose exactly that comparator. Suppose the selection itself has deterministic overhead* $d_k \geq 0$*. Under the coupling that uses the* same *comparator realization on the baseline branch,*

$$\mathbb{E}[C_B - C_{\text{mix}}] \geq \alpha \left(4k - \frac{14}{3}\right) - d_k.$$

If $d_k = 0$*, then*

$$\mathbb{P}(C_B > C_{\text{mix}}) \geq \alpha \left(1 - \frac{56}{N(N-1)}\right).$$

Proof. Write $I \in \{0, 1\}$ for the selection indicator. Under the stipulated coupling, $C_B - C_{\text{mix}} = I\Delta_k - d_k$. Independence implies $\mathbb{E}[I\Delta_k] = \alpha\mathbb{E}[\Delta_k]$. With $d_k = 0$, the event $\{I = 1, J \geq 7\}$ implies strict saving; its probability is $\alpha\mathbb{P}(J \geq 7)$. □

This finite mixture is not a proof of nonstarvation or completeness for an infinite proof-search policy. Those properties require an independently protected fair channel or a separate scheduling theorem.

16.5 Quantitative bridge to the existing locality formula

Theorem 16.5 (N-28: locality–comparison identity bridge)**.** *Let* Λ_k *be the* pointwise *locality surplus proved by CM-2 for the same family, namely*

$$\Lambda_k = 1 - \frac{1}{3k+2}, \qquad N = \frac{2}{1-\Lambda_k}.$$

Then, under the distinct IAE-20 two-candidate charged comparator,

$$\mathbb{P}(\Delta_k > 0) \geq 1 - \frac{28(1-\Lambda_k)^2}{1+\Lambda_k}, \qquad \mathbb{E}[\Delta_k] \geq \frac{4}{3(1-\Lambda_k)} - \frac{22}{3}. \tag{4}$$

Proof. Substitute $N = 2/(1-\Lambda_k)$ into (2); its probability correction becomes $56/[N(N-1)] = 28(1-\Lambda_k)^2/(1+\Lambda_k)$. Also $k = [1/(1-\Lambda_k) - 2]/3$, so $4k - 14/3 = 4/[3(1-\Lambda_k)] - 22/3$. □

This is a *common-parameter identity*, not a universal principle that a larger geometric locality surplus causes faster certification. The feature-induced Hilbert prefix and the frozen random comparator are essential ingredients.

16.6 Proof-status ledger, shell discipline, and Eureka/IAE

Item	Sufficient proof ingredients	Publication status
N-25	IAE-20 pointwise ledger and elementary random stopping law	Conditional theorem; proposed former CM-25 label, no certified +5
N-26	IAE-20 pointwise ledger, deterministic overhead and finite tail bound	Conditional theorem; proposed former CM-26 label, no certified +6
N-27	IAE-20 pointwise ledger, independent selection coin, explicit coupling	Conditional proposition; elementary, no certified +6
N-28	CM-2 plus IAE-20 law and algebraic substitution	Conditional cross-model bridge; low-parent candidate, not universal geometry-to-speedup theorem

Catalogue policy. N-25–N-28 are *provisional Section 16 identifiers*, not CM-25–CM-28 promotions inside the original retained catalogue. In particular, N-25 does not require CM-21 as a mathematical parent merely to put it at a nominal shell +5, and N-26 and N-27 need not be put behind N-25 to manufacture a shell +6. The only certified shell profile of the unchanged Section 13 catalogue remains +1 through +4. A fresh locally irredundant parent ledger could be audited later, but nothing here certifies an outer-shell depth or worldwide novelty.

Why keep Eureka and IAE? Their defensible role is an explicit *discovery protocol*: generate a structural feature or conditional assertion, challenge it with verifier-independent proof checks and counterexamples, revise the cost model, and retain only statements with auditable derivations. The historical IAE-20 record may explain *how* the two-candidate benchmark was reached, but no prompt is a proof rule, an empirical comparative trial, or a priority certificate. The earlier single-target six-unit claim remains conditional despite the later theorem. Historical details in Appendix D keep their original qualified status.

Next prepublication test. Reproduce the two explicitly specified algorithms with a shared indexed representation and itemize all unmatched runtime charges. Compare them not only against the frozen random comparator but also against a competitor using the public two-action feature. Do not infer a general SAT improvement from this family alone.

H6. Established local-update result

[
\boxed{
\text{If each variable occurs in at most }\Delta\text{ clauses, then one SAT assignment requires only }O(\Delta)
\text{ encoded-state updates rather than }\Theta(m)\text{ full rebuilding.}
}
]

H7. Established forced-action concentration result

For a nontrivial class of simplified CNF states with a unique unit clause, a theoremhood-independent structural
↪ feature can isolate the verifier-checkably right unit-propagation action inside a Hilbert-local region with
↪ positive locality surplus.

H8. Established non-forced immediate-gain concentration result

For a nontrivial class of non-unit-forced CNF states, a visible one-step clause-deletion-gain feature can place
↪ the unique action maximizing immediate certified clause reduction inside a Hilbert-local region with positive
↪ locality surplus.

Do not spend the Eureka hint merely rediscovering H6, H7, or H8.

MULTI-STEP REGIME

Restrict attention to non-forced search states (F).

At minimum:

no unit clause may force the next assignment;

no other frozen local propagation rule may already determine the next action.

To separate this conjecture from H8, prefer a class in which immediate one-step gain does not uniquely identify
↪ the retrospectively best branch.

A particularly clean regime is one in which two or more candidate actions tie on all frozen immediate
↪ simplification features used in H8.

Freeze the candidate action grammar

[
A(F)=\{(x,b): x\text{ is currently unassigned in }F,\ b\in\{0,1\}\}.
]

RETROSPECTIVE MULTI-STEP COST

Before running the experiment, freeze a certified settlement-cost functional (C_F(a)).

For each candidate action (a\in A(F)), (C_F(a)) is the total charged cost of settling the descendant state
↪ produced by taking (a) first and then running the same frozen verifier-gated fallback settlement procedure.

The charged cost may include, according to a declaration fixed before the run:

candidate expansions;

verifier calls;

encoding/update work;

memory;

lemma generation;

BANK operations;

Hilbert navigation overhead;

wall-clock-independent abstract step cost;

or another explicitly frozen reproducible cost measure.

The cost evaluator must be the same for every candidate action at the state.

The evaluator must return a total comparable value for every candidate action. This requirement may be met in either of two frozen ways:

(a) the fallback settlement procedure is proved to terminate on every descendant in the declared benchmark class; or

(b) a finite retrospective evaluation budget and a deterministic timeout/failure penalty are fixed before the run, with every unsettled branch receiving that same predeclared penalty convention.

No action may be omitted from comparison because its descendant is difficult, slow, or unsettled.

A ``retrospectively right next action'' is the unique action minimizing (C_F(a)) after application of a frozen theoremhood-independent deterministic tie-breaking rule declared before the run.

The tie-breaking rule is part of the evaluation protocol and may not use satisfiability, successful-branch information, retrospective costs beyond detecting a tie, or any equivalent solution information.

BASE-PROTOCOL FREEZE BEFORE EUREKA

Before the Eureka Scout is invoked, freeze all protocol components that define the problem independently of any proposed geometry:

the admissible state class;

the benchmark distribution;

the candidate-action grammar;

the settlement-cost functional and its total-evaluation convention;

the fallback settlement procedure;

all canonical simplifications;

the deterministic tie-breaking rule;

all retrospective evaluation rules;

and all external-information and verifier rules.

These base-protocol items may not be altered in response to the Eureka hint or to later retrospective outcomes during the same experiment.

EUREKA SCOUT

You may use exactly one Eureka Scout.

The Scout is invoked before the final pre-evaluation freeze and before any retrospective settlement cost, minimizing-action identity, successful descendant branch, or equivalent outcome information is computed.

The Scout's job is to give exactly one new, nontrivial hint that may help AMLD+0 settle the conjecture.

The Scout must not give:

the complete proof;

a proof outline;

the final settlement;

or a complete SAT algorithm.

The hint should contribute exactly one new structural idea, invariant, decomposition, feature, coordinate system, neighborhood construction, quotient, multi-step proxy, or enrichment mechanism that is not already supplied above.

```
In particular, look for a theoremhood-independent visible feature that could correlate with lower retrospective
↪  multi-step settlement cost even when immediate clause-deletion gain is tied or uninformative.

Do not merely restate:

that Hilbert curves preserve locality;

H6 local-update reuse;

H7 unit propagation;

H8 immediate clause-deletion gain;

or the instruction to benchmark the system.

Do not define a feature using the retrospective answer label.

Do not assume that geometric closeness automatically implies multi-step SAT-search relevance.

The Scout may challenge any working hypothesis except an explicitly stated established result or a base-protocol
↪  item already frozen before Scout invocation.

Record the Scout's hint exactly as given before AMLD+0 acts on it.

After the hint is recorded, AMLD+0 may use only pre-branch, theoremhood-independent information to instantiate,
↪  refine, or reject the hinted feature, coordinate system, neighborhood construction, invariant, quotient,
↪  proxy, or enrichment mechanism. This construction stage must occur before any retrospective outcome
↪  information is computed.

The resulting feature map, rational/Hilbert representation, and neighborhood family must then be included in the
↪  final pre-evaluation freeze. After that freeze they cannot be changed during the current experiment.

The hint is advisory only.

It is not certified knowledge merely because the Scout proposed it.

Any theorem-producing use of the hint must independently pass the verifier.

The Scout may be invoked only once. AMLD+0's later iterations may reason about, test, reject, refine, or abandon
↪  that single hint, but they may not obtain additional Eureka hints.

FINAL PRE-EVALUATION FREEZE

After the single Eureka hint has been recorded and any theoremhood-independent pre-branch construction inspired
↪  by it has been instantiated, but before any retrospective settlement cost (C_F(a)), minimizing-action
↪  identity, successful descendant branch, or equivalent outcome information is computed, freeze all remaining
↪  experiment components, including:

the structural feature map;

the rational/Hilbert representation;

the finite Hilbert ordering and all deterministic collision/tie handling in that ordering;

the Hilbert-local neighborhood family;

all preprocessing allowed to influence online geometry or guidance;

and all other components whose later selection could depend on retrospective outcome information.

No frozen component may subsequently be selected, modified, tuned, filtered, or discarded because of observed
↪  retrospective (C_F(a)) values, minimizing-action identities, successful branches, or equivalent labels.

Any such modification begins a new independently declared experiment and requires a new base-protocol freeze, a
↪  new final pre-evaluation freeze, and no reuse of retrospective labels as online guidance.

A state class may not be retrospectively chosen because the desired enrichment was already observed inside it.

CRITICAL NON-LEAKAGE RULE
```

The values (C_F(a)), the identity of the minimizing action, successful future branches, and any equivalent outcome label are evaluation data only.

They may be computed retrospectively for certification of the benchmark result, but they may not be supplied to:

the encoding;

the feature map;

the Hilbert coordinates;

the neighborhood definition;

the online controller;

the Eureka Scout;

or AMLD+0's online search guidance.

The online geometry must be computed entirely from information available before the candidate branch is settled.

HILBERT-LOCAL NEIGHBORHOODS

For each state (F), the frozen representation induces a finite deterministic Hilbert ordering of the candidate actions in (A(F)). If multiple actions map to the same Hilbert cell or key, their order must be resolved by a frozen theoremhood-independent deterministic rule, for example exact syntactic identity.

A Hilbert-local region (G(F)\subseteq A(F)) is a nonempty proper contiguous block in that frozen finite Hilbert ordering, selected by a frozen theoremhood-independent rule using only information available before any candidate branch is retrospectively settled.

A neighborhood family may contain multiple predeclared block scales or centers, but the rule selecting the region used for evaluation at a state must itself be frozen before retrospective evaluation and may not depend on (C_F(a)), winning-action identity, successful branches, or equivalent labels.

LOCALITY SURPLUS

For each benchmark state (F), let

$$
N(F)=|A(F)|,
\qquad
g(F)=|G(F)|,
$$

and let (a^*(F)) be the unique retrospectively right next action determined by the frozen cost functional and deterministic tie-breaking rule.

Over the explicitly declared benchmark distribution, define

$$
p=\Pr[a^*(F)\in G(F)]
$$

and define the frozen geometric baseline

$$
\beta=\mathbb{E}\left[\frac{g(F)}{N(F)}\right].
$$

Define locality surplus by

$$
\Lambda=p-\beta.
$$

If the region fraction (g(F)/N(F)) is constant over the declared state class, say equal to (g/N), this reduces to

$$
\Lambda=p-\frac{g}{N}.
$$

A claimed nontrivial locality region must satisfy, for every evaluated state,

[
0<g(F)<N(F).
]

The action space must contain at least two genuinely competing candidate actions.

The admissible state class must be specified by a theoremhood-independent rule before retrospective evaluation. ↪ It may not consist merely of a single hand-selected successful instance or be defined by the fact that the ↪ proposed geometry succeeds on its members.

Prefer an explicitly parameterized family of states when a theorem-level result is claimed.

EMPIRICAL VERSUS THEOREM-LEVEL SUCCESS

A measured empirical value

[
\Lambda>0
]

on a finite benchmark is benchmark evidence only unless one of the following holds:

the declared finite class and distribution have been exhaustively evaluated under the frozen protocol, so that ↪ the claimed frequency statement is exactly certified for that declared finite benchmark; or

a verifier-checkable mathematical argument establishes the claimed probability or frequency inequality for the ↪ stated class/distribution.

Do not promote sample enrichment, exploratory measurements, or post-selected observations into a theorem-level ↪ statement.

If statistical estimation rather than exhaustive evaluation is used, report it as empirical evidence with its ↪ sampling procedure and uncertainty, not as settlement of the mathematical conjecture.

CONJECTURE

There exists a nontrivial explicitly defined class of non-forced 3CNF search states, a computable ↪ theoremhood-independent Hilbert-compatible feature geometry, a frozen candidate-action grammar, a frozen ↪ retrospective certified settlement-cost functional, and a reproducible family of Hilbert-local neighborhoods ↪ such that

[
\Lambda>0
]

for at least one nontrivial regime, where (\Lambda) is the benchmark locality surplus defined above.

Equivalently:

Even when no move is forced and immediate one-step simplification does not already identify the best branch, ↪ visible pre-branch structure contains enough information that retrospectively lower-cost SAT branches are ↪ enriched in some Hilbert-local region beyond that region's share of the candidate-action space.

The claimed enrichment must not be produced by hiding:

a SAT solver;

a satisfiability oracle;

a stored satisfying assignment;

a successful branch;

the retrospective cost labels (C_F(a));

the identity of the winning/minimum-cost action;

target-specific advice;

hidden preprocessing outcomes;

or any equivalent solution information.

All encoding, feature extraction, Hilbert mapping, neighborhood construction, search, verification, memory, preprocessing, update, retrospective evaluation, and comparison costs are charged.

ALLOWED OUTCOMES

A run may produce one of the following kinds of outcome.

Full settlement

A full settlement requires either:

a verifier-checkable proof of the conjecture for an explicitly defined nontrivial class satisfying the frozen rules; or

a verifier-checkable refutation of the conjecture as stated.

Only these outcomes, or another logically equivalent terminal certificate, may be reported as settlement.

Certified partial progress

The following are valuable certified results but are not automatically full settlement:

a precise sufficient condition for positive multi-step locality surplus;

a strictly weaker verifier-checkable statement supported by the frozen hypotheses;

a certified obstruction, limitation, impossibility result, or reduction that materially narrows the conjecture.

A partial result may be promoted to full settlement only if a verifier-checkable argument shows that it logically settles the stated conjecture.

Do not relabel a weaker theorem, benchmark success, suggestive correlation, sufficient condition, or failed search as settlement.

TRUST BOUNDARY

Maintain a strict distinction among:

$$
\text{certified knowledge}
\neq
\text{unverified candidates}
\neq
\text{failed/rejected directions}.
$$

No generated conjecture, heuristic, correlation, representation, or Scout proposal enters certified knowledge until independently verifier-checked.

A candidate may guide search without being treated as true.

Failed and rejected approaches should remain recorded when they contain information useful for avoiding repeated work.

IMPORTANT ANTI-BAILOUT RULE

Bailing out merely because AMLD+0 or the Scout suspects that this conjecture is equivalent to, implies, would resolve, or is closely related to P versus NP or another famous open problem is forbidden.

A suspicion of equivalence to a famous unresolved problem is not itself a settlement certificate.

Do not replace the requested investigation with statements such as:

``This would solve P versus NP.''

``This appears equivalent to a famous open problem.''

``Therefore no further progress can be made.''

``This cannot presently be answered because the related problem is open.''

If an actual reduction, implication, or equivalence is discovered, it may be recorded as a result, but AMLD+0 must distinguish a verifier-checkable reduction from resemblance or suspicion.

AMLD+0 must continue pursuing remaining admissible routes unless the frozen settlement protocol supplies a legitimate terminal certificate.

EXTERNAL-INFORMATION RULE

Do not look up a solution, retrieve a known proof, use an external SAT solver, invoke an external ATP, or use external sources outside the frozen admitted context.

No hidden external result may enter the certified state.

RESOURCE BOUNDS AND TERMINATION

The following bounds apply to the AMLD+0 settlement run itself.

The Eureka Scout may be called exactly once.

After the Scout hint has been recorded, AMLD+0 may perform at most 20 materially distinct settlement/search iterations, including any pre-evaluation construction iteration used to instantiate or reject the hint before the final freeze.

For this purpose, an iteration is one materially distinct investigation step that evaluates, develops, rejects, proves, tests, or substantially reformulates a candidate route. Merely rewriting the same idea does not create a new iteration.

The run has a primary hard time limit of 240 seconds if wall-clock enforcement is available.

Stop immediately when the first applicable terminal condition occurs:

Settlement: a valid full-settlement certificate has been obtained.

Time: 240 seconds have elapsed, if enforceable.

Iteration cap: 20 AMLD+0 settlement iterations have been completed.

Stagnation: 5 consecutive iterations produce neither meaningful progress nor a materially new search direction.

Cycle: the process has returned to substantially the same discovery state or policy without new progress.

Verifier stop: continuing would require treating an unverified result as established.

No admissible continuation: no materially distinct permitted operation can presently be identified under the frozen rules.

If stagnation appears before the five-iteration limit, AMLD+0 should first attempt a major change of representation, decomposition, invariant, or proof strategy rather than immediately stopping.

The resource limits constrain the search procedure; they do not weaken the verifier standard.

Failure to settle within the resource bounds is not a refutation of the conjecture.

AMLD+0 PROCEDURE

After receiving the single Eureka hint:

record the hint exactly as given;

place it initially in the unverified candidate state;

using only pre-branch theoremhood-independent information, instantiate, refine, or reject any proposed feature, invariant, coordinate system, neighborhood construction, quotient, proxy, or enrichment mechanism that will be used in the current experiment;

complete the final pre-evaluation freeze before computing any retrospective branch cost or outcome label;

```
independently investigate the hint and the frozen construction;

retain only verifier-certified theorem-producing consequences as certified knowledge;

reject or quarantine unsupported claims;

adapt subsequent AMLD+0 proof/search investigation using certified results, useful unverified candidates, and
↪  informative failures, but do not adapt any frozen experimental component;

preserve both freeze stages and the non-leakage rules throughout;

continue until one of the declared stopping conditions occurs.

Do not alter the frozen benchmark, geometry, Hilbert ordering, neighborhood family, cost functional, candidate
↪  grammar, tie-breaking rule, or evaluation rules in response to retrospective success or failure during the
↪  same experiment.

A potentially useful alteration may be proposed for a future experiment, but it must not be silently incorporated
↪  into the current frozen run.

FINAL REPORT

When the run terminates, report separately:

the final status: SETTLED-PROVED, SETTLED-REFUTED, CERTIFIED-PARTIAL-PROGRESS, or UNSETTLED;

the exact Eureka Scout hint as originally supplied;

the strongest verifier-certified result obtained;

the best surviving unverified idea, if any;

all newly certified lemmas or consequences;

the most important failed or rejected directions;

whether any empirical locality surplus (\Lambda=p-\mathbb{E}[g(F)/N(F)]) was measured and, if so, whether it was
↪  exhaustive, theorem-level, or merely sample evidence;

the most promising unresolved direction;

the terminal stopping condition;

a compact record of the major AMLD+0 adaptations made during the run.

Do not claim settlement unless the declared full-settlement criterion has actually been satisfied.

Do not confuse positive benchmark evidence with theorem-level settlement.

Do not confuse inability to settle within the resource limit with refutation.

Do not promote the Eureka Scout's hint to certified knowledge merely because it was useful.

Preserve the verifier trust boundary throughout.
```

B IAE Speedup-Theorem Provenance and Release History

The original Preliminary Release 1.5 speedup theorem arose from a later IAE / AMLD+0 run targeted specifically at the remaining overhead obstruction. The frozen target asked whether the already-certified F_k Hilbert-locality theorem could be converted into a fully charged net search-speedup theorem by proving a treatment overhead smaller than the available gross rank margin. Under the conditional-uniformity model then used, that margin was $3k + 1$.

The single Scout hint was:

> Do not construct the entire Hilbert ordering. Compile the already-proved residual-cover characterization into a lazy two-cell Hilbert prefix: fetch the two certified residual-cover-positive actions directly from the exact-address action index, expose only those cells, and construct the remainder of the Hilbert schedule only if the prefix fails.

The hint is recorded as discovery provenance. It is not an axiom. The theorem in Section 4 is certified by the explicit indexed-access hypothesis, cost ledger, certified two-action block, random-baseline rank calculation, and direct inequality proof. Preliminary Release 1.7 strengthens the main theorem by using the deterministic bound $R_H \leq 2$ instead of assuming conditional uniformity inside the block. No exhaustive external priority claim is made without a separate literature audit.

C Consequence-Mining Protocol and Release History

Preliminary Release 1.6 introduced the formal Consequence Mining protocol. The Release 1.7 audit found that calling the twenty-two-item list a normalized "closure" and calling its multi-parent counts globally minimal were stronger claims than the supplied enumeration justified. Release 1.7 therefore recasts the object as a selected normalized consequence catalogue with explicit retained parent sets. The displayed parent count witnesses a low-fan-in derivation but is not asserted to be globally minimum absent exhaustive derivation search.

The audit also surfaced two immediate one-parent limit consequences permitted by the stated grammar but absent from the earlier list: $\Lambda_k \to 1$ and $E[R_{\text{random}}] \to \infty$. They are now retained as CM-19 and CM-20. The catalogue therefore contains twenty-four displayed consequences: twenty with one- or two-parent retained derivations and four with three-parent retained syntheses. The catalogue remains intentionally non-exhaustive.

D Repair Report and Release Audit

Status separation

- **Historical frozen-run status: CERTIFIED–PARTIAL–PROGRESS.** The supplied source does not contain a verifier-checkable record that the admissible family, benchmark distribution, fallback, cost functional, and tie-break were frozen before the Eureka Scout. The original run therefore must not be cited as a protocol-certified `SETTLED--PROVED` IAE experiment.
- **Mathematical structural-theorem status: PROVED.** The Multi-Step Structural Concentration Theorem is established directly for every $k \geq 2$ under the explicitly stated lexicographic fallback-cost model, independently of the missing historical timestamp/order evidence.
- **IAE speedup status: PROVED UNDER THE STATED COST MODEL.** The Release 1.7 main theorem uses the certified F_k locality result, the Indexed Exact-Address Access Hypothesis, the six-operation primitive ledger, $R_H \leq 2$, and the random-baseline expectation. Conditional uniformity inside the two-action block is optional and used only for the sharper refinement.
- **Consequence-mining status: RETAINED CATALOGUE, NOT EXHAUSTIVE CLOSURE.** Each listed theorem has a displayed parent set of size at most three. The displayed size certifies an explicit low-parent derivation; global minimum fan-in is not claimed for the multi-parent entries.

Exact Eureka Scout hint

> Look for a residual-cover feature: after a hypothetical first action, test whether the remaining clauses share a single visible literal that could satisfy the whole residual formula in one further non-forced move. Compare this with branches whose residual clauses split across independent supports.

The hint is retained as provenance and is not promoted to certified knowledge.

Repairs made through Preliminary Release 2.2

The audited revision retains the earlier structural and Release 1.7 repairs, preserves the local consequence-genealogy presentation added in Release 1.10, and incorporates the self-contained path-calibration package added in Release 1.11. Preliminary Release 1.12 removes the formerly imported cross-paper consequence chain so the Consequence Mining section remains internal to this manuscript; it also adds theorem-graph curvature, the geometric dictionary, the curvature-profile settlement, the conjecturing-to-settlement architecture in Section 14, and the new future-work items stated in the open-research section. Preliminary Release 1.13 adds selected nonredundant theorem-geometry results from the author's earlier papers and re-proves the imported statements in the present notation.

Preliminary Release 2.1 integrates the GBG/GBG$^+$ formal foundation into Part I as an operational substrate rather than as a replacement for the exact-addressing framework. The new material: (i) identifies rule-complete proof states as the correct objects for exact operational addressing; (ii) proves a match-proof/verifier-path correspondence; (iii) proves continuation sufficiency and an operational-address adequacy criterion; (iv) introduces exact move-labeled history addresses and proves lossless history reconstruction plus exact endpoint-sensitive history counting; (v) separates raw addressed history multiplicity from essential-route quotient multiplicity; (vi) adds the board-player $P_0 = B^\star$ and proves the unit player-label geodesic $d_{\mathrm{label}}(r, 0) = r$; (vii) proves a passive GBG$^+$ projected-cost isometry and settlement-preservation result; and (viii) specializes the scheduler/verifier separation principle to active GBG$^+$ architectures. None of these claims identifies player-label distance, address distance, or navigation distance with intrinsic proof distance without a separately declared bridge.

Preliminary Release 2.2 makes only local mathematical repairs and does not restructure the manuscript. In particular, it: (i) removes the unnecessary minimum-cost hypothesis from the saturated-path exactness criterion and lets saturation certify geodesicity directly; (ii) separates uniqueness of an optimum from positivity of the directional gap; (iii) makes the positive isolation-radius hypothesis do the necessary work in the doorway zero-residual proof; (iv) restricts the settlement-congruence conclusion to quotient completeness unless a separate representative-canonicalization mechanism is supplied; (v) requires BANK macro reuse to preserve the hypotheses, side conditions, provenance, and replayable support of the stored derivation; (vi) distinguishes the undirected curvature metric d_u from the directed legal path distance d_{legal}; (vii) supplies a direct proof of the leaf-amplified curvature formula; and (viii) rewrites CM-10 as a liminf bound and CM-22 as a per-instance geometry-invariance statement.

1. It preserves the separation between the directly proved structural theorem and the historically incomplete frozen-run record.

2. It retains the explicit lexicographic fallback cost (B, U, E), deterministic exact-identity tie-break, common charged overhead term $H(F_k)$, and feature-induced Hilbert-block disclosure.

3. It retains the extension of the structural theorem to every $k \geq 2$ and the distribution-free family bound $\Lambda \geq 7/8$.

4. It repairs the rational-address reversibility language by defining the canonical representative set $S_{\mathbb{Q}} = s(\mathbb{Q})$ and stating the genuine bijection $s : \mathbb{Q} \leftrightarrow S_{\mathbb{Q}} : r|_{S_{\mathbb{Q}}}$. The noninjective surjection $r : \mathbb{N} \twoheadrightarrow \mathbb{Q}$ is no longer described as globally invertible.

5. It requires a genuinely bijective rational enumeration when a reversible WFF–rational-address correspondence is asserted; a section-equipped surjection is used only for canonical representatives.

6. It retains the strengthened Exact Addressing recoding-invariance theorem and the no-speedup-by-addressing corollary.

7. It strengthens the Lazy Hilbert speedup theorem by removing conditional uniformity inside the certified two-action block from the main statement. The theorem now uses $R_H \leq 2$ and proves the robust gross expected rank lower bound $3k + 1/2$ and net charged margin $3k - 11/2 > 0$ for every $k \geq 2$.

8. It retains the older conditional-uniformity result only as an optional refinement, recovering $E[R_H] = 3/2$, gross expected rank advantage $3k + 1$, and net margin at least $3k - 5$.

9. It removes "Novel" from the speedup-theorem heading and treats IAE as discovery provenance rather than external historical-priority evidence.

10. It repairs Consequence Mining from an asserted finite normalized closure to a selected normalized catalogue. The catalogue no longer claims exhaustiveness.

11. It replaces unsupported global minimum-fan-in assertions by explicit retained derivation fan-in d^{-}_{ret}. For every displayed theorem, the listed parent set witnesses a derivation with at most three parents; stronger global minimality is left open unless separately proved.

12. It adds the two audit-surfaced one-parent limit consequences $\Lambda_k \to 1$ and $E[R_{\mathrm{random}}] \to \infty$, expanding the selected catalogue from twenty-two to twenty-four entries.

13. It renames CM-17 to "without assuming address-metric proximity," making clear that the proof does not require metric closeness rather than claiming that the paired addresses are provably far apart.

14. It updates release-history wording so results introduced in Preliminary Releases 1.5 and 1.6 are described historically rather than as newly introduced in Release 1.7.

15. It adds PDF title, author, subject, and keyword metadata.

16. It separates exact-address distance d_A from intrinsic proof-geodesic distance by writing the latter as d_{geo}, eliminating the prior notation collision.

17. It repairs the rectilinear calibration denominator by taking Λ_A only over positive-cost edges, explicitly requiring zero-cost edges to have zero coordinate displacement, and assuming $0 < \Lambda_A < \infty$.

18. It makes the Lazy Hilbert cost semantics explicit: every ordinary candidate inspection is one normalized candidate-inspection-equivalent unit, so per-candidate processing is counted exactly by the rank terms, while only fixed common state-level setup costs cancel; the six-unit term is treatment-specific overhead in that frozen model.

19. Releases 1.9 and 1.10 experimented with an imported cross-paper shell chain; Release 1.12 removes that chain from this manuscript while retaining the local CM shell coordinates and genealogy annotations.

20. Release 1.11 incorporated the path-calibrated bridge theorem package directly into Part I, including coordinate/proof-distance calibration, target potentials, geodesic certificates, separator bounds, bottleneck certification, re-addressing invariance, isometric trading, and map-versus-road separation.

21. Release 1.12 adds the unit-edge Ollivier–Ricci theorem-graph curvature definition, curvature profiles, negative-curvature burden, and a geometric dictionary separating intrinsic proof geometry from navigation geometry.

22. Release 1.12 strengthens Section 14 with the conjecturing lifecycle, earlier theorem-space conjectures, and the AMLD+0 settlement of the Geodesic Curvature-Profile Conjecture, including the counterexample, infinite failure family, universal burden bound, and Curvature–Potential Geodesic Bridge.

23. Release 1.12 adds future research targets for curvature-profile prediction, restricted universal navigation, AMLD+0 installation instructions, and safety-constrained differential-diagnosis hypothesis generation.

24. Release 1.13 imports finite face-continuous Hilbert probe geometry, including the axis-parallel probe law and exact finite-cell coverage, from the repaired MAPQUEST foundations.

25. Release 1.13 adds a seven-theorem approximate-geodesic package: local 2ε regret, telescoping cumulative regret, the wrong-move slack barrier, selective exactness, a relative-approximation criterion, the BANK-shortcut relative-precision effect, and shortcut-induced resolution rescaling.

26. Release 1.13 adds premise-doorway isolation and address distortion; cross-fiber Lipschitz and bi-Lipschitz transport; verifier-safe quotient obstruction and exact geodesic quotient results; settlement-congruence reduction; and a verified-lemma macro-shortcut theorem.

27. Release 1.13 adds observed essential route multiplicity and the heterogeneous cartographic signature $\Sigma_{\mathrm{cart}} = (H_0, \mu, \tau, \beta, \rho)$, explicitly separating shortest-path geometry from redundancy, traffic, bottlenecks, and representation sensitivity.

28. Every newly reproduced proof in the Release 1.13 import package is explicitly marked "Proved by Sol."

29. Preliminary Release 2.2 strengthens the Saturated-Path Exactness Criterion so the saturation hypotheses themselves certify geodesicity; no prior minimum-cost assumption is required.

30. Preliminary Release 2.2 repairs the directional-gap sentence: a unique minimizer may have zero gap in an infinite successor set, so positive gap is now a separate explicit hypothesis.

31. Preliminary Release 2.2 repairs the doorway zero-residual proof by using the stated positive isolation radius, not injectivity alone.

32. Preliminary Release 2.2 narrows settlement-congruence reduction to quotient completeness and states separately that one-representative implementations require canonicalization or transition redirection.

33. Preliminary Release 2.2 adds the context/support-closure hypothesis needed for verifier-safe BANK macro reuse.

34. Preliminary Release 2.2 separates the undirected curvature metric d_u from directed legal proof distance d_{legal} and updates the curvature burden and bridge statements accordingly.

35. Preliminary Release 2.2 supplies the missing direct Wasserstein calculation for the leaf-amplified failure family.

36. Preliminary Release 2.2 rewrites CM-10 as a lim inf lower bound and CM-22 as per-instance theorem-DAG invariance across a family with growing discovery-cost separation.

Preliminary Release 2.3 scientific-hypothesis recovery and formalization

Preliminary Release 2.3 preserves the existing section order and adds only local material to the scientific-hypothesis-generation program. A nearly lost fictional differential-diagnosis development record is restored as a protocol-calibration pilot. The release does not treat that run as clinical evidence: the generator was blind to the later reference answer, but the clue presenter knew it and the reveal schedule was not shown to have been precommitted. The run is therefore labeled controller-blind and presenter-not-blind.

The mathematically retained consequences are: (i) a frozen observational-signature relation separating evidence fit from explanatory refinement; (ii) Theorem 14.5, showing that observational equivalence prevents an evidence-only evaluator from certifying explanatory maximality; (iii) a public-history-measurable basin detector; (iv) a one-use mechanism-changing depth-charge policy with an explicit discriminator; (v) the charged one-step value criterion $V(p_D - p_0) > C_D - C_0$; and (vi) an evidence-presentation anti-leakage requirement for future held-out benchmarks. No claim of medical efficacy, clinical validation, or autonomous diagnosis is made.

Final Atlas audit checklist

Audit item	Release 2.3 finding
3CNF family and candidate count $N(F_k) = 4 + 6k$	PASS
Three-way immediate clause-deletion tie	PASS
Fallback comparison for all candidate types	PASS
Residual-cover equivalence for every $k \geq 2$	PASS
Unique deterministic tie-break	PASS
Total comparable cost without hidden large-M assumption	PASS
Feature-induced Hilbert contiguity stated without overclaim	PASS
Nine-state exact locality arithmetic	PASS
General bound $\Lambda \geq 7/8$ on distributions supported on $k \geq 2$	PASS
Canonical rational representative bijection $\mathbb{Q} \leftrightarrow S_{\mathbb{Q}}$	PASS
Noninjective $r : \mathbb{N} \twoheadrightarrow \mathbb{Q}$ no longer treated as an inverse pair	PASS
Reversible WFF–rational correspondence requires a genuine bijection	PASS
Strengthened Exact Addressing recoding-invariance theorem	PASS
No-speedup-by-addressing corollary under exact transport	PASS
Address distance d_A and proof-geodesic distance d_{geo} use distinct notation	PASS
Rectilinear lower-bound calibration has no 0/0 edge ratio and assumes $0 < \Lambda_A < \infty$	PASS
Lazy extra-cost ledger $C_H(k) \leq 6$ under declared calibration	PASS
Each ordinary candidate inspection is one normalized unit; only fixed common setup costs cancel	PASS

Audit item	Release 2.3 finding
Distribution-free certified-block rank bound $R_H \leq 2$	PASS
Robust net expected charged margin $3k - 11/2 > 0$ for all $k \geq 2$	PASS
Conditional-uniformity refinement separated from the main theorem	PASS
Feature-induced rather than exogenous Hilbert geometry stated explicitly	PASS
Satisfiability preservation under uniform variable renaming	PASS
Transported WFF-address SAT locality $\Lambda_{\mathrm{SAT}} = 2p(1-p) > 0$	PASS
CM-17 wording does not assert unproved metric nonproximity	PASS
Consequence Mining described as a selected catalogue, not exhaustive closure	PASS
Twenty-four retained interesting theorems with explicit parent sets	PASS
Four three-parent syntheses not mislabeled globally minimum	PASS
Obvious one-parent limit consequences CM-19 and CM-20 included	PASS
Retained shell numbers and ancestor/descendant genealogy printed beside local CM theorems	PASS
Path-calibrated bridge theorem package included with full theorem statements and proofs	PASS
Finite face-continuous Hilbert probe law and exact cell coverage imported and reproved	PASS
Fixed-target geometric proof-distance specialization included	PASS
Doorway isolation threshold and address-distortion theorem included	PASS
Approximate-geodesic seven-theorem regret/precision package included	PASS
Cross-fiber Lipschitz and bi-Lipschitz transport included	PASS
Verifier-safe quotient obstruction, exact geodesic quotient, and settlement congruence included	PASS
Verified-lemma macro shortcut theorem preserves theoremhood while permitting horizon reduction	PASS
Route multiplicity and cartographic signature separated from intrinsic distance	PASS
Newly reproduced import-package proofs explicitly marked Proved by Sol	PASS
Coordinate/proof-distance lower bounds explicitly require finite edge calibration	PASS
Target-potential, geodesic, separator, and bottleneck certificates are proved internally	PASS
No citation to an unavailable unpublished path-first manuscript is required for the new theorem package	PASS
Theorem-graph Ollivier–Ricci curvature defined separately from proof distance	PASS
Positive/near-zero/negative curvature interpreted only as local neighborhood regimes	PASS
Six-vertex curvature-burden counterexample stated with exact W_1 values	PASS
Leaf-amplified failure family and off-path-neighborhood obstruction scoped correctly	PASS
Universal burden bound $0 \leq B_-(P) \leq 2\lvert P\rvert$ proved internally	PASS
Curvature–Potential Geodesic Bridge stated as sufficient, not minimal	PASS
Saturated-path criterion certifies geodesicity without assuming minimum cost	PASS
Unique optimum no longer asserted to imply positive directional gap on infinite successor sets	PASS
Doorway zero-residual proof uses positive isolation radius rather than injectivity alone	PASS

Audit item	Release 2.3 finding
Settlement-congruence theorem claims quotient completeness; representative reduction requires extra canonicalization	PASS
Verified-lemma macro shortcut requires replayable context, side conditions, provenance, and support	PASS
Undirected curvature metric d_u is separated from directed legal distance d_{legal}	PASS
Leaf-amplified curvature formula now has a direct Wasserstein proof	PASS
CM-10 uses a liminf lower bound; CM-22 states per-instance geometry invariance	PASS
The conjecturing section separates conjecturing $K_t \to c$ from settlement $c \to x$	PASS
Eureka hint remains unverified until verifier-supported consequences enter BANK	PASS
Recovered fictional differential-diagnosis run is labeled controller-blind / presenter-not-blind	PASS
Evidence-fit and explanatory-completeness objects are formally separated	PASS
Basin detector is frozen and measurable from public history only	PASS
Depth charge is one-use, mechanism-changing, and requires an explicit discriminator	PASS
Future held-out benchmarks require precommitted or mechanical evidence presentation when the presenter knows the hidden mechanism	PASS
Clinical validation or autonomous diagnostic efficacy from the recovered pilot	NOT CLAIMED
Cross-paper S1.1–S6.1 chain removed from the local consequence-mining section	PASS
External historical-priority claim	NOT CLAIMED
Historical pre-Scout base-freeze certificate	NOT ESTABLISHED; NOT CLAIMED
Arbitrary-3CNF or general SAT speedup claim	NOT CLAIMED
Polynomial-time SAT or $P \neq NP$ conclusion	NOT CLAIMED

Final audit conclusion

HISTORICAL RELEASE 2.3 INTERNAL AUDIT: PASS WITH THE NOTED REPAIRS ON THE EXPLICITLY AUDITED CLAIMS. Release 2.4 performs editorial separation and renumbering; no fresh external formal-verifier audit is claimed.

The earlier repairs remain in force: address distance and intrinsic proof distance use distinct notation; coordinate lower bounds require explicit calibration; the Lazy Hilbert ledger charges per-candidate work through rank while canceling only fixed common setup costs; and the consequence-mining catalogue is finite and selected rather than an asserted exhaustive closure. Release 1.12 added discrete curvature and the conjecturing/hypothesis-generation architecture without collapsing those layers into proof distance or certification. Release 1.13 extended Part I with selected theorem-geometry results recovered from the author's earlier manuscripts: exact finite Hilbert probe geometry, address-isolation certificates, cross-fiber metric distortion, verifier-safe quotient geometry, certified macro shortcuts, bounded-error geodesic regret, BANK-dependent precision, and structural route/cartography coordinates. Preliminary Release 2.1 added the GBG/GBG$^+$ operational substrate and re-proved the integration theorems stated in Section 11. Preliminary Release 2.2 preserves that architecture and incorporates the local mathematical repairs listed above: strengthened saturation-based geodesic certification, corrected gap

and isolation logic, narrower quotient-representative claims, explicit macro-support conditions, separated curvature/proof distances, the missing leaf-family proof, and repaired asymptotic consequence wording. Preliminary Release 2.3 preserves all of those repairs and adds the recovered scientific-hypothesis pilot only under a stricter audit classification: controller-blind, presenter-not-blind, nonclinical, and not a controlled validation. Its mathematical contribution is the explicit separation of observational fit from explanatory refinement, together with a frozen basin-triggered one-depth-charge protocol and evidence-presentation anti-leakage requirements. The 2.3 checkpoint continues to distinguish exact WFF identity from exact rule-complete-state identity, reconstructs move-labeled histories exactly, preserves raw history multiplicity, and proves projected-cost invariance for the passive board-player extension. These repairs do not broaden the family-specific scope of the F_k speedup theorem and do not establish a general SAT, universal-navigation, universal metric-embedding, or curvature-predicts-difficulty result.

Preliminary Release 2.4 editorial relocation ledger

The source paper's old Sections 10, 11, 15, 17, 19, 20, 21, and 22 are not missing. Old Section 10 forms a one-section standalone PDF, *The Central Composition Problem*; the other seven sections form *IAE, Opaque Search, and Formalized Scientific Hypothesis Generation.* Old Section 23 has been regenerated as the open directions section of this release. Main-text theorem sections were reorganized for a result-first reading order, and LaTeX automatically renumbers local theorem counters and all label-based cross-references. The original scientific protocol audit remains historical, not retroactive evidence of precommitment.

E Universal Proof Horizons and Exact-Address Geometry

Release 3.1: fully replaced and repaired appendix

The old Appendix E from Preliminary Release 3.0 has been superseded. The complete, newly typeset Appendix E.1-E.5 and Theorems E.2-E.9 start on the next page. This preserves the original historical Release 2.4 editorial ledger above without silently changing its contents.

Appendix E (Repaired): Universal Proof Horizons and Exact-Address Geometry

Exact Addressing · Preliminary Release 3.3

Brian Tenneson
Mathematical repair and editorial preparation assisted by OpenAI ChatGPT

October 7, 2026

Scope and status. This is a self-contained replacement for Appendix E of Preliminary Release 3.0, preserving its five-subsection structure and theorem numbering E.2–E.9. It gives mathematical proofs within the stated hypotheses, not a certificate from a formal proof assistant. Statements attributed to the separate Universal Proof-Horizon Operator (UPO) v1.2 are used as stated assumptions or source-context, not independently reverified here. All costs, representations, and search policies are typed separately. In particular, Appendix E does *not* certify Section 4's end-to-end runtime claim without that section's additional common-infrastructure and cost-calibration assumptions.

E.1 The two notions of minimum and the necessary interface

Fix an effectively encoded deductive theory T, an effective sentence indexing $n \mapsto \sigma_n$, and a total decidable proof-checking predicate $V(p,n) \in \{0,1\}$. Let $c : \mathbb{N} \to \mathbb{N}$ be total computable. Assume *effective properness*: for each $b \in \mathbb{N}$, the set $\{p : c(p) \le b\}$ is finite and there is a terminating algorithm returning exactly this set. Write $P_n = \{p : V(p,n) = 1\}$. For $P_n \neq \varnothing$ let

$$h(n) = \min_{p \in P_n} c(p), \qquad H(n) = \min\{p \in P_n : c(p) = h(n)\}.$$

The second minimum is the ordinary numerical ordering of proof-code integers. Both are *undefined* for $P_n = \varnothing$; a graph-theoretic distance to an unreachable target, by contrast, is $+\infty$.

Independently fix a directed legal proof-state multigraph $G_n = (S_n, E_n)$, initial state s_n, nonempty target set $C_n \subseteq S_n$, and edge weights $w : E_n \to [0,\infty)$. Let Γ_n be its finite s_n-to-C_n legal paths, and define $d_w(s_n, C_n) = \inf_{\gamma \in \Gamma_n} \sum_{e \in \gamma} w(e)$, with $\inf \varnothing = +\infty$. A graph path, a WFF, a proof-code integer, and an address are different typed objects.

Definition E.1 (Two-way cost-matched realization)**.** *A cost-matched realization for sentence σ_n consists of* total assignments on their stated domains $F : P_n \to \Gamma_n$ *and* $J : \Gamma_n \to P_n$ *satisfying* $\mathrm{cost}_w(F(p)) = c(p)$ *for every* $p \in P_n$ *and* $c(J(\gamma)) = \mathrm{cost}_w(\gamma)$ *for every* $\gamma \in \Gamma_n$*. No injectivity, computability, or canonical-code preservation is assumed unless separately stated. These identities imply integral cost for every target-reaching path, not necessarily for every unrelated graph path.*

Theorem E.2 (Proof-horizon/path-distance identification)**.** *Under Definition E.1, $P_n \neq \varnothing$ if and only if $\Gamma_n \neq \varnothing$. If nonempty, $d_w(s_n, C_n) = h(n) \in \mathbb{N}$. If empty, $h(n)$ remains undefined whereas $d_w(s_n, C_n) = +\infty$.*

Proof. The maps in both directions give equivalence of nonemptiness. For each proof code its cost occurs as a path cost and for each target path its cost occurs as a proof cost; thus the two attained-cost sets coincide. A nonempty subset of $\mathbb{N}$ has a least element, equal to the graph infimum and the proof-code minimum. The empty case follows by definitions. □

Remark. Only a one-way proof-to-path realization yields $d_w \le h$; only path-to-proof yields $h \le d_w$. Exact equality requires both. Cost preservation must include any length/size/administrative encoding conventions: equal line counts do not entail equal bit lengths.

E.2 Exact re-addressing transports the horizon, not necessarily the chosen code

Theorem E.3 (Cost-preserving proof-code transport)**.** *Let (T, V, c) and (T', V', c') be effective presentations. Let $\tau : \mathbb{N} \to \mathbb{N}$ and $f : \mathbb{N} \to \mathbb{N}$ be computable bijections with computable inverses. Assume for all p, n,*

$$V(p, n) = V'(f(p), \tau(n)), \qquad c'(f(p)) = c(p).$$

Then $P'_{\tau(n)} = f[P_n]$. Consequently their theorem-index domains correspond and $h'(\tau(n)) = h(n)$ whenever defined. Effective properness transfers from c to c'. The selected least numerical proof code need not satisfy $H'(\tau(n)) = f(H(n))$.

Proof. The first equality and surjectivity of f identify precisely the accepted-code sets. The second identifies their attained costs. Further, $\{q : c'(q) \le b\} = f[\{p : c(p) \le b\}]$; compute the finite source set and map its members under f. For the final statement, a computable permutation need not preserve the numerical order between distinct tied minimizers. □

Remark (Explicit tie-break counterexample). Suppose $P_n = \{1, 2\}$ and $c(1) = c(2) = 1$ (extend c to an effectively proper cost, e.g. $c(p) = p$ for $p \ge 3$). Choose f swapping 1 and 2 and transport verifier and cost. Then $H(n) = 1$, but $H'(\tau(n)) = 1 \ne f(H(n)) = 2$.

Remark. A pure exact re-addressing cannot lower the minimum cost when the source/target cost functions are transported. It may change code order, lookup cost, memory footprint, or discovery time; these are different measurements.

E.3 Potentials provide certified lower bounds and optimality tests

Let a *consistent target potential* be $p : S_n \to [0, \infty)$ with $p(t) = 0$ for $t \in C_n$ and $p(x) \le w(e) + p(y)$ for every legal edge $e : x \to y$. To call a proposed potential *certified*, require a mathematically valid argument or a verifier-checkable certificate covering *all relevant edges*; empirical spot checks do not establish the universal inequality.

Theorem E.4 (Integral horizon squeeze)**.** *Assume Definition E.1, a consistent target potential p, and an exhibited verifier-valid path $\gamma : s_n \rightsquigarrow C_n$ with integer cost U. Then*

$$\boxed{\lceil p(s_n) \rceil \le h(n) \le U.}$$

If $\lceil p(s_n) \rceil = U$, both the path and its realized proof code are minimum-cost. This does not imply that the code is the UPO numerical tie-break minimizer.

Proof. Sum the edge consistency inequalities along any target-reaching path and use the terminal value zero to obtain $p(s_n) \le \mathrm{cost}(\gamma')$. By Theorem E.2, the infimum is the integer $h(n)$. The exhibited path realizes a proof of cost U. The rounding statement follows. □

Theorem E.5 (Calibrated-coordinate horizon certificate)**.** *Suppose $H : S_n \to \mathbb{R}^d$ and $0 < L < \infty$ satisfy $\|H(x) - H(y)\|_1 \le L w(e)$ on* every *legal edge $e : x \to y$ (in particular, zero-cost edges have zero displacement). Define*

$$p_H(x) = \frac{\mathrm{dist}_1(H(x), H(C_n))}{L}, \quad \mathrm{dist}_1(z, A) = \inf_{a \in A} \|z - a\|_1.$$

Then p_H is consistent. If a target-reaching cost-U path exists under Definition E.1,

$$\boxed{\left\lceil \frac{\mathrm{dist}_1(H(s_n), H(C_n))}{L} \right\rceil \le h(n) \le U.}$$

Proof. Distance to a nonempty subset is 1-Lipschitz, even if that subset is not closed. Therefore $p_H(x) - p_H(y) \le \|H(x) - H(y)\|_1 / L \le w(e)$; it vanishes on targets. Apply Theorem E.4. □

Remark. This is an *operationally available* numerical bound only if its lower estimate and the edge calibration can be computed or soundly certified. An arbitrary coordinate is not an admissible heuristic merely because its points are called Hilbert or rational addresses.

Theorem E.6 (Canonical minimum-code recovery after certified pruning)**.** *Assume effective properness, a certified integer lower bound $b \le h(n)$, and a verified code of cost $U \ge b$. Then a terminating algorithm returns the* same canonical code $H(n)$ *as exhaustive UPO search while checking no proof code with $c(p) < b$: enumerate each finite exact-cost layer $j = b, \dots, U$, sorting each by numerical proof-code order, and return the first code with $V(p, n) = 1$.*

Proof. No accepted code has cost below b; a supplied accepted code has cost at most U. Effective properness enumerates each bounded layer exactly. The first accepted layer therefore has minimum cost, and sorting that layer yields its least numerical code. □

Remark (Accounting). The procedure is terminating because the *upper proof is supplied.* This does not make the UPO partial function a total theorem decider. Cost to obtain the lower bound, upper proof, exhaustive layer enumeration, sorting, and verification must be charged in comparisons of search methods. The pruning theorem alone is not a net speedup claim.

Remark (Small valid example). In the graph $s \to a$ (cost 2), $a \to t$ (cost 1), $s \to b$ (cost 1), $b \to t$ (cost 4), $s \to t$ (cost 5), set $p(s) = 5/2$, $p(a) = 1$, $p(b) = 3/2$, $p(t) = 0$. All inequalities hold; the s, a, t path has cost 3 and $\lceil p(s) \rceil = 3$, proving it optimal. It says nothing about computation of this potential or numerical tie-break of proof codes.

E.4 Passive GBG+, finite layers, and the policy boundary

Theorem E.7 (Passive projection and the effective-properness boundary)**.** *Suppose a passive GBG+ extension has zero-cost administrative steps, every passive target-reaching history projects to a legal ordinary target-reaching history preserving the ordered ordinary moves, and every ordinary history has a passive lift of equal ordinary-move count from the declared initial tag to a target in the declared target fiber. Then the minimum projected ordinary-move count is the same in both systems. If the ordinary graph is effectively finitely branching and ordinary moves have unit cost, breadth-first*

search finds a shortest target-reaching ordinary history when a target is reachable. *These facts do not imply effective properness for a code cost that counts only ordinary moves on* all raw passive histories.

Proof. Projection and lift produce the opposite inequalities between the two path minima. At each finite ordinary depth, finite effective branching makes the search layer finite and enumerable. Reachable targets are found in increasing depth; absence of a target need not be decidable. For the final assertion, let $s \xrightarrow{\epsilon} s$ be a zero-cost administrative self-loop and $s \to t$ have cost one. The histories $\epsilon^m(s \to t)$ for all $m \geq 0$ are distinct raw histories of identical cost one, making the corresponding cost sublevel infinite. □

Remark. A canonical normal form deleting administrative loops, or a separately specified effectively proper code-size cost, can restore finite code layers, but needs its own proof and may change the objective being minimized. The passive projection statement does not apply to an active board whose legal moves change the projected road system.

Theorem E.8 (Policy invariance of horizons)**.** *Fix a proof-code universe, verifier V, cost c, and target sentence σ_n. Two search policies which only reorder or select queries to the* same *verifier over the* same *admissible code universe have identical theorem domain and minimum-cost horizon $h(n)$. Their discovery time, resource consumption, and probability of discovering a proof within a finite budget may differ.*

Proof. The minimum is defined by V, c, and the admissible code set, not by its enumeration policy. □

Remark (Compatibility with main-text Sections 3–4). The F_k concentration construction selects a feature-induced two-action prefix; this is not a Hilbert embedding discovered independently. The displayed $3k - 11/2$ bound is valid *inside the normalized inspection-ledger model* only when the baseline and treatment genuinely share every omitted preprocessing cost, the family recognizer and symbolic shortcut are charged, and the six treatment operations meet their stipulated calibration. Section 3's eager all-candidates feature computation cannot simply be treated as common to Section 4's lazy feature computation unless both competitors really execute it or an explicit accounting proof shows why it cancels. No end-to-end wall-clock advantage follows without these conditions.

Theorem E.9 (No universal sound complete non-theorem decider)**.** *If $\{n : T \vdash \sigma_n\}$ is undecidable, no total computable procedure can always output, for each n, either a verifier-accepted proof of σ_n or an object accepted by a sound decidable checker as certifying $T \nvdash \sigma_n$, with one of those outcomes guaranteed for every input.*

Proof. Running the proposed total procedure and distinguishing the two types of accepted outputs would decide the theorem-index set, a contradiction. □

Remark. This leaves open decidable restrictions, per-instance nonreachability certificates, and incomplete procedures that may diverge.

E.5 Bidirectional consequence ledger and audit boundary

UPO-side object	Exact-Addressing-side object	Joint consequence and exact boundary
Proof verifier, proper cost	Proof-state weighted paths	E.2: equality of minima requires a *two-way cost-matched realization*; source-only encoding implies merely one inequality.
Proof-code renaming	Reversible WFF / state addressing	E.3: transported minimum cost invariant; canonical numerical code not invariant under arbitrary permutation.
Cost minimization	Target potential, geometry	E.4–E.5: a certified potential plus a matching cost objective yields integer-rounded lower bound; no geometric estimate without calibrated edges.
Finite code layers	Proof search and pruning	E.6: supplied upper proof and certified lower bound permit finite canonical search in $[b, U]$, but only as a conditional algorithm.
Shortest inference paths	Passive GBG+ projection	E.7: ordinary move minimum invariant under path lifting; raw zero-cost administrative histories can violate effective properness.
Proof-domain definition	Hilbert / F_k scheduling	E.8: policy affects discovery costs, not optimal proof-code horizon; Section 4's advantage remains cost-model-relative.
Undecidable theorem domain	Geometric nonreachability certificates	E.9: no total sound, complete non-theorem decider on an undecidable theory.

Cross-paper implication discipline. E.2–E.9 are elementary mathematical consequences of their explicitly displayed assumptions. Their correctness does not certify the existence, computability, or efficient construction of the required cost-matching maps or globally consistent potentials for arbitrary proof systems. No theorem here proves that shortest encoded proofs have shortest inference length, that an arbitrary exact address is a proof-distance coordinate, or that Hilbert scheduling is faster in the absence of the declared feature and calibrated ledger.

Release 3.1 correction record. This appendix replaces the original E.1–E.5 content without changing their order: (i) separates horizons, numerical proof codes, and graph paths; (ii) isolates the exact bidirectional interface; (iii) clarifies partial-versus-infinite conventions; (iv) gives a concrete recoding tie-break example; (v) identifies integrality and effective-certification dependencies; (vi) distinguishes canonical code recovery from optimal path recovery; (vii) conditions BFS on target reachability; (viii) exposes zero-cost raw-history improperness; (ix) adds an explicit warning about main-text preprocessing accounting. This is a manuscript-level mathematical repair, not an external formal verification.

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Appendix F: Capstone Theorem

Certified Geometric Discovery and Amortized Search Advantage

Exact recoding, complete verifier-gated discovery, and a fully charged expected-cost separation

Brian Tenneson
Research working draft; document preparation assisted by OpenAI ChatGPT

October 13, 2026 | Exact Addressing 3.3.1 working integration

Abstract

We formulate and prove a *conditional composition theorem* for exactly addressed automated theorem provers. The theorem combines (i) invariance of theoremhood and shortest-proof layers under rule-preserving recoding; (ii) eventual verifier-certified discovery by an explicitly constructed, protected exhaustive proof enumerator; and (iii) strict fully charged *expected* discovery-cost improvement on an eligible family of finite certificate searches with a target-blind informative neighborhood and bounded implementation overhead. We give the precise rank-saving identity, a net charged-cost lower bound, and an amortized threshold of order $\log N$ related tasks. The result does *not* assert that arbitrary Hilbert navigation is complete or faster, nor does it establish that an independently interesting infinite benchmark family already meets the quantitative hypotheses. Those are outstanding verification obligations.

F.1 Purpose and relationship to the preceding conjecture

The earlier capstone asked whether five broad assumptions about exact recoding, verifier soundness, fairness, geometric locality, and Hilbert prioritization implied four conclusions: theoremhood invariance (T1), shortest-proof-layer invariance (T2), finite certified discovery (T3), and fully charged search advantage (T4). A counterinterpretation of the displayed assumptions exposed the distinction between *fairness among enabled actions* and *generation of all proof steps*. It also exposed the missing link between positive locality surplus and a strict advantage after guidance costs are charged.

Here we deliberately *revise both hypotheses and conclusions*. We keep T1–T4 as named objectives, but derive T3 from a constructed complete enumerator and restrict T4 to a quantitatively specified class. This is an explicitly labeled working-edition appendix; the preceding manuscript text and section order are unchanged.

F.2 Formal systems, search architecture, and charged benchmarks

Definition F.2.1 (Formal system and proof layers)**.** Let $F = (x, y, z)$ be an effectively presented formal system, where y is a set of well-formed formulas and z is a computably enumerable family of finitary inference rules. Let $\Gamma \subseteq y$. A proof is a finite sequence in which each line is a hypothesis or follows by an applicable inference rule from earlier lines. Write

$$\Gamma \vdash_F \varphi, \qquad \|\varphi\|_F(\Gamma) := \min\{|\pi| : \pi \text{ proves } \varphi \text{ from } \Gamma\},$$

with value ∞ when there is no proof, and $\mathrm{Layer}_F(\Gamma, \varphi, n)$ for $\|\varphi\|_F(\Gamma) = n$. No formation of WFFs from connectives is assumed.

Definition F.2.2 (Verifier-gated guided architecture)**.** The architecture H consists of an exact addressing map, a geometry-guided certificate-testing subroutine, and a *protected universal proof enumerator* U. An independently specified verifier V checks proofs relative to F and Γ. A certificate is committed to the BANK only after V accepts it. The universal enumerator receives an unbounded sequence of protected service steps, irrespective of decisions made by the geometric subroutine. The architecture emits a theorem if either accepted search channel finds a certificate.

Definition F.2.3 (Matched finite-candidate task)**.** A task has $N \geq 2$ distinct candidates, exactly one of which is an accepted certificate for the designated statement. Its position J is unknown to the search policy. Testing any candidate, including its verifier check and associated rejection/acceptance handling, costs one *charged comparison unit* in the common implementation model. The baseline B tests candidates in a uniform random order. The guided strategy tests a predeclared neighborhood G first and then the complement, in each block using an order independent of the hidden accepting identity. Both strategies use the same sound verifier and candidate universe.

The benchmark is a distribution of q related tasks with common candidate-count N. The target-blind geometry, index maintenance, and selection processes are included in the guided cost. A protected universal service quantum may be performed *between*, rather than during, these finite subroutine calls; identical fixed universal service work for the two competitors is fully charged to both. This makes the subroutine cost comparison well defined without presuming that universal enumeration takes the same number of steps under different early-stopping policies.

F.3 The six revised hypotheses

The hypotheses have intentionally non-overlapping functions.

H1. Exact rule-preserving recoding. There is an effective bijective syntactic recoding t that carries each F-WFF to its F^t counterpart and preserves and reflects hypotheses and every legal inference-rule application. The recoded target is $t(\varphi)$ and the recoded premises are $t[\Gamma]$.

H2. Sound and positively complete verification. For finite candidate proof certificates c, the effective verifier halts and accepts exactly the valid F-proofs of their stated targets from Γ. Invalid certificates are never committed to BANK. This is a certificate-checking property, not a promise that the guided search will find a certificate.

H3. Protected complete enumeration. The proof presentation and hypotheses are effective enough to enumerate all finite candidate proof codes. The architecture explicitly runs a dovetailed enumerator of these codes and grants it infinitely many protected service steps in any infinite run. Every valid finite proof code is therefore eventually checked and, if accepted, emitted. For a finite list of benchmark tasks the protected enumerator continues after the list ends; for infinitely many finite tasks it receives at least one service quantum between consecutive tasks.

H4. Target-blind, quantified informative locality. For each benchmark task i, before discovering its accepting candidate a computable permitted-information rule defines a neighborhood G_i of size $1 \leq g_i \leq g_0 < N$. Conditional on the permitted structural data, the unique accepted candidate lies in G_i with probability $p_i \geq p_0 > 0$. Its position is uniform within the neighborhood conditional on inclusion and uniform within the complement conditional on exclusion. This is a statement about an *auditable task distribution*, not hidden knowledge of the accepted candidate.

H5. Constructively bounded guidance ledger. There is a specified data structure and algorithm (not just a claim that a patch is small) whose *total additional charged work* over the q tasks satisfies

$$D_q \le a_0 N L_N + q\big[(a_1\beta_0\Delta_0 + a_2 g_0)L_N + a_3\big], \qquad L_N := \lceil \log_2(N+1) \rceil, \tag{1}$$

with nonnegative constants a_0, a_1, a_2, a_3, and bounded update parameters β_0, Δ_0 independent of N. The ledger includes initial encoding/index construction, Hilbert-address computations, neighborhood acquisition, all changed records and their dependent index entries, scheduling and stopping checks, and nonshared memory/service overhead. One possible route is an $O(N \log N)$ indexed build and $O(\beta_0\Delta_0 \log N + g_0 \log N)$ work per task; these bounds *must be verified for the chosen implementation.*

H6. Matched charged comparison. Both policies operate over identical tasks, candidate certificates, verification semantics, proof system, and comparison units. Their common setup and fixed protected-universal service is separately recorded and cancels only when actually identical. No proof discovery, target lookup, geometric precomputation, or stopping cost is provided for free.

F.4 The new capstone theorem

Theorem F.4.1 (Certified geometric discovery and amortized search advantage)**.** *Suppose H1–H6 hold for a formal system F, premises Γ, the architecture H, and an eligible family of q finite-candidate benchmark tasks of common size N. Then:*

T1. Theoremhood invariance. *For all $\varphi \in y$,*

$$\Gamma \vdash_F \varphi \quad \Longleftrightarrow \quad t[\Gamma] \vdash_{F^t} t(\varphi).$$

T2. Exact shortest-proof-layer invariance. *For all $\varphi \in y$ and $n \in \mathbb{N}$,*

$$\mathrm{Layer}_F(\Gamma, \varphi, n) \quad \Longleftrightarrow \quad \mathrm{Layer}_{F^t}(t[\Gamma], t(\varphi), n).$$

T3. Finite certified discoverability. *For every $\varphi \in y$, if $\Gamma \vdash_F \varphi$, then the guided architecture eventually emits a certificate accepted by V:*

$$\Gamma \vdash_F \varphi \implies \exists m < \infty \; \exists c \, [\mathrm{Emits}_{\mathsf{H}}(\varphi, m, c) \wedge V(c, \varphi)].$$

T4. A strict net charged expected-cost advantage whenever the explicit inequality holds. *Writing C_{B} and C_{H} for the* total *matched charged cost of the q tasks, we have*

$$\boxed{\mathbb{E}[C_{\mathsf{B}} - C_{\mathsf{H}}] \;\ge\; \frac{q}{2}(p_0 N - g_0) - a_0 N L_N - q[(a_1\beta_0\Delta_0 + a_2 g_0)L_N + a_3].} \tag{2}$$

Hence, if the right-hand side is positive, then $\mathbb{E}[C_{\mathsf{H}}] < \mathbb{E}[C_{\mathsf{B}}]$.

Furthermore, if $p_0 > 0$, $g_0, \beta_0, \Delta_0, a_0, \ldots, a_3$ are independent of N, and

$$K > \frac{2a_0}{p_0}, \tag{3}$$

then there exists N_0 such that for every $N \ge N_0$ and every integer $q \ge K L_N$, the strict expected-cost advantage holds.

F.5 Proof

Proof of Theorem F.4.1. **T1 and T2: recoding.** For every finite proof $\pi = \langle \pi_1, \ldots, \pi_k \rangle$ of φ from Γ, H1 carries each hypothesis line and each inference step to a valid proof line in F^t. Thus $t(\pi)$ proves $t(\varphi)$ from $t[\Gamma]$ in k lines. The inverse recoding transports proofs back and preserves k. These maps give a length-preserving bijection between proof sets, proving T1 and equality of their minimum proof lengths, including the common value ∞. Thus T2 holds for every n. This recovers the core conclusion of *Depths of a Simulation*, Version 45, Theorem 4.8, without requiring connectives in WFFs.

T3: protected enumeration. Let $\Gamma \vdash_F \varphi$ and choose any finite proof certificate c of φ. Since the proof-code enumeration in H3 is exhaustive, c occurs at some finite enumeration index j. Since U receives infinitely many service steps in an infinite run, it reaches index j at a finite global stage m. By H2 the verifier accepts c; the emission protocol then emits φ with this certificate. Equivalently, the same conclusion follows by induction over the finitely many proof lines. Hence T3 holds even if the geometric subroutine repeatedly makes unproductive choices.

T4: expected candidate-rank saving. Fix task i, let $g = g_i$ and $p = p_i$. A uniformly randomized baseline tests its accepting candidate at expected rank $(N+1)/2$. Guided search tests the neighborhood first. If the accepting candidate lies in that neighborhood, its expected rank is $(g+1)/2$. Otherwise the guided search performs g rejected tests and then finds the accepted candidate at mean complementary rank $(N-g+1)/2$, so its overall expected rank is

$$\begin{aligned}\mathbb{E}[R_{\mathsf{H},i}] &= p\frac{g+1}{2} + (1-p)\left(g + \frac{N-g+1}{2}\right) \\ &= \frac{N+g+1-pN}{2}.\end{aligned}$$

Therefore, without any independence assumption among the tasks,

$$\mathbb{E}[R_{\mathsf{B},i} - R_{\mathsf{H},i}] = \frac{p_i N - g_i}{2} \;\geq\; \frac{p_0 N - g_0}{2}.$$

Linearity of expectation and the one-unit charged test convention yield total *gross* savings of at least $q(p_0 N - g_0)/2$. Identical common costs cancel only by H6. Subtract the additional guidance ledger D_q bounded by H5 to obtain (2). If its right-hand side is positive, the expected net-cost inequality is strict.

Amortized threshold. Define $A := a_1\beta_0\Delta_0 + a_2 g_0$ and $L = L_N$. The right side of (2) is

$$q\left[\frac{p_0 N - g_0}{2} - AL - a_3\right] - a_0 NL.$$

For all sufficiently large N the bracket is positive and equals $(p_0/2)N - O(\log N)$. If $q \geq KL$, it follows that the expression is at least

$$L\left\{K\left[\frac{p_0 N - g_0}{2} - AL - a_3\right] - a_0 N\right\}.$$

The coefficient of N in braces is $Kp_0/2 - a_0 > 0$ by (3), while the remaining terms grow only as $O(\log N)$. It is therefore strictly positive for all sufficiently large N. This proves the last assertion. □

F.6 Numerical illustration and scope

Choose, only to illustrate the algebra, $N = 10{,}000$, $q = 100$, $g_0 = 2$, $p_0 = 1/2$, $\beta_0 = 3$, $\Delta_0 = 2$, and $a_0 = a_1 = a_2 = a_3 = 1$. Then $L_N = 14$. Formula (2) yields

$$\mathbb{E}[C_{\mathsf{B}} - C_{\mathsf{H}}] \;\geq\; \underbrace{249{,}900}_{\text{gross rank savings}} - \underbrace{140{,}000}_{\text{initial index}} - \underbrace{11{,}300}_{\text{task overhead}} \;=\; \boxed{98{,}600}.$$

This is an *arithmetic demonstration*, not empirical evidence that an existing system has achieved $p_0 = 1/2$ with two preferred candidates or these particular machine costs. The probability and work bounds must be justified for the same task family and implementation before the conclusion can be applied.

Remark F.6.1 (T3 is not identical to weak fairness). Scheduling every action that remains enabled does not ensure that all proof-producing actions are ever generated or enabled. H3 instead prescribes a concrete complete proof enumeration and protected service allocation. T3 therefore holds independently of whether geometry is useful.

Remark F.6.2 (Proof length versus search cost). T2 is an invariance result about the shortest number of proof *lines*. T4 concerns a distinct operational quantity: the *charged expected number and cost of candidate inspections*. Exact recoding alone implies no search speedup.

Remark F.6.3 (Probability is over a declared benchmark distribution). T4 proves expected cost under H4's conditional-uniformity model. It does not imply worst-case speedup, an advantage on every instance, or an advantage on an unspecified distribution. If accepted-candidate locations are adversarial rather than sampled as specified, the displayed expectation may not apply.

Remark F.6.4 (The index-overhead obligation is substantive). A bound of $O(\Delta)$ changed clause records does not by itself prove $O(\Delta \log N)$ total updates. The costs of exact identities, Hilbert coordinates, all dependency updates, and the verifier must be analyzed. In particular, earlier examples with a highly informative two-action neighborhood may have unbounded variable occurrence, and therefore need not satisfy fixed Δ_0.

F.7 Open certification obligations

Before describing this as a certified result for a substantive existing solver family, the following tasks remain:

1. **Concrete family.** Identify a genuinely useful infinite family of proof-search or certificate-search tasks with a predeclared, target-blind neighborhood and rigorously proved $p_i \geq p_0$ and conditional rank distributions.

2. **Implement H5.** Supply complete code or pseudocode for the exact-address construction, geometry calculations, balanced-index build, dependent-record updates, and stopping protocol, deriving $a_0, \ldots, a_3$ in the same cost units as each verifier inspection.

3. **Verify uniform bounds jointly.** Establish fixed g_0, Δ_0, β_0 *on the same family*. A locality result from one example and an update bound from another do not suffice.

4. **Verifier integration.** Demonstrate that the protected enumerator's rule enumeration is complete, that checking is decidable and sound for the chosen formal presentation, and that no geometric decision can starve protected service.

5. **Independent audit.** Mechanically check the rank and algebra lemmas and independently test any claimed numerical ledger. No external ATP/AMLD+0 certificate for the full composition is claimed in this draft.

F.8 Provenance and interpretation

The proof-transport component is a reuse of Brian Tenneson's *Depths of a Simulation*, Version 45, especially Theorem 4.8. The finite-proof-search motivation is related to the user's effective proof-horizon and certified-settlement work. The locality and update hypotheses are informed by the user's Exact Addressing and Hilbert Addressing research. The *new contribution proposed here* is their explicit, verifier-safe composition with a fully charged amortized expected-rank inequality. The composition is mathematically conditional; it must not be cited as a settled universal algorithmic speedup or as proof of a nontrivial-family existence theorem.

References

[1] B. Tenneson, *Depths of a Simulation*, Version 45, especially Theorem 4.8 (proof transport and proof-length invariance), 2026.

[2] B. Tenneson, *Global Proof Horizons and Logical Doorways: Certified Settlement*, working paper 0.1, 2026.

[3] B. Tenneson, *Hilbert Addressing of 3CNF Well-Formed Formulas*, working source, 2026.

[4] B. Tenneson, *Exact Addressing*, preliminary working research versions, 2026.

Draft status. Conditional theorem with a written proof; unrestricted capstone conjecture replaced; benchmark realization and external verifier certification are open.
Manuscript policy. Integrated as Appendix F in the Exact Addressing 3.3.1 working edition; Sections 1–16 and Appendices A–E are unchanged.